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# **TITLE: PREDICTING DENGUE HOTSPOTS IN MALABON CITY THROUGH AI AND INTEGRATED WEATHER-CASE ANALYSIS**

A Capstone Project Presented to the Faculty of  
College of Computer Studies City of Malabon University

City of Malabon

Presented to the Faculty of the College of Computer Studies

CITY OF MALABON UNIVERSITY

Malabon City

In Partial Fulfillment of the Requirements for the Degree  
of Bachelor of Science in Information Technology

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**OCTOBER 2025**



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## APPROVAL SHEET

This Research entitled **“PREDICTING DENGUE HOTSPOTS IN MALABON CITY THROUGH AI AND INTERGRATED WEATHER-CASE ANALYSIS”** was prepared and submitted by **Conejero, Ivan F., Gabatbat, Jame Matthew, Litaban, Strawberry C., Natividad, Jeffrey John S., Oliveros, Ammaron C.**, in partial fulfillment of the requirement for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval for ORAL EXAMINATION.

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### **DECLARATION OF ORIGINALITY**

The undersigned hereby declares that this thesis, titled “Predicting Dengue Hotspots in Malabon City Through AI and Integrated Weather-Case Analysis”, submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology at City of Malabon University, is an original work.

This thesis, to the best of our knowledge, contains no material previously published or written by another person, nor does it contain any material which has been accepted for the award of any other degree or diploma at this university or any other educational institution, except where due acknowledgment is made in the text. All sources, including books, journal articles, websites, and any other research materials, have been fully and properly referenced in accordance with the university’s citation guidelines. We have also appropriately acknowledged the contributions of all individuals who have assisted in the completion of this work, including supervisors, research participants, and any others.

We understand that plagiarism and other forms of academic dishonesty are serious offenses and may result in disciplinary action, including but not limited to, a failing grade for this thesis, suspension, expulsion, or withdrawal of the Bachelor of Science in Information Technology degree awarded to us.

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Above all, the researchers give thanks to Almighty God, whose guidance and blessings made the completion of this research possible.

The completion of this thesis is a proof to the collective support of everyone mentioned above. While the mistakes remain our own, the successes reflect their contributions. Thank you.



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## **DEDICATION**

We, the researchers, sincerely dedicate this capstone project to our beloved parents and families, whose unwavering support, sacrifices, and encouragement have been our constant source of strength and inspiration throughout this journey. Your faith in us has provided the motivation and perseverance to accomplish this milestone.

We also dedicate this work to our adviser, Professor Cristielle-Love A. Flores, and our mentors, to whom we extend our deepest gratitude for your invaluable guidance, patience, and support. Your knowledge and dedication have significantly contributed to the completion of this study and to our development as students and future professionals.

To our friends and classmates, who have shared the challenges and successes of this endeavor, we are truly grateful for your cooperation, understanding, and encouragement.

Above all, we offer this work to Almighty God, whose divine guidance, wisdom, and strength have enabled us to accomplish this project.



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## **ABSTRACT**

Research Title: Predicting Dengue Hotspots in Malabon City Through AI and Integrated Weather-Case Analysis

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Dengue fever continues to be one of the most pressing public health challenges in the Philippines, especially in urban areas such as Malabon City, where population density and environmental conditions increase vulnerability to outbreaks. Traditional dengue monitoring systems often rely on delayed reporting and manual data analysis, which limit the ability of local health authorities to implement timely and preventive interventions. To address this issue, the study developed a predictive web-based system titled Predicting Dengue Hotspots in Malabon City Through AI and Integrated Weather-Case Analysis. The system integrates artificial intelligence (AI) with real-time and historical data to forecast potential dengue hotspots and provide an evidence-based approach for disease surveillance and management.

The main objective of the study was to design and implement a system capable of analyzing dengue case records in conjunction with meteorological data such as rainfall, temperature, and humidity to determine areas at high risk of dengue transmission. The system was developed using Python and Django



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frameworks and utilized machine learning algorithms trained on datasets obtained from the Department of Health (DOH) and the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA). Evaluation was conducted by three IT professionals using ISO 25010 quality standards to assess reliability, usability, and functionality.

Results showed that all major system features—including authentication, dashboard visualization, hotspot prediction, weather integration, and report generation—performed effectively and produced accurate outputs. The evaluators rated the system as “Good” across all testing criteria, confirming its stability and responsiveness. Overall, the findings indicate that the Dengue Hotspot Predictor System is a reliable, user-friendly, and efficient tool that can assist local government units in predicting and mitigating dengue outbreaks through data-driven and proactive decision-making.

*Keywords: Dengue Prediction, Artificial Intelligence, Weather Data Integration, Disease Surveillance, Public Health Informatics*



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## **Chapter 1**

### **INTRODUCTION**

Dengue virus infection remains a global health problem, particularly in tropical countries such as the Philippines. Tens of thousands of cases each year combine for a persistent threat that has pushed our health care systems — and our communities — to the brink, sparing no age group: Infants, older adults, and young children have suffered significant effects. The City of Malabon belongs to urban areas with very high population density and is classified as a flood-prone area because of its inadequate channel systems. These are the areas where mosquito-borne diseases, including dengue, far outnumber other infectious illnesses. Early detection, prevention, and prompt response were the best ways to mitigate the impact of dengue outbreaks, the Health Department further said. But the current methods — like case reports — may be a step behind the actual course of disease spread. They remain reactive rather than proactive.

Recent technologies such as Artificial Intelligence (AI) and data analytics for disease surveillance and prediction are receiving increasing attention for addressing these issues. Results from several other countries have been encouraging too, and crucially, a range of other influences — among them dengue cases, rainfall, humidity, temperature, and population density — are used in the prediction model. Now, with new technology, we can draw on large datasets to spot signals of emerging outbreaks — trends and patterns that indicate an outbreak is imminent. This is particularly true in the era of climate change, where continued



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weather perturbations may extend the breeding season and create conditions for yet another outbreak.

The proponents of this capstone project propose developing an AI-based dengue hotspot predictor for Malabon City using machine learning and an integrated analysis of historical dengue case data and weather variables. The system aims to provide a forecast of each barangay's dengue risk in Malabon City by analyzing 4 years of historical data from the city health department, along with current and forecasted meteorological data sourced from reliable weather APIs. User-friendly interface that Malabon city's health officials or barangay leaders can use to view dynamic dengue hotspot maps and make data-informed decisions on local health intervention needs, vector control activities, and other related matters.

Even though flooding is a common cause of mosquito population explosion, AI-enabled prediction and real-time/forecasted environmental data have yet to be incorporated into a cost-effective web-based setup that can be tailored for local requirements, such as our implementation in Malabon City (a high-risk area for mosquito-borne diseases associated with frequent flooding). This study is consistent with the vision and aims of public health surveillance and disaster preparedness among BHPs in urban communities in the Philippines. Laying the foundation of this capstone project on the powerful Django Framework means our system will be strong, responsive, and secure — something that every community stakeholder really needs. The system development life cycle (SDLC) structured



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methodology should also be incorporated in the project design, which is critical to ensuring proper planning, testing, deployment, and implementation.

After all, the importance of this capstone is to prevent our LGUs from responding in a reactive manner and to issue a proactive health response. By employing predicted dengue hotspots in advance, the CDS can support an informed decision-making process to take preventive measures or reduce the potential impact of the spread. Furthermore, this work may also expand the existing literature on the use of machine learning to address health-related issues in the Philippines, which is lacking.

The capstone project team hopes to demonstrate that judicious application of technology at the right place on a local level can be an effective way to resolve endemic health problems. It is designed not only as an academic endeavor but as a contribution to our society. A framework that incentivizes innovation to benefit and safeguard vulnerable communities.

### **The Problem and Its Background**

Dengue fever continues to be an alarming health concern in the Philippines. Most rainy seasons bring many outbreaks that fill hospital beds and might overwhelm health workers. Many are coming down with the illness, mainly children and older people. Malabon City in northern Metro Manila is notorious as a flood-prone area, at low elevation and close to the sea. Its crowded population, buildings, and houses make it a pleasant place for the mosquitoes that transmit



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dengue — *Aedes aegypti*, the yellow fever mosquito, included — to grow in numbers. In the rainy season, when water supplies from flooding and poorly drained systems give mosquitoes a place to flourish and lay eggs.

In a joint statement with the Malabon City Health Office, the Department of Health said dengue is likely to rise during heavy rains and when humidity is high. And these attacks continue to plague the health centers and disrupt any sense of normalcy for those living nearby. It also creates financial and emotional strain on a resident. Despite regular clean-up drives and information campaigns by the barangay, cases of dengue still persist as a serious health concern in Malabon City.

This capstone project aims to develop an AI web tool that identifies dengue risk areas in Malabon City by analyzing weather data and historical dengue case data. The study will collect and use dengue records and weather information from the past 4 years in Malabon. The researchers will then build a machine learning model trained to integrate different datasets to predict each barangay's risk level. Finally, the web application for this capstone project will be developed using the Django framework.

This app will let users view an interactive map showing real-time dengue risk information. It also includes parts that can be used by health workers and barangay staff, so they can see the important info in their areas. The system will try to follow the ISO/IEC 25010 standard for software quality, but the focus is more on how well it works, whether it's easy to use, fast, and whether it can be fixed or



moved when needed. In the Philippines, dengue is one of the most serious health problems that many health workers continue to work to address and prevent. They make extra efforts to stop outbreaks and keep the community safe. Lastly, the study will determine whether the system is truly helpful to local health officials when planning dengue prevention and to identify early which areas may pose a dengue risk.

### **Related Literature and Studies**

This chapter reviews the literature on recent and previous works to justify the existence of the proposed system. The paper is organized into five primary sections relevant to how the system will be developed and ultimately applied: database management, user interface design, data analysis, system implementation, and machine learning. In each section, we describe how other scientists have used these tools for health surveillance and, in particular, for disease forecasting during the last 4 years. By reviewing past studies, this chapter identifies best practices, technology standards, and gaps that this research aims to address. The review of related studies serves as the basis for the proposed system, both in theory and in practice. This will support the goal of predicting dengue hotspots in Malabon City by combining historical dengue cases and weather data.



## Database Management

Ghosh et al. (2023) also introduced a mixed ARIMA–ANN model with the number of dengue cases as an endogenous variable and precipitation as an exogenous variable. Relational databases and timestamped MySQL records were of the utmost importance for historical accuracy. This shapes our system design, since we are getting both barangay-level dengue rates and the monthly weather data in Django and PostgreSQL, to make accurate, auditable predictions.

Barboza et al. (2022) presented a more accurate weather-enhanced dengue prediction using models that were saved on a standard PostgreSQL schema, and ETL has been scheduled. Their core principles of clean data alignment and efficient querying have motivated our implementation of equal normalization and ETL (i.e., barangay-level input storage for model retraining and query speed).

The study by Pham et al. 2020 used an LSTM–Random Forest pipeline based on a weather-epidemiological database, with underlying API-based data ingestion and cleaning that included timestamp alignment. Their process guides our own data ingestion and data preprocessing pipeline in Django that defines the automatically updating system, structured, validated input of the monthly barangay forecast graphs we plan to generate.



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Morgan et al. (2021) established a geo-epidemiologic data model integrating climatic, socio-economic, and demographic factors for enhanced risk assessment by elastic data associations. This frames how we design our schema—initially to weather and dengue data, but with potential for extension to include localized demographic or environmental inputs as needed in the context of barangay-level prediction.

Silva & Perez (2023) developed a Django-based dengue alert system with daily climate and case data ingestion, database normalization, and rollback features for data accuracy. Their backend architecture closely aligns with ours, reinforcing our plan to implement rollback-capable migrations and structured data validation to ensure reliability in barangay-level forecasting.

Although the current literature provides a good groundwork for dengue prediction and health data systems, some challenges in their local adoption, user interaction, and system scalability at the barangay level are still present. The majority of the previous studies (Ghosh et al., Barboza et al., Pham et al.) were based on national or regional level datasets, which were not real-time-integrated with visualization or localized health applications. Furthermore, techniques such as the one established by Morgan et al. investigated variable integration in depth but did not specifically focus on usability for frontline health officers. Differing from those, our work is specifically tailored to work at the barangay level, making it a 1)



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barangay-granularity-level predictive model as well as an application in combining machine learning forecasts and mobile-responsive map-based dashboards for both public users and barangay health personnel admins. Our system employs Django and PostgreSQL systems, with automatic ETL pipelines, rollback safety, and modular schema design, not only to meet the technical levels of previous works but also extend existing work in terms of hyperlocal accuracy (vs block level), improved public interactivity, and some admin-oriented tools for decision making that make it a suitable candidate for implementing in resource challenged urban settings like Malabon City.

### **User Interface**

From Singh et al. (2022), we noticed that a user-friendly application interface for the real-time monitoring of COVID-19 could be implemented with React Native. The use of tabbed dashboards, color-coded and responsive layouts, became our veto to deploy Django with Bootstrap for mobile access to barangay-level dengue alerts for maps and banners. Their usability reports directly influence our emphasis on ease of use and engagement.

Lopez & Chan (2021) developed a typhoon information system that also provides real-time maps and progressive disclosure to ensure the user can get all information without everything cramping the interface. We take their understanding of color hierarchy and alert framing to help guide our UI strategy—teasing dengue



threats at a glance, while still allowing deeper exploration through expandable layers and GPS-based barangay search filters for localized forecasts.

Narayan & Patel (2023) focused on minimalism and trust in a health risk dashboard, where cluttered interfaces led to reduced user confidence. Their clear structuring and explanation of the risk score help us define a simple and educational UI containing charts, severity indicators, and a glossary. Their research supports the importance of visual clarity and trust to users for predictive systems such as ours, particularly for both non-experts and decision-makers.

Huang et al. (2020) presented a hospital dashboard with card-based designs and color-coded health alerts to increase glanceability. Their real-time visualizations and ability to toggle between various data types inform our interface, which displays barangay-level risks in graphs showing green, yellow, and red risk levels, as well as ways to filter risk by time or location. Their architecture aligns with our target API for fast, data-driven turnarounds for barangay health personnel.

Using Leaflet. JS and OpenStreetMap, Reyes & De Guzman (2022), developed a web-to-user interface with an interactive map. They used tooltips and modals to aid in the comprehension of complicated data without interface clutter. We use nearly identical methodology for dengue risk display—GeoJSON maps,



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popups, severity modals—to surface predictions in a manner that they can be easily acted upon by the public users. and LGUs of Malabon.

Although previous systems have effectively utilized interactive and informative dashboards, most were geared towards national crises or an institutional setting; they did not maintain the hyperlocal focus and public-facing capabilities that are necessary for barangay-level public health tools. Studies such as those of Singh et al. and Lopez & Chan visually communicated well, but they neither combined geospatial data and predictive modeling into a platform for disease surveillance localized to the village-level. Unlike Narayan & Patel and Huang et al., who are more concerned with clarity and response as they continue work in clinical or administrative environments, they are almost absent from public interaction. Unlike any other, it combines forecasting with geospatial mapping alongside interactivity and public accessibility in a single mobile-friendly web application. It was developed for dengue communication at the barangay level. It likewise utilized color coding to indicate the level of risk, implemented filters for predictive maps, gave glossary aids, and conducted administrative views to help translate predictive insights for citizens and LGU officials. This full-featured solution fills an important gap in local epidemic response capabilities - the intersection of data science and health communication - which directly affects community access.



## Data Analysis

According to Li et al. (2021), the authors analyzed the features of the dengue outbreak in Southern China against the background of the environment. They found that the seasonality pattern and changes in temperature are the primary predictors of dengue outbreaks. They used spatiotemporal mapping and visualization using Python, and that got us thinking about linking up the monthly case data we had in Malabon with weather datasets on rainfall and humidity to further unveil segment trends over time at the barangay level.

Ahmad et al. (2022), in analyzing dengue transmission in the urban setting of Pakistan, showed a strong association of cases with rainfall and humidity. The impact these two weather factors have on case counts is what interests us. The application of interpolation and smoothing methods to their noisy data to help us focus on our core analysis was inspirational for our preprocessing pipeline. A good database is vital to analytics and forecasting.

Morales & Teng (2020) performed barangay-level clustering of dengue data in Metro Manila, uncovering patterns of hotspot persistence through five years of DOH records. Their use of unsupervised learning methods like K-means shapes our own system's analytical layer, where recurring case trends per barangay will be monitored to detect emerging risks over time.



Bui et al. (2023) explored disease overlaps using a hybrid model and multiple inputs—rainfall, mobility, and population density. They applied PCA and correlation heatmaps to refine their feature set, reducing redundancy. This guides our selection of impactful predictors (e.g., rainfall vs. temperature) to improve model accuracy while minimizing noise in barangay-level forecasts.

Salazar & Reyes (2021) investigated the lagged relationship between weather variables and dengue occurrences in the Visayas region. They were able to argue that the weather conditions that existed one to two months ago strongly predicted the dengue outbreaks. Such a finding has made the incorporation of lag variables into our model almost indispensable since it has already been established that rainfall and humidity from previous months can be used as early warning indicators of coming spikes in dengue risk.

Nonetheless, a lot has been learned from past studies regarding the interplay of environmental and time factors in the case of dengue outbreaks, but most of the studies were retrospective, and thus they missed the opportunity for real-time, geographically targeted analysis. Li et al. and Ahmad et al. gave strong evidence for environmental correlations; however, they did not engage communities through the development of interactive systems, like most of the research. Morales & Teng's barangay-level clustering in Metro Manila identified persistent hotspots but did not go further in their dissemination of findings through



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predictive tools or alert systems. Just as Bui et al. and Salazar & Reyes upped the ante in methodological rigor via feature selection and lag analysis, their work also did not transition into operational, forecast-driven platforms for frontline use. Our program, however, not only conducts analytical modeling but also produces a fully operational, barangay-level forecasting web tool that utilizes real-time weather and dengue data, takes lagged effects into account, and applies clustering insights. By integrating the cleaned and normalized inputs into a predictive dashboard, we are able to provide the public and health administrators with early warning capabilities—thus, effectively bridging the gap between scholarly work and real-world public health.

### **System Implementation**

Ocampo & Miranda (2022) built a Django-MySQL health monitoring system for clinics in Quezon City, featuring SMS alerts, weekly reports, and cloud-based role access. Their focus on timely data delivery directly informs our own system's architecture using the Django Framework to store and visualize barangay-level dengue forecasts through dashboards. Their predictive health systems show that even LGUs lacking resources can effectively run these systems given the right training and infrastructure.



In Taiwan, Liu et al. (2020) developed an AI-based disease surveillance tool with real-time analysis of regional outbreaks using Django, TensorFlow, and Celery. Their system guides our own setup in combining Django with background tasks and PostgreSQL to run AI-driven dengue predictions. Their work validates the feasibility of deploying ML models in public health tools that inform local decision-making rapidly and accurately.

Rosales & Tagle (2023) deployed a barangay-level disaster response dashboard using AWS, real-time reports, and SMS alerts during typhoon season. Although disaster-focused, their success in training LGU users to interpret real-time data directly supports our plan to build a dengue tool that's accessible to barangay officers. Their system shows that with proper interface design and onboarding, even non-technical personnel can operate web-based early warning tools effectively.

Kang et al. (2021) developed a modular flu alert system using Flask, background tasks, and API-linked components for prediction and visualization. While Flask is lighter than Django, their separation of ingestion, processing, and forecasting workflows shapes our own design—where model training, prediction scheduling, and frontend display are modular but tightly integrated. This modularity ensures scalability and maintainability for barangay-level deployment. Santos & Dela Cruz (2022) launched a school health system in Manila using Django,



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MongoDB, and real-time alerts through email and SMS. Their use of role-based dashboards and QR logins for health monitoring reinforces our own approach in designing an intuitive, multi-role dengue prediction system with automated notifications. Their project confirms the viability of using Django to communicate health risks quickly and clearly within community settings.

While previous systems successfully demonstrated the feasibility of web-based health monitoring and alert tools, many were either general-purpose, domain-specific (e.g., disaster response or school health), or lacked predictive forecasting capabilities. Systems like those of Ocampo & Miranda and Santos & Dela Cruz proved that Django can support real-time alerts and role-based dashboards, but their implementations did not extend to machine learning-driven prediction or dynamic geospatial visualizations. Liu et al.'s use of TensorFlow offered advanced analytics but operated at a regional scale rather than at the hyperlocal barangay level. Meanwhile, Rosales & Tagle and Kang et al. showcased modular and user-trainable systems but focused on reactive response rather than proactive risk forecasting. In contrast, capstone incorporates all other things, fine-tuning these fundamentals by integrating hotspot predictions using AI, automated backend modularized systems, and customizable local visualizations in color-coded maps, all built within a Django-PostgreSQL architecture web structure designed specifically for remote barangay access. This closes the gap between epidemic forecasting and practicality, providing local government units with a fully



operational and analytical data system as an early warning tool, combining its epidemiological precision and public health relevance.

### **Machine Learning**

Abeywardena et al. (2021) used Random Forest and Gradient Boosting to predict dengue cases in Sri Lanka with 89% accuracy, using weather and historical case data. Their use of barangay-like zones and feature importance analysis confirms our choice to apply Random Forest for barangay-level prediction in Malabon, focusing on temperature and rainfall as key predictors.

Tashman & Arif (2020) used LSTM models in dengue forecasting in Indonesia, capturing seasonal phenomena and overall patterns in the data, and achieving 92% accuracy. LSTM and lagged weather-enhanced feature predictive models give promise for future system upgrades. Preprocessing, however, will require the use of structured time-series arrangements.

Dizon & Palma (2022) employed SVM and correlation-based feature selection in creating an AI model that predicted the barangays in Tacloban that are affected by typhoons and floods. Their use of sanitized data and barangay-level disaggregation of data in the predictive model strengthens our choice of preprocessing, which involves feature reduction as well as the use of open data so that the model is able to meet our target of high predictive accuracy.



Zhang et al. (2023) conducted an outbreak prediction study on malaria using climate and vegetation data with XGBoost. Their use of SHAP for model interpretability and dashboards for visualization informs our plan to integrate Random Forest or XGBoost with explainable AI features. Their study proves ensemble models can deliver actionable local health forecasts when paired with a clear interface.

Using Decision Trees and geospatial flood data, Garcia & Yabut (2021) constructed a barangay-level leptospirosis predictor in Metro Manila, attaining an accuracy of 87%. Their successful use of visual alerts and time-aware training confirms our approach to integrating forecast-driven maps and temporal modeling for proactive dengue risk management at the barangay level.

Although prior studies have demonstrated the predictive strength of machine learning models like Random Forest, LSTM, and XGBoost in infectious disease forecasting, most implementations remained limited in scope—either focusing on regional data or omitting real-time deployment and interpretability. Abeywardena et al. and Garcia & Yabut utilized high-performing models for localized prediction but lacked integration with dynamic web dashboards or explainable outputs for end-users. Tashman & Arif achieved high accuracy using LSTM, but did not explore its use in interactive systems for LGU decision-making.



Dizon & Palma emphasized barangay-level granularity, yet their application remained static, not offering ongoing updates or user-facing forecasts. Meanwhile, Zhang et al. introduced model interpretability via SHAP but applied it only to malaria risk and not within a user-accessible tool. In contrast, our capstone project unifies these innovations, leveraging ensemble models like Random Forest or XGBoost, barangay-level forecasting, explainable AI (via feature importance or SHAP), and a Django-based interface that delivers real-time, understandable risk maps. This comprehensive integration transforms theoretical accuracy into practical action—empowering LGUs in Malabon with both predictive precision and accessible health insights.

### **Synthesis**

The comprehensive literature review conducted in this study highlights the importance of data-driven systems, especially those based on AI and web technologies, in improving public health surveillance and response. The combination of structured data management, user-friendly user interface (UI) design, solid data analytic methods, user-friendly system deployment, and AI-driven (advanced) machine learning (ML) models is a primary ingredient of current predictive health systems, which coincides with the goals of this study.

The studies reviewed here demonstrate that integrating health and environmental data with AI forecasting tools can help address public health challenges. Some studies by Ghosh et al. (2023), Barboza et al. (2022), and Pham



et al. (2020) discuss the need to use databases and structured data in order to ensure this information is correct when predicting outbreaks. They also employ an automatic approach for data, but the majority of their work concentrates on large areas (national or regional level), not small ones. In our project, we've made it smaller and more focused by looking at it at the barangay level to provide local insights appropriate for Malabon City. In addition, while previous works in the literature used data on a daily or weekly basis, our system can provide monthly forecasts for wise and early planning to prevent dengue, which can be of assistance to LGUs.

The work on user interface by Singh et al. (2022), Lopez and Chan (2021), and Reyes & De Guzman (2022) highlights that a user's trust in the system to use it requires a responsive design, easy navigation, and clear visuals. Their system relied mostly on black/white alert or maps, but what started as simply tightly integrating our work into this environment became a robust series of color usage with an interactive dengue risk map, layers that can put more data, and different dashboards for which user you are, like a public member, a health worker, or an admin. We'll also add features, such as tooltips and a step-by-step walkthrough of the details, to help non-technical users understand the predictions better — something that wasn't always present in previous systems.

Django and Celery-based cloud servers can also play a significant role in improving the performance and reliability of real-time health systems, as Liu et al. (2020), Rosales and Tagle (2023), and Santos and Dela Cruz (2022) have shown.



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Through their study, they showed the utility of these technologies in improving system performance and data management. But most of the systems they created were for specific categories, like schools or small organizations, and were more geared toward following up on problems than anticipating them. Our system goes beyond that by adding AI-based forecasting instead of just monitoring, and by defining background tasks to automate both data ingestion and prediction deliveries. It's also user-friendly, interpretable, and operable by BHW without phasing the need for technical assistance – a form of upgraded community-based empowerment.

In terms of AI modeling, studies by Abeywardena et al. (2021), Tashman and Arif (2020), and Zhang et al. (2023) showed how models like Random Forest, LSTM, and XGBoost can be effective in predicting disease trends. Most of these studies focused mainly on accuracy, but only a few tried to explain how their models made predictions or adjusted them for local use. In our project, we improved this by using a more transparent approach—specifically, a Random Forest model with SHAP values—to make the results easier to understand. The model is also customized for barangay-level decision-making, combining both past dengue cases and live weather data. It also gives confidence levels in its predictions, so local health workers can plan more wisely.

Overall, while our study is built on many past works, it goes a step further by focusing on smaller communities, using integrated forecasting, explainable AI, and a role-based interface. These features aim to make advanced machine



learning more useful and practical for local health offices, helping them respond better to dengue cases in their areas.

### **Theoretical Framework**

The capstone project, “Predicting Dengue Hotspots in Malabon City Through AI and Integrated Weather-Case Analysis,” is made from different ideas that mix artificial intelligence, data analysis, and public health. The framework mainly explains how the system will work to predict where dengue might happen in some barangays in Malabon.

### **Epidemiological Triad Theory**

Epidemiological Triad Theory consists of three primary components that researchers can use to understand how dengue spreads between the virus and people in their region. In order to make predictions, they need to consider a few things at once: what the weather is like, how the virus behaves, and what sorts of people live in an area. Our machine scans old dengue case files and also queries weather data, how much it rains , and how hot it gets, to predict what could lead to an outbreak. It's not always easy to figure out, but combining all of this can help show which areas may get more cases.



### **Theory of Predictive Analytics**

This theory is about how organizations can take the data they already have and turn it into knowledge that helps them guess what might happen next. Our machine learning model uses old dengue case records and weather information to try to figure out which areas might become high-risk for dengue. Information can help the Local Health Department create effective and evidence-based prevention strategies to achieve desirable community results.

### **Information Systems Success Model (DeLone and McLean, 2003)**

System success occurs through six important dimensions, which include System Quality, Information Quality, Service Quality, Use, User Satisfaction, and Net Benefits. The framework guarantees system development produces precise forecasts while maintaining user-friendliness, decision support, and public health advantages for local government units. The evaluation of system usefulness, together with accessibility and reliability, will be conducted through this particular model.

### **The Decision Support System**

This theory is generally about how computer systems might help people to do a better job of such decision-making using data. A DSS is not necessarily trying to think for people. It's only information that can assist (conventionally from multiple sources and ordered, if not always accurate). It's there to help people sort out



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problems and decide what needs doing, especially when the decisions are made in a hurry and have to use the best available information, as happens with public health work.

In this study, the project entitled “Predicting Dengue Hotspots in Malabon City Through AI and Integrated Weather-Case Analysis” is aligned with DSS theory. By integrating Historical dengue case data and weather case data like rainfall, temperature, and humidity, the system helps flag which areas are more vulnerable to dengue outbreaks. This gives local health workers and barangay health officials the opportunity to devise and undertake plans for preventive interventions for potential outbreaks early on.

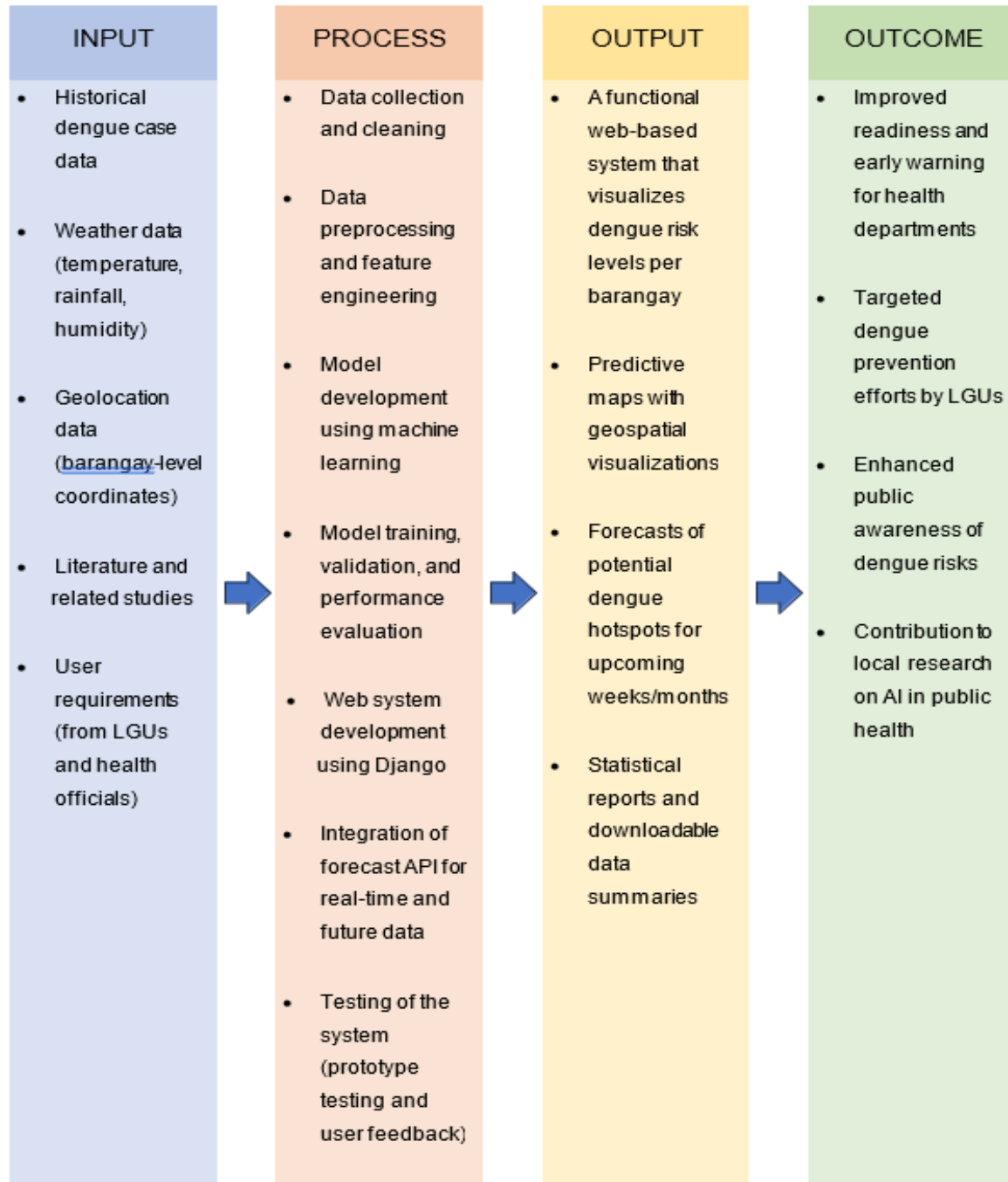
As a Decision Support System, it identifies areas with higher numbers of dengue cases, examines the progression of cases over time, and analyzes weather patterns to identify possible correlations with outbreaks. This allows for streamlined allocation of resources and targeted intervention planning in order to enhance dengue prevention and control in Malabon City.



## Conceptual Framework

**Figure 1.1**

### *Conceptual Framework Design*





This capstone applies the Input-Process-Output-Outcomes (IPOO) model to illustrate the conceptual framework. In the Input stage, the study collects historical dengue case records, weather information, and location data, which form the foundation for the prediction models. The Process stage covers the key activities of the study, such as data preprocessing and building the machine learning models, while also implementing the system using the Django web framework. The system further connects to forecast APIs to provide real-time and upcoming weather information. In the Output stage, the system provides a fully working web-based platform that shows dengue risk levels for all barangays in Malabon City, with geospatial maps and reports that users can download. In the Outcome stage, it highlights the wider social benefits, like better health planning, quicker responses from local government units, and greater public awareness of dengue prevention. Overall, this framework allows the study to use artificial intelligence to address public health problems through decisions based on data.

### **Statement of the Problem**

The Philippine government has implemented a regular dengue prevention project, but the disease remains a significant health issue in urban flooding areas such as Malabon City. Current surveillance systems do not have the capacity to generate real-time localized predictions of outbreaks in time to prevent disease from spreading widely. There is a climate and weather feature used for hatching



malaria mosquito larvae that needs prediction in order to have an advanced warning of dengue-prone areas.

The capstone project aims to fill this gap by creating a Malabon City-specific AI-based dengue prediction system.

To guide the development and evaluation of the system, the study seeks to answer the following capstone study questions:

1. Does integrating historical dengue case data with weather data help predict dengue hotspot locations within Malabon City?
2. What machine learning model is most effective in forecasting barangay-level dengue risks?
3. What system design features enable real-time, user-friendly, and accurate visualizations for both health officials and local residents?
4. What is the expected level of precision in forecasting dengue case patterns in 1–2 weeks ahead?
5. What benefits do local government units gain from deploying the system to enhance their early-response capabilities and dengue outbreak planning?



## Hypothesis/Assumptions

### Hypothesis

The assumptions of this study are:

- **Null Hypothesis ( $H_0$ ):** There is no significance of the predictive model using historical dengue cases and weather variables (precipitation, temperature, humidity) to the recorded dengue cases in the barangays of Malabon City.
- **Alternate Hypothesis ( $H_1$ ):** There is a predictive model using AI that correctly correlates recorded dengue cases in the barangays of Malabon City, together with historical weather data (precipitation, temperature, humidity) and historical records of dengue cases, and thereby accurately predicts high-risk areas.

### Assumptions

This capstone study is based on the following assumptions:

- **Infrastructure and Reliability of Sources:** The historical weather data from government agencies and previous instances of the disease in the Malabon City Health Office are complete and reliable data sources corresponding to the period of the study.
- **Weather Variables:** Rainfall, temperature, and humidity in the weather are key variables in the reproduction of the *Aedes aegypti* mosquito, which, in turn, facilitates the spread of the disease in the area.



- **Geo-Consistency:** Geographic and demographic conditions of Malabon City's twenty-one barangays were similar between the data collection period and the data analysis.
- The developed web application can be used efficiently by health workers and barangay officials due to their sufficient digital applications, skills, and internet availability.
- The employed machine learning algorithms can predict future risk levels accurately and with acceptable margins of error by generalizing historical data.

### **Significance of the Study**

The research holds importance because it tackles the ongoing rising health threat of dengue fever in the Philippines by presenting a modernized technological approach. The predictive system that uses AI and integrated weather-case analysis to identify dengue hotspots in Malabon City creates expected advantages for various sectors:

### **Local Government Units (LGUs) and Health Departments**

The system functions as an important decision-support platform that offers early alerts and risk visualizations to help barangay health workers and city health officers. The system helps health officials use their resources better and perform



focused interventions and implement preventive actions at the right time, which leads to lower dengue outbreak effects.

### **Community Members and Residents**

Local government units now deliver real-time and forecasted dengue risk details, which motivates citizens to perform precautionary measures, including removing mosquito breeding areas and seeking early medical care for improved community health resistance.

### **Academic and Research Communities**

The research adds new knowledge to the expanding collection of local studies that examine artificial intelligence in disease prediction. The research proves successful integration of public health information with environmental variables to solve problems specific to particular communities.

### **Future Researchers**

The project serves as an applied example that demonstrates the development process for predictive web applications through Django and machine learning. Developers who want to build comparable solutions for health or environmental challenges can use this project as a development guide.



Finally, this study exists as an answer to an existing requirement of society. The implementation of modern technologies for regional health issues leads toward sustainable and predictive disease prevention methods throughout the Philippines.

### **Scope and Limitations**

This research is focused on creating a web-based prediction platform that uses artificial intelligence along with weather-case analysis to detect possible dengue outbreaks in Malabon City. Four years of past dengue case records, together with matching weather parameters such as rainfall, temperature, and humidity, will help the system produce risk predictions for individual barangays. The application will use Django as its framework, and it will provide a simple interface that enables users to access present and future dengue risk assessments across the city. Barangay health personnel, together with barangay officials, health workers, and residents who want to track dengue activity in their areas, will use this system.

However, the study is subject to several delimitations that define its scope and limitations. The research is geographically confined to Malabon City, and its findings are not intended to be generalized to other cities or provinces. The system focuses solely on predicting dengue cases, thereby excluding other vector-borne and infectious diseases. Although the system generates predictions based on the integration of health case records and meteorological data, the quality and consistency of these data sources may vary. Furthermore, the study utilizes only



four years of dengue case data, as provided by the Malabon City Health Department, which may limit its capacity to identify long-term trends or patterns in disease transmission.

Additionally, due to financial constraints, the researchers were unable to subscribe to premium weather data services. As a result, the Open-Meteo API used in this study can only provide weather forecasts of up to two weeks, thereby limiting the temporal range of the predictive model. The use of third-party APIs for weather data retrieval may also restrict customization in terms of data precision and update frequency. Lastly, it is important to emphasize that the predictions generated by the system are probabilistic in nature and should be interpreted as risk indicators rather than absolute forecasts.

Despite these limitations, the system provides valuable insights that can support evidence-based decision-making, improve dengue surveillance, and strengthen proactive public health intervention efforts within Malabon City.



### Definition of Terms

To establish clarity and consistency throughout this study, the following key terms are defined as they are used within the context of the research:

- **Artificial Intelligence (AI):** A field of computer science concerned with creating systems capable of performing tasks that typically require human intelligence. In this study, AI is employed to analyze complex datasets involving dengue case reports and weather patterns to predict future outbreaks.
- **Barangay:** The smallest administrative unit in the Philippines. For this study, dengue case data and predictive risk levels are collected and analyzed on a per-barangay basis to provide more localized insights.
- **Climate-Sensitive Diseases:** Diseases such as dengue that are influenced by climatic conditions like temperature, rainfall, and humidity. These variables directly affect mosquito population dynamics, which is critical to this research's predictive modeling.
- **Confidence Score:** A numerical indicator that represents the level of certainty or reliability of each prediction produced by the model. Higher



confidence scores suggest stronger agreement between input data and the predicted outcome, helping users assess the credibility of the forecast.

- **Data Integration:** Refers to the process of combining dengue case data and meteorological information from different sources into a unified system for analysis. In this study, data integration ensures that both health and weather datasets are synchronized and compatible, allowing the prediction model to generate accurate and meaningful results.
- **Data Preprocessing:** The technique of cleaning, transforming, and organizing raw data into a usable format before analysis or machine learning. This step is essential in ensuring data quality and model accuracy in the system.
- **Django:** A Python-based web development framework used to build secure, scalable, and maintainable web applications. It is used in this project to create the backend and frontend of the dengue prediction system.
- **Ensemble Model:** In machine learning refers to a method that combines multiple individual models to produce a more accurate and robust prediction than any single model could achieve alone.



- **Epidemiological Data:** Information related to the incidence, distribution, and control of diseases. In this context, it refers to the historical dengue case records used to train and validate the prediction model.
- **Forecast API:** A type of Application Programming Interface that provides future weather data for a specific location. It allows developers to programmatically access weather predictions, such as temperature, rainfall, humidity, wind speed, and other meteorological variables for upcoming hours or days.
- **Forecast Horizon:** Refers to the period of time into the future for which the system can generate predictions. In this study, the forecast horizon is limited to two weeks due to API constraints, which define how far ahead the system can predict potential dengue hotspots.
- **Forecasting Model:** A computational tool or algorithm that predicts future events based on past and present data. In this study, the forecasting model estimates the likelihood of dengue outbreaks in the near future.
- **Geospatial Visualization:** The representation of data that is spatially referenced, often using maps. The system uses this technique to display barangay-level dengue risks on an interactive city map.



- **Hotspot:** A geographic area identified as having a higher-than-average concentration of dengue cases. The system predicts such hotspots to enable early interventions.
- **Interactive Map:** A visual component of the system that displays dengue risk levels for each barangay using geographic markers and color-coded indicators. The interactive map allows administrators and users to easily interpret spatial data, view real-time risk updates, and make data-driven decisions for targeted interventions.
- **Lag Variable:** a variable that represents the past value of another variable in a time series dataset. It is used to capture time-dependent patterns, like how previous weather conditions or dengue cases influence current or future values.
- **Machine Learning (ML):** A subset of AI that enables computers to learn patterns from data and improve prediction or classification tasks without being explicitly programmed. The system uses ML to recognize dengue outbreak trends from weather and case data.



- **Meteorological Data:** Data related to weather conditions, such as rainfall, humidity, temperature, and wind speed. These are integrated into the predictive model to identify correlations with dengue outbreaks.
- **Open Data:** Publicly available datasets that can be freely used, modified, and shared. In this research, open data sources such as weather APIs and health department statistics form the basis of the training dataset.
- **Predictive Analytics:** The practice of using statistical and machine learning techniques to analyze current and historical data to make forecasts about future outcomes. It is the core methodology used in the dengue hotspot prediction system.
- **Prototype Testing:** The process of evaluating an early version of a system to identify bugs, test performance, and refine functionalities before final deployment. This ensures that the system functions as intended under real-world conditions.
- **Risk Classification:** A method of categorizing the predicted dengue risk levels of each barangay based on computed probabilities. The classification uses color-coded legends—green for low, yellow for moderate, orange for



high, and red for very high—to help administrators and health officials interpret and visualize the severity of potential outbreaks.

- **Risk Mapping:** A technique used to visualize areas with varying levels of health risk. In this study, dengue risk levels are color-coded and mapped by barangay to enhance decision-making.
- **Supervised Learning:** A machine learning approach where the model is trained on a labeled dataset (data with known outcomes). This method is used in the system to teach the algorithm how to associate input features (e.g., weather) with dengue case trends.
- **Validation Metrics:** Statistical tools (such as precision, recall, and accuracy) used to evaluate the performance of a predictive model. These are used in the study to assess the reliability of the forecasts.
- **Web-Based System:** A software application that operates through a web browser without requiring local installation. The system developed in this study is fully accessible online for users such as health officers and barangay leaders.



## **Chapter 2**

### **METHODS**

This chapter outlines the research methodology used to conduct the study. It details the research design employed, explaining the approach taken to investigate the research problem. The chapter then describes the population and sampling techniques used to select the study participants, followed by a description of the research instrument and the data gathering procedure. It also specifies the data analysis and statistical treatment methods applied to the collected data. Finally, the chapter addresses the ethical considerations that were observed throughout the research process to ensure the integrity and safety of the study.

#### **Research Design**

In this study, the researchers chose a Quantitative Predictive Research Design to forecast dengue hotspots in Malabon City efficiently. This sort of methodology aims at collecting and analyzing numerical data, which is made up of meteorological elements such as temperature, precipitation, rainfall chance, flooding risk, wind speed, wind direction, and humidity. Furthermore, the researchers can also develop an AI Model for historical dengue case records for each barangay. Through this data-driven research, the proponents hope to discover patterns and predictions of high-risk areas in the city, as well as provide



a strong foundation that is based on statistical and computational perception of the disease to enhance the accuracy of prediction hotspots.

The quantitative approach during preprocessing, together with training and evaluation stages, utilizes measurable elements, including dengue incidence data and weather parameters, which establish important statistical connections between these variables. The modified dataset served as training material for a predictive model, which used Random Forest or Gradient Boosting methods to identify dengue risk levels in barangays by assigning them low, medium, or high-risk classifications.

Additionally, system assessment is also governed by the ISO/IEC 25010 standard, which defines software quality characteristics. Quality will be evaluated according to software reliability, functionality, universal design, efficiency, and maintainability, as well as portability that satisfies technical and user requirements.

This design allows the researchers to actually build an implementable and scalable AI-based tool to aid the Malabon City Health Department in dengue surveillance and community health planning. It facilitates informed decision-making with real-time and predictive information presented in an easy-to-use interface built on the Django web framework.



## Software Architecture

The software architecture behind the “Dengue Hotspot Prediction System Using AI and Integrated Weather-Case Analysis” is thoughtfully designed to support continuous improvement. Rather than being a rigid system, it works like a smart, learning ecosystem constantly evolving as it gathers more data and feedback. It all begins with the Weather API, which regularly sends in real-time weather updates such as rainfall, temperature, and humidity. This live data is stored in the central database, along with dengue case reports from various barangays and predictions generated by the AI. By keeping all these data sources in one place, the system can better understand how weather patterns influence dengue outbreaks across Malabon City.

At the heart of the system is the AI Model Server, which runs powerful algorithms like Random Forests and Neural Networks to analyze the combined weather and case data. These models are trained to identify patterns and forecast where future outbreaks might happen. The process is overseen by the Web Server, built using the Django framework, which acts like the system’s control tower. It handles all requests from users, manages model training schedules, and ensures the AI predictions are always updated based on the newest information.

Admins play a key role in managing the system—they take care of uploading new data, checking for errors, and kicking off training sessions whenever needed. Meanwhile, **health and public officials** use the platform to



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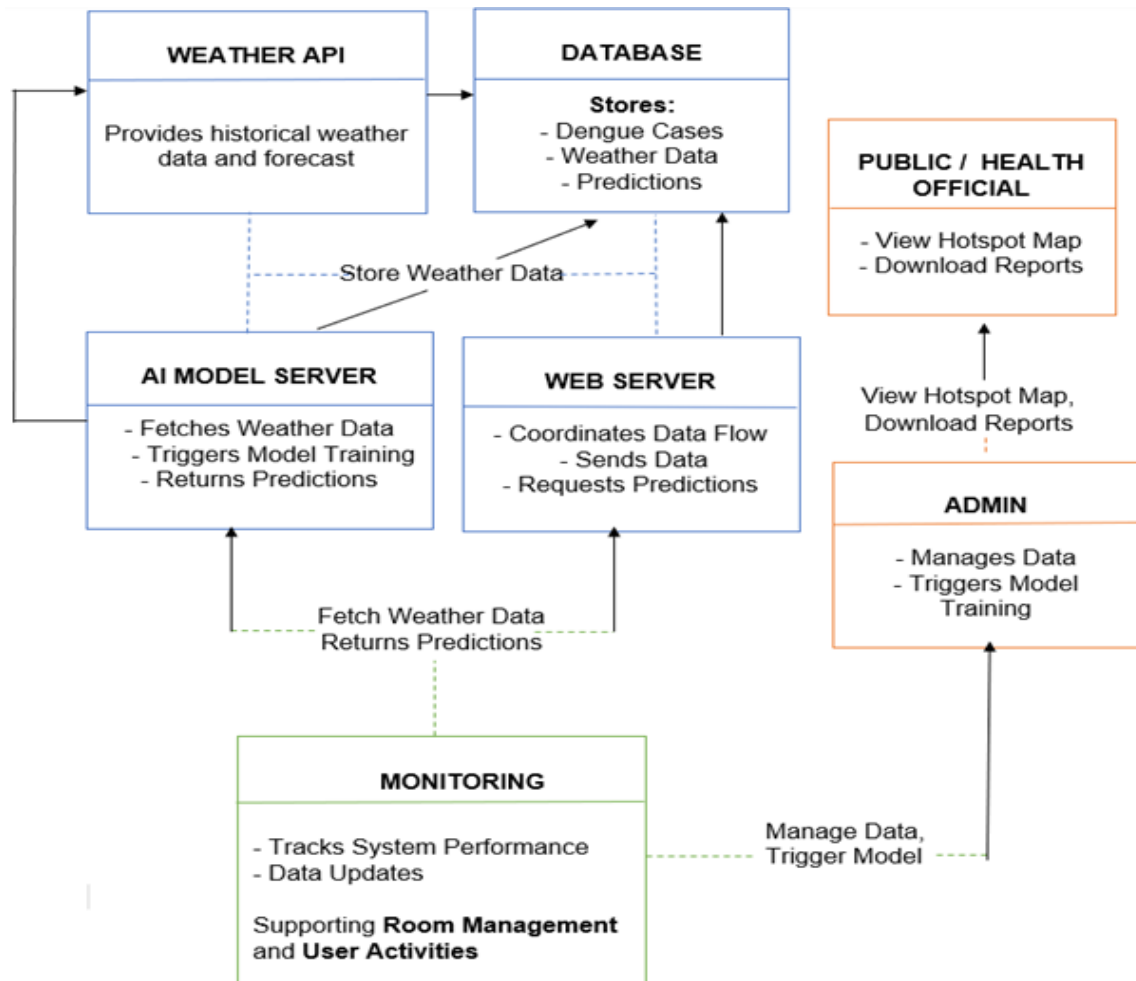
access risk maps, track case trends, and download customized reports. These reports help them make smarter decisions, like where to deploy medical teams or launch cleanup drives. The system is also equipped with monitoring tools to make sure everything runs smoothly in the background from checking API connections to tracking system performance.

One of the most important parts of this architecture is its feedback loop. After health officers observe the real conditions in the field, they provide insights on whether the predicted hotspots matched actual dengue cases. This feedback isn't ignored—it's fed directly back into the AI model, helping it to become more accurate over time. It's a cycle of learning and improvement that keeps going until the system consistently delivers the best value possible.

In the end, this architecture isn't just about technology—it's about building a reliable partner for Malabon City's health officials. By turning data into timely and actionable insights, the system helps protect communities and supports the city's goal of reducing dengue cases before they spread out of control.

**Figure 2.1**

*Software Architecture Diagram*





## Software Development Methodology

The researchers adopted an **Iterative Incremental Development Methodology** to ensure that the Dengue Hotspot Prediction System remains flexible, user-focused, and responsive to real-world needs. This approach is particularly well-suited for complex, data-driven projects like ours, where requirements may shift and understanding deepens over time. Rather than building the entire system all at once, the project is divided into smaller, manageable phases, each one allowing the team to develop, test, and refine specific features. Every iteration produces a functional version of the system, starting from a basic prototype and gradually evolving in complexity as new insights and user feedback are gathered.

In the early phases, the focus is on establishing a solid foundation. This includes integrating real-time data from a trusted weather API, uploading historical dengue case data, and training the initial version of the AI model using machine learning algorithms such as **Random Forest** and **Neural Networks**. With each cycle, the model's performance is evaluated and adjusted to improve accuracy and responsiveness. At the same time, the **user interface** and **GIS mapping tools** are developed incrementally. These components are shaped by feedback from local government health officers and community experts, ensuring that the system is practical, easy to use, and aligned with actual on-the-ground needs in Malabon City.

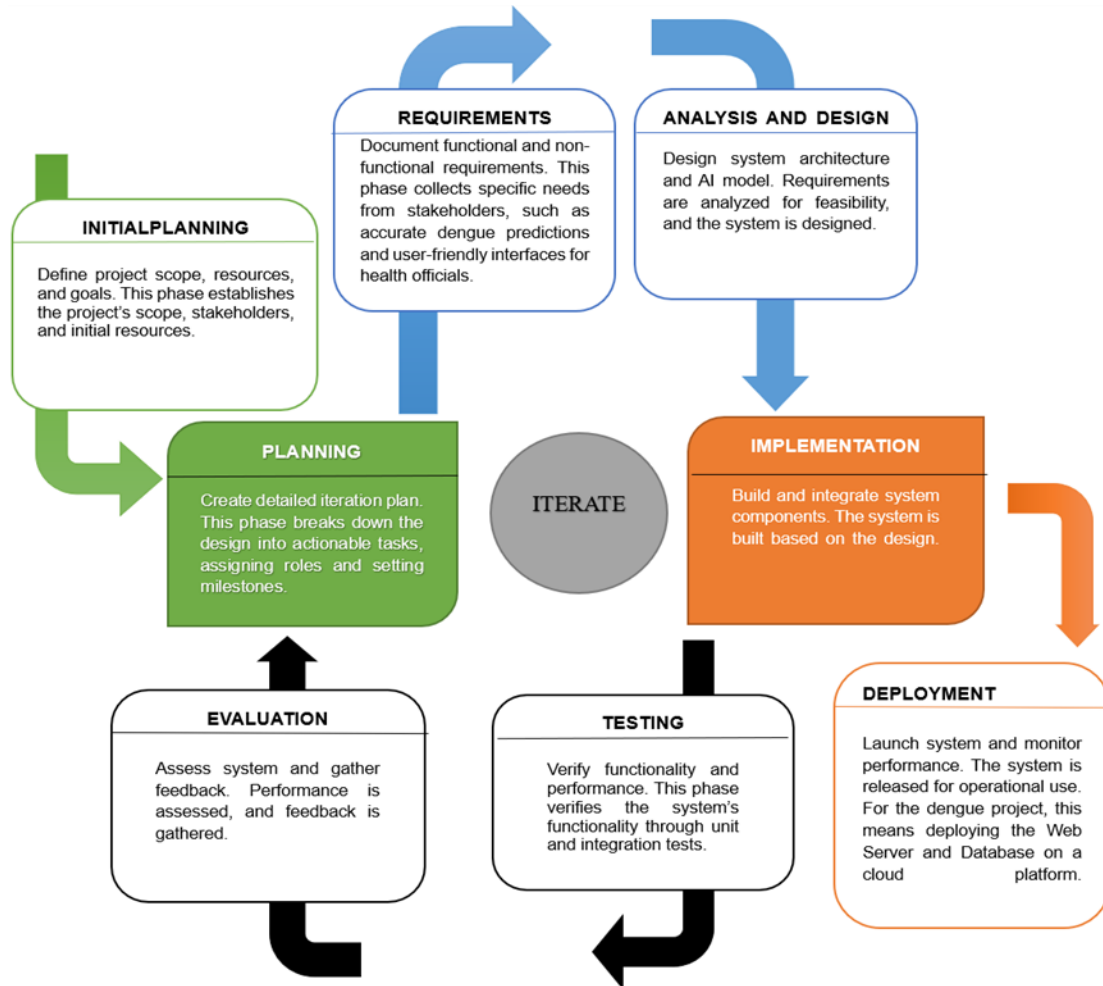


One of the greatest strengths of this methodology is its adaptability. Because every cycle includes testing and feedback, the team can respond quickly—whether it's refining prediction results, simplifying user interactions, or resolving technical issues. Engaging end-users throughout the development process not only leads to a more effective system but also builds confidence and trust in its real-world application.

In the end, this iterative and incremental approach supports more than just technical growth it fosters continuous learning for both the system and the people who rely on it. With each improvement, the platform becomes more accurate, more intuitive, and more aligned with the goal of **predicting dengue hotspots in Malabon City**. It ultimately serves as a reliable decision-support tool to help local health authorities act faster, allocate resources wisely, and protect communities more effectively.

**Figure 2.2**

*SDLC Iterative Incremental Model*



The iterative model above functions as a cyclical software development method and project management framework specifically designed for the capstone project which builds a "Dengue Prediction System Using AI and Weather Data". The development process uses repeated cycles of planning and development and



testing and evaluation to achieve continuous improvement which helps the system adapt to new requirements and challenges. Each stage receives an in-depth description together with its summary which explains its part in the entire development sequence. Initial Planning establishes project foundations while the system undergoes Requirements gathering and Analysis and Design for system design.

The system development begins after Detailed Planning which precedes Implementation. The system evaluation process uses testing to confirm operational capabilities and evaluation to assess performance results and collect user response. The system enters deployment phase and the central Iterate function operates through repeated cycles which use acquired knowledge to make system improvements. The capstone project persists with its iterative development until it reaches its established targets while remaining aligned with project timelines and goals.

### **Data Flow Diagram**

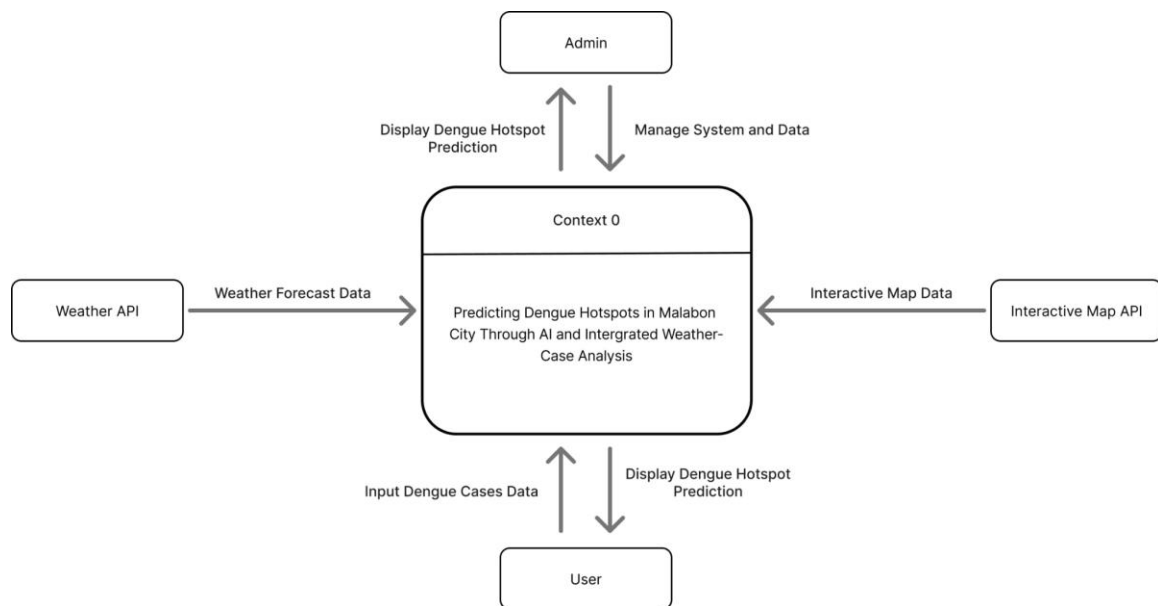
A Data Flow Diagram (DFD) is a graphical representation that illustrates how data moves through a system, showing the flow of information between processes, data stores, and external entities. In this study, the DFD serves as a vital tool for visualizing how dengue-related data, weather forecasts, and predictive analytics interact within the proposed system. It depicts how information is



gathered, processed, and displayed through modules such as the Admin Panel, Prediction System, Weather Forecast, and User Dashboard. The diagram is structured across multiple levels, beginning with the Context Diagram (Level 0), which provides an overview of external interactions, followed by Level 1 and Level 2, which break down internal processes in greater detail. Through this hierarchical structure, the DFD offers a clear and comprehensive understanding of the system's data flow, ensuring that each component contributes effectively to accurate dengue prediction and timely information dissemination.

**Figure 2.3**

*Data Flow Diagram Level 0*



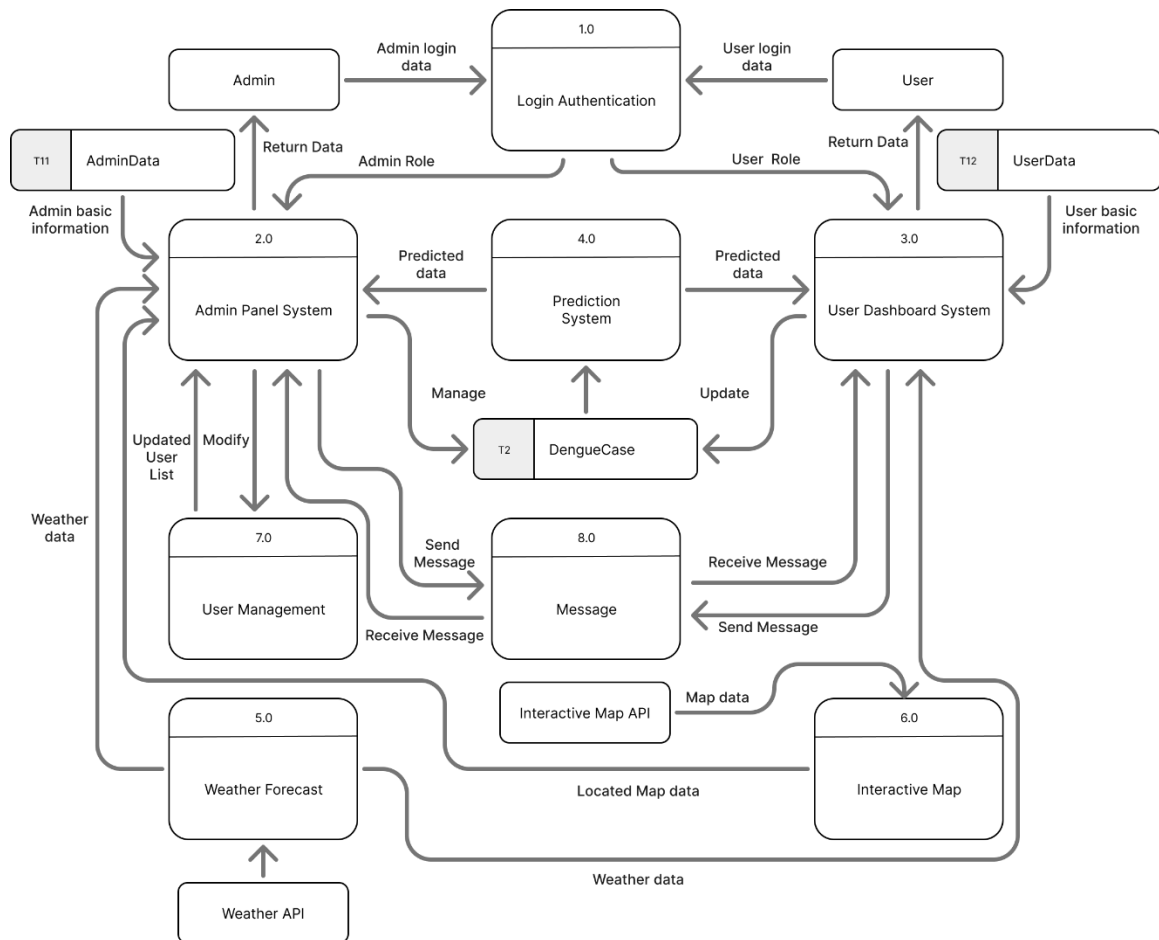


The Context Level 0 Data Flow Diagram provides a holistic representation of the proposed system, “Predicting Dengue Hotspots in Malabon City Through AI and Integrated Weather-Case Analysis.” This diagram illustrates the system as a single process that interacts with various external entities through defined data exchanges. The primary function of the system is to analyze dengue case data and weather conditions to generate predictive insights identifying potential dengue hotspots across Malabon City. The admin serves as a system manager responsible for maintaining datasets, overseeing operations, and receiving analytical outputs for monitoring and validation purposes.

The User contributes by inputting dengue case data into the system and accessing the generated hotspot predictions through an interactive interface. In addition, the system integrates data from external sources such as the Weather API, which supplies real-time meteorological information vital to the predictive analysis, and the Interactive Map API, which facilitates the visualization of hotspot predictions in a geographical format. Collectively, the Context Level 0 DFD delineates the system’s operational boundary, its interactions with external actors, and the overarching data flow that supports its predictive functionality. This diagram establishes the conceptual foundation for the detailed decomposition presented in the subsequent levels of the data flow model.

**Figure 2.4**

*Data Flow Diagram Level 1*



The Level 1 Data Flow Diagram presents a detailed decomposition of the main process identified in the Context Level 0 diagram for the system “Predicting Dengue Hotspots in Malabon City Through AI and Integrated Weather-Case Analysis.” This level illustrates the internal structure of the system by dividing it into six interconnected subprocesses, each representing a key functional component.



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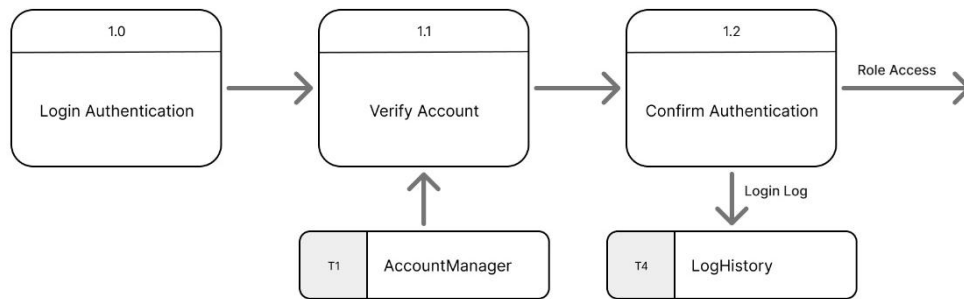
The Login Authentication (Process 1.0) serves as the initial entry point for both Admin and User entities, verifying credentials and assigning appropriate access roles. Upon successful authentication, the Admin Panel System (Process 2.0) allows administrators to manage system operations, including account maintenance, dengue case data, and message handling through the associated data stores: AccountManager (T1), DengueCase (T2), and Message (T3).

The Prediction System (Process 4.0) functions as the core analytical component of the project, utilizing input from the Weather Forecast (Process 5.0) to integrate meteorological data with dengue case records for predictive modeling. The results generated by this process are subsequently transmitted to the User Dashboard System (Process 3.0), where users can access and visualize dengue hotspot predictions in an interactive format. The Interactive Map (Process 6.0) supports this functionality by providing spatial data used for visual representation of hotspot areas. Overall, the Level 1 DFD illustrates the system's major subsystems, their interconnections, and data flows, emphasizing how each component collaborates to deliver AI-driven predictions and interactive mapping for dengue monitoring. This diagram provides a comprehensive view of the internal operations that underpin the high-level process established in the Context Level 0 DFD.



**Figure 2.5**

*Data Flow Diagram Level 2 Subprocess 1.0*



The Level 2 Data Flow Diagram further decomposes the Login Authentication (Process 1.0) identified in the Level 1 DFD, providing a detailed view of the internal processes involved in verifying user credentials and granting system access. This level illustrates how authentication is systematically managed within the system to ensure secure access control for both administrative and user roles. The process begins with the Login Authentication module receiving login credentials from the user, which are then validated through Verify Account (Process 1.1). This sub-process interacts with the AccountManager (T1) data store to retrieve and compare stored account information against the provided input. Once the account credentials are confirmed as valid, the system proceeds to Confirm Authentication (Process 1.2), which finalizes the login process, determines the user's access role, and logs the event into the LogHistory (T4) data store for monitoring and audit purposes. The process concludes by granting Role

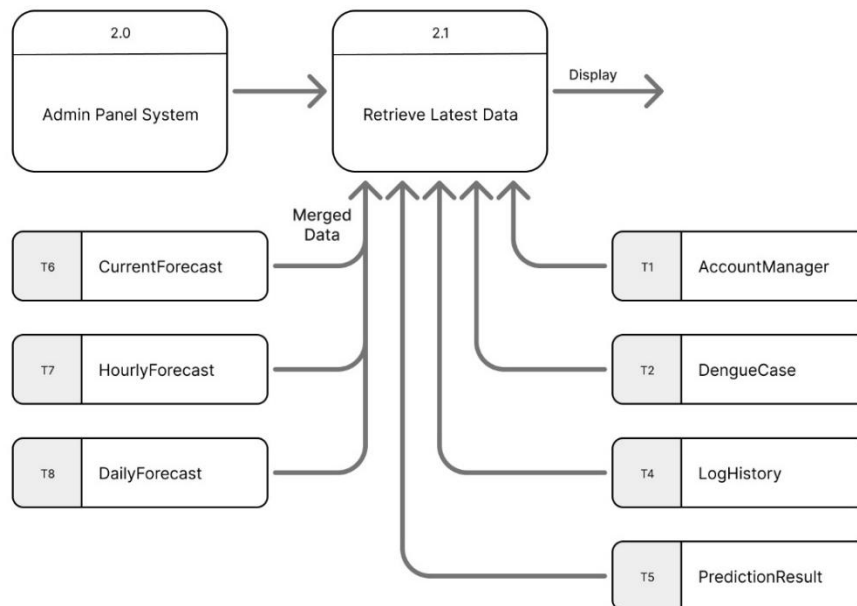


Access to the authenticated user, directing them to their respective system interface based on assigned privileges.

Overall, the Level 2 DFD provides a comprehensive breakdown of the authentication workflow, emphasizing data validation, logging, and access control as key security mechanisms. This decomposition highlights how the system ensures the integrity of user sessions and protects sensitive data through structured and traceable authentication procedures, reinforcing the reliability and accountability of the entire predictive system.

**Figure 2.6**

*Data Flow Diagram Level 2 Subprocess 2.0*





The Level 2 Data Flow Diagram provides a detailed decomposition of the Admin Panel System (Process 2.0) identified in the Level 1 DFD. This level illustrates how the administrative interface retrieves and consolidates essential datasets from various system components to support data monitoring, management, and decision-making. The process begins with the Admin Panel System invoking the Retrieve Latest Data (Process 2.1) sub-process, which aggregates and synchronizes information from multiple internal data stores. These include the AccountManager (T1) for user and administrator records, DengueCase (T2) for case monitoring and statistics, LogHistory (T4) for system activity tracking, and PredictionResult (T5) for predictive analytics outcomes.

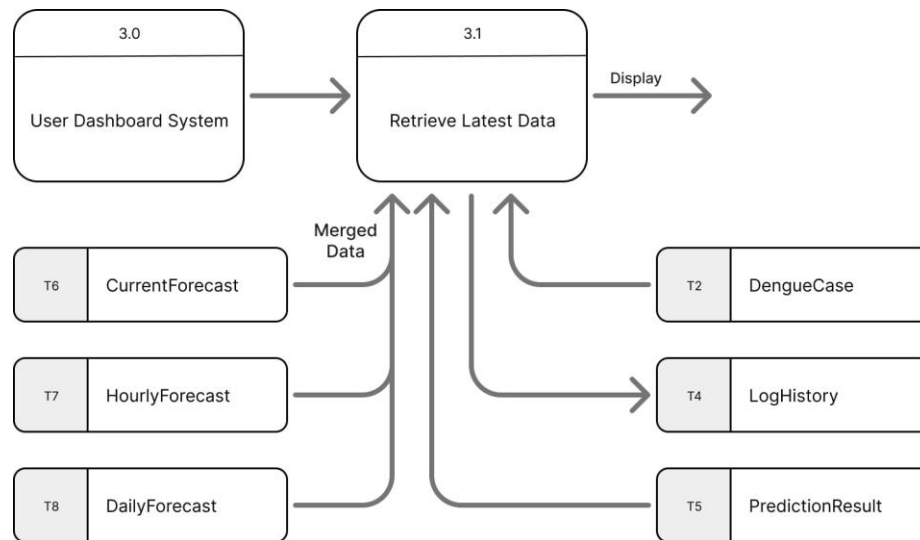
In addition to these internal data sources, the system also integrates external environmental data through CurrentForecast (T6), HourlyForecast (T7), and DailyForecast (T8) data stores. These provide real-time and time-based weather updates essential for analyzing and predicting dengue outbreak patterns. The retrieved information is merged into a unified dataset, ensuring consistency and accuracy before being displayed on the administrative dashboard. This consolidated view enables the administrator to oversee system performance, manage user activities, and assess predictive trends effectively.

Overall, the Level 2 DFD for the Admin Panel System emphasizes the integration of operational, environmental, and predictive datasets to support comprehensive administrative control. It demonstrates how the system ensures

data synchronization and accessibility, thereby enhancing decision support and maintaining the reliability of the dengue prediction framework.

**Figure 2.7**

*Data Flow Diagram Level 2 Subprocess 3.0*



The Level 2 Data Flow Diagram (DFD) for Process 3.0 provides a comprehensive representation of the internal process flow within the User Dashboard System, emphasizing how data is retrieved, integrated, and presented to end-users in a meaningful format. This level of detail illustrates the system's role as the primary interface for user interaction and information dissemination. The main subprocess, 3.1 Retrieve Latest Data, is responsible for gathering and merging information from multiple internal and external sources to ensure that



users consistently receive accurate, real-time insights. These sources include DengueCase (T2), which stores reported dengue case data; LogHistory (T4), which records detailed logs of user activities and system access; and PredictionResult (T5), which holds the outputs generated by the system's predictive model. Additionally, the subprocess integrates environmental data obtained from CurrentForecast (T6), HourlyForecast (T7), and DailyForecast (T8), which provide up-to-date weather information such as temperature, humidity, and rainfall—factors that significantly influence dengue case trends.

After the necessary data has been collected, the system consolidates and processes it into a unified dataset, ensuring consistency and eliminating redundancies across various data sources. This integrated dataset is then transformed into structured visual outputs displayed within the User Dashboard Interface. Through this interface, users can view, analyze, and interpret dengue case trends, predictive analytics, and relevant environmental data in an intuitive and interactive format. By combining multiple data streams into a single visualization layer, the system empowers users—whether they are citizens, health workers, or decision-makers—to better understand and respond to potential dengue hotspots in their area.

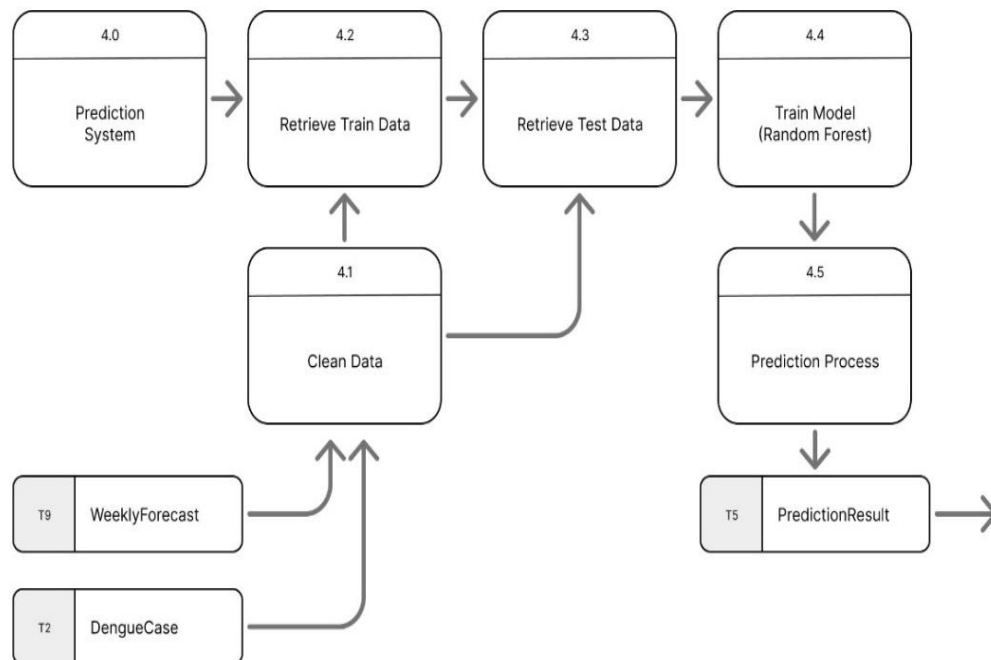
Overall, this Level 2 DFD effectively illustrates how the User Dashboard System functions as the central visualization component of the Dengue Hotspot Predictor. It demonstrates the system's emphasis on real-time data integration, transparency, and user accessibility, ensuring that critical information is



continuously available to support informed decision-making and proactive community action. By transforming complex analytical outputs into clear, user-friendly visualizations, the dashboard bridges the gap between predictive modeling and practical application, promoting awareness and supporting dengue prevention and monitoring efforts across affected areas.

**Figure 2.8**

*Data Flow Diagram Level 2 Subprocess 4.0*



The Level 2 Data Flow Diagram for Process 4.0 presents a detailed view of the internal operations of the Prediction System, which is responsible for generating dengue hotspot predictions based on integrated weather and case



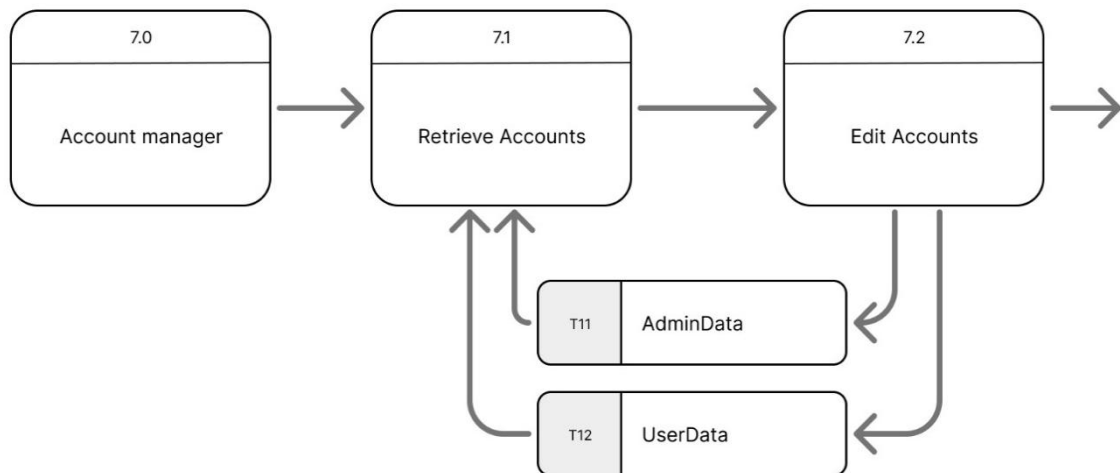
data. This process begins with 4.1 Clean Data, which retrieves raw inputs from WeeklyForecast (T9) and DengueCase (T2) data stores. The data cleaning phase ensures that missing values, inconsistencies, and noise are removed to improve the accuracy and reliability of subsequent modeling processes. Once the data is refined, it is passed to 4.2 Retrieve Train Data and 4.3 Retrieve Test Data, where the system divides the dataset into separate subsets for training and validation purposes.

The 4.4 Train Model (Random Forest) process represents the core of the machine learning component, in which the Random Forest algorithm is trained using the prepared training dataset. This step allows the model to learn complex patterns and relationships between environmental variables and dengue case occurrences. After training, the 4.5 Prediction Process applies the trained model to the test dataset or live input data to generate forecasts of potential dengue hotspots. The resulting predictions are then stored in the PredictionResult (T5) data store for access by both the Admin Panel and User Dashboard systems.

Overall, this Level 2 DFD emphasizes the systematic workflow of data preprocessing, model training, and predictive analysis that underpins the dengue prediction framework. It highlights the intelligent use of integrated datasets and the application of the Random Forest algorithm to produce accurate, data-driven insights that can support timely public health decision-making and intervention strategies.

**Figure 2.9**

*Data Flow Diagram Level 2 Subprocess 7.0*



This Level 2 Data Flow Diagram provides a detailed breakdown of the Account Manager (Process 7.0) sub-process, which is a component of the broader Admin Panel System. This diagram focuses specifically on the functionalities for managing user and administrator accounts within the system.

The process is initiated when an administrator accesses the Account Manager. The core function of this module is handled by the Retrieve Accounts sub-process. This sub-process queries the system's data stores, specifically the AdminData (T11) and UserData (T12) repositories, to fetch a complete list of account information.

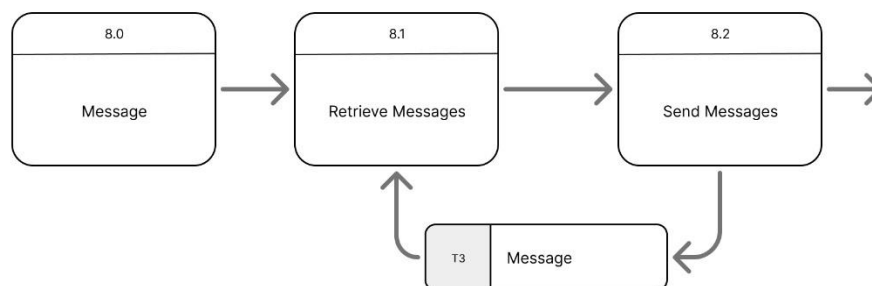


After recovery of the accounts, the administrator can make changes using the Edit Accounts sub-process. This can be used, for example, to update account settings (e.g., permissions or user info). Any modifications are then atomically committed to the respective data stores, guaranteeing that both AdminData and UserData stores have up-to-date information.

In short, this Level 2 DFD helps to explain the two primary operations of account management: first, requesting existing account records from distinct data stores for both administrators and users; secondly, offering the ability to edit and update those records to preserve the integrity and security surrounding access control on the system.

**Figure 2.10**

*Data Flow Diagram Level 2 Subprocess 8.0*





This Level 2 DFD presents a detailed picture of the Message process, which demonstrates how the system will handle transmissions and receptions.

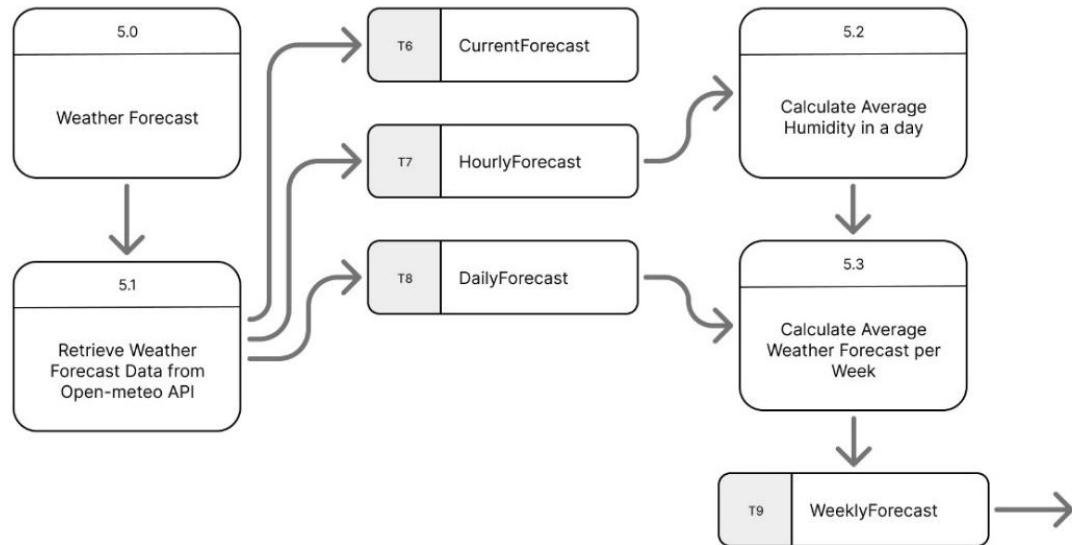
The whole process revolves around a data store called Message. The operation starts with the sub-process Retrieve Messages (Process 8.1). This subprocess retrieves existing message data from the Message data store to allow users (or an administrator) to view communication history and received messages.

This 2nd main function is the sub-process Send Messages (Process 8.2). This module generates and sends new messages. When a new message is created, it is persisted in the Message data store, thus creating an audit trail as well as delivering the message to the recipient.

In brief, in this Level 2 DFD, the two core processes of the messaging system – reading old messages from storage and adding new outgoing messages to be deposited in storage –are shown, which gives a complete communication cycle within the system.

**Figure 2.11**

*Data Flow Diagram Level 2 Subprocess 5.0*



The Level 2 Data Flow Diagram for Process 5.0 illustrates the detailed internal workflow of the Weather Forecast subsystem, which is responsible for collecting, processing, and summarizing meteorological data to support the dengue prediction model. The process begins with 5.1 Retrieve Weather Forecast Data from Open-Meteo API, where real-time weather data such as temperature, rainfall, and humidity are gathered from the Open-Meteo API. This data is then distributed to three data stores: CurrentForecast (T6), HourlyForecast (T7), and DailyForecast (T8), which organize the retrieved data according to temporal granularity.

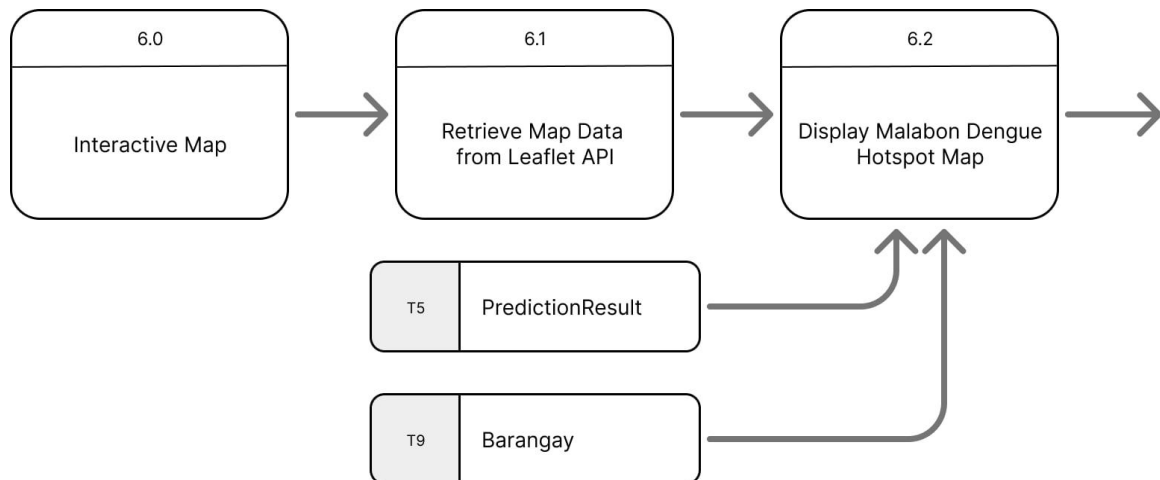


The 5.2 Calculate Average Humidity in a Day process takes inputs from the hourly and daily forecasts to compute daily average humidity levels. This calculation serves as an essential feature for modeling dengue occurrence, given the influence of humidity on mosquito breeding and survival rates. Following this, 5.3 Calculate Average Weather Forecast per Week aggregates the daily weather data into weekly summaries, capturing broader environmental patterns that may affect disease transmission trends. The output of this stage is stored in the WeeklyForecast (T9) data store, which provides structured, time-based meteorological information used by the Prediction System (Process 4.0) for training and forecasting.

Overall, this Level 2 DFD highlights the systematic approach of acquiring and transforming raw weather data into meaningful aggregated insights. It ensures that both short-term and long-term weather patterns are captured, enabling more accurate and temporally relevant dengue prediction outcomes. This modular design allows for scalability and facilitates future integration with additional meteorological sources or predictive analytics models.

**Figure 2.12**

*Data Flow Diagram Level 2 Subprocess 6.0*



The Level 2 Data Flow Diagram for Process 6.0 illustrates the internal process of the Interactive Map Module, which is responsible for visualizing dengue hotspot areas within Malabon City. The process begins with the retrieval of base map data from the Leaflet API, a web mapping library that provides accurate geographical boundaries, street layouts, and mapping layers. The retrieved geographic data is then combined with information from PredictionResult (T5), which stores the system's generated dengue risk predictions, and Barangay (T9), which provides localized area identifiers and administrative details. These datasets are integrated and processed to produce the Malabon Dengue Hotspot Map, which visually represents areas predicted to be at higher risk of dengue outbreaks based on the most recent forecast and analytical results.



Once the data is processed and rendered, the system dynamically overlays dengue risk indicators on the city map, using color-coded markers or gradient layers to represent varying levels of risk intensity. Users can interact with the map through zooming, clicking, or hovering over specific barangays to view detailed information such as recent case counts, environmental conditions, and risk probabilities. This interactive visualization not only enhances user engagement but also allows for more precise monitoring of hotspot trends across different timeframes. By combining spatial data with predictive analytics, the map transforms raw data into a comprehensible visual tool that promotes situational awareness and supports targeted intervention planning.

Overall, the Interactive Map Module serves as a vital communication and decision-support tool within the Dengue Hotspot Predictor System. It bridges the gap between data analytics and public health action by presenting complex predictive outputs in a visually intuitive and user-friendly manner. Through real-time integration of geographic, epidemiological, and environmental data, the module enables users—from local health officials to community members—to identify potential outbreak zones, prioritize response efforts, and implement preventive measures more effectively. This process highlights the system's commitment to accessibility, transparency, and proactive dengue management within Malabon City.

## System and Program Flowchart

**Figure 2.13**

### *System Flowchart*

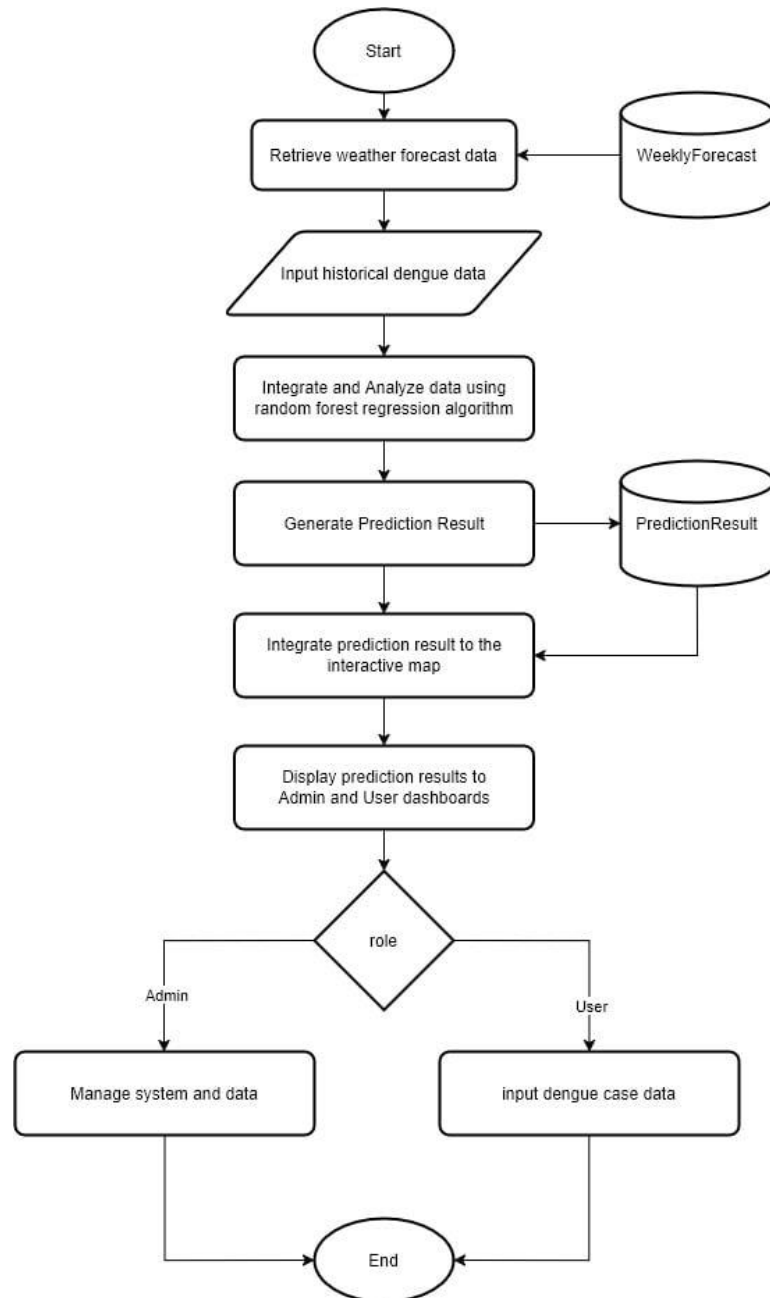




Figure 2.11 illustrates the System Flowchart for the Dengue Hotspot Prediction Process, which presents the step-by-step sequence of operations executed by the system from data retrieval to prediction generation and result dissemination. The process begins with the system establishing a connection to the database, ensuring that all required datasets are accessible before any analytical task is performed. Once connected, the system retrieves the historical dengue case data and weather forecast information, which serve as the primary input variables for the prediction model. The retrieval of these datasets guarantees that both temporal disease patterns and relevant environmental conditions are incorporated into the analysis.

After obtaining the necessary datasets, users are allowed to update the weekly dengue case data to ensure that the database reflects the most recent field reports. The updated records are then stored back into the dengue case database, maintaining data consistency and supporting future analyses. The system subsequently integrates and analyzes the data using the Random Forest Regression algorithm, a machine learning technique designed to identify patterns and correlations between weather parameters and dengue incidences. This computational process enables the system to generate reliable and data-driven predictions regarding potential dengue hotspots within Malabon City.



Upon completion of the analysis, the system generates the prediction results and stores them in the database for record-keeping and reference. The predicted outputs are then integrated with the interactive map, allowing visual representation of the forecasted hotspots. This integration provides users and administrators with an intuitive interface to view spatial patterns and assess high-risk zones effectively. The system subsequently displays the prediction results on both the Admin and User dashboards, ensuring transparency and accessibility of information.

At this stage, the process diverges based on the user's role. Administrators are permitted to manage the system and its data, including overseeing updates, configurations, and model performance. Meanwhile, regular users are responsible for inputting new dengue case data on a weekly basis to sustain the model's predictive accuracy. The process concludes once the data management or input tasks are completed. Overall, this systematic workflow ensures that dengue prediction, visualization, and data maintenance are performed efficiently and collaboratively, promoting proactive public health management through the integration of artificial intelligence and real-time data analytics.

**Figure 2.14**

*Program Flowchart for Administrator Login*

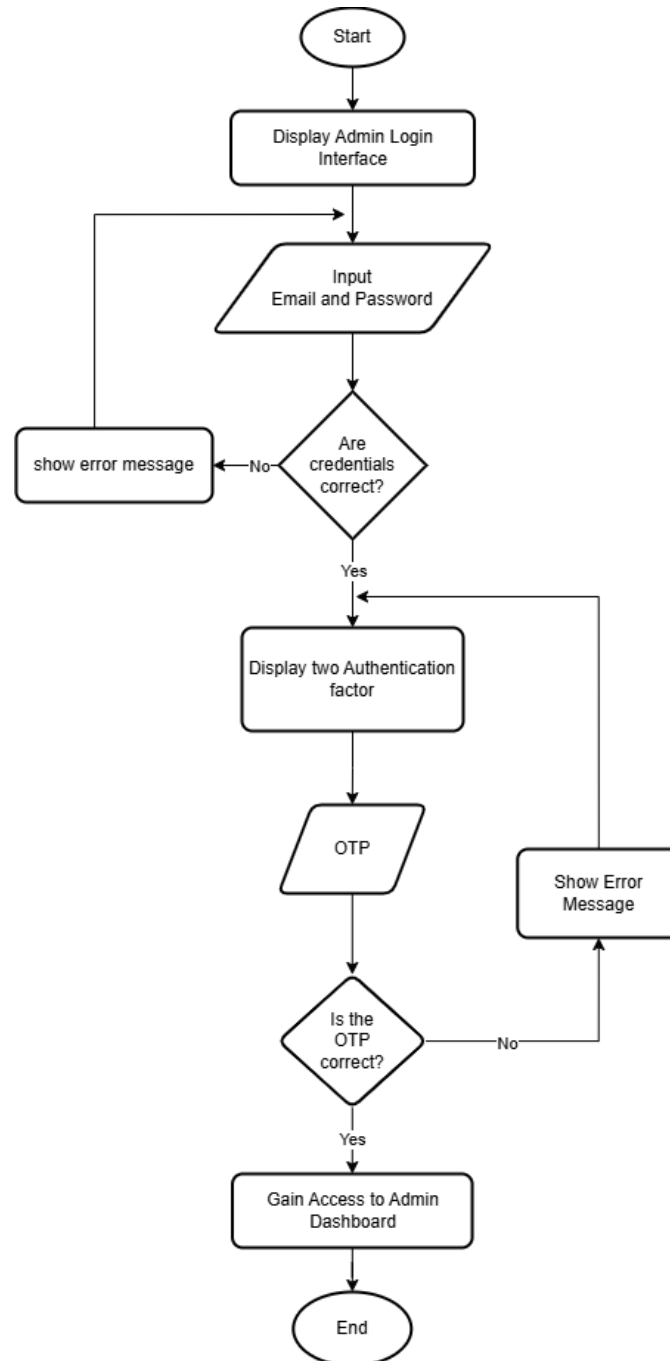




Figure 2.12 illustrates the System Flowchart for the Administrator Login Process, which outlines the step-by-step sequence of operations that occur when an administrator attempts to access the system. The process begins with the display of the Admin Login Interface, where the administrator is prompted to input valid login credentials, specifically an email address and password. This serves as the initial stage of the authentication mechanism, ensuring that every login attempt is initiated through a controlled and secure entry point.

Once the administrator submits the credentials, the system performs a validation process by cross-checking the entered information against the stored records in the database. This verification step ensures that only registered administrators with legitimate credentials are granted access. If the system detects incorrect information, such as an invalid email or password, it immediately displays an error message notifying the user of the failed attempt. The administrator is then prompted to re-enter the correct credentials, maintaining the integrity of the system by preventing brute-force or unauthorized access attempts. This error-handling loop reinforces the platform's security and reliability by ensuring that access remains strictly limited to verified users.

When the login credentials are successfully authenticated, the process proceeds to the One-Time Password (OTP) verification stage, which serves as an additional layer of security through two-factor authentication (2FA). In this step, the system generates a unique OTP and sends it to the administrator's registered email address. The administrator must then input the received code into the



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verification interface to confirm their identity. If the OTP entered is incorrect or expired, the system issues an alert and prompts the user to re-enter the valid code, thereby preventing potential unauthorized access even in cases where login credentials might have been compromised.

Once the OTP is correctly verified, the administrator gains full access to the Admin Dashboard, marking the successful completion of the authentication process. From this dashboard, the administrator can manage data, oversee system functions, and monitor overall system activity. This structured and multi-layered login procedure ensures that only authorized personnel can access sensitive system components. By combining credential-based verification and OTP authentication, the system strengthens its defense against unauthorized access, data breaches, and security vulnerabilities, thereby enhancing both the confidentiality and integrity of the entire platform.

**Figure 2.15**

*Program Flowchart for Administrator Dashboard*

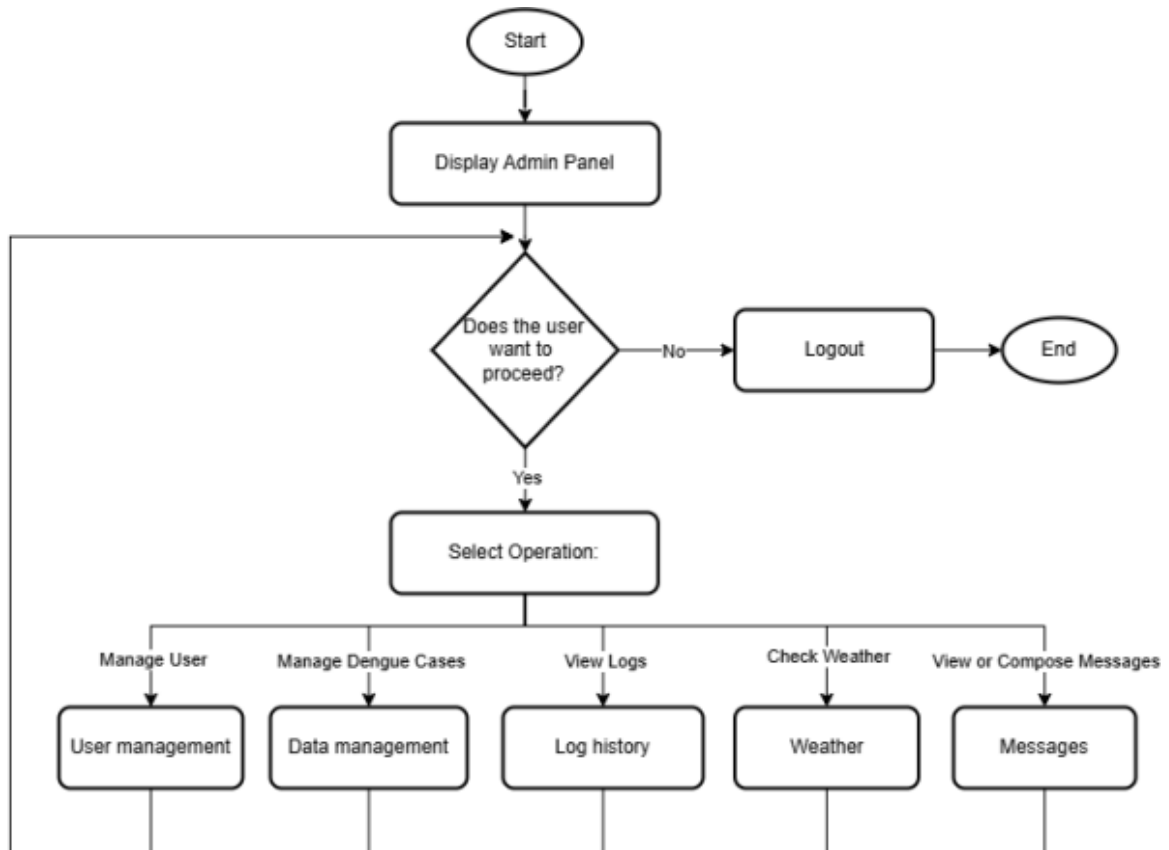


Figure 2.13 illustrates the System Flowchart for the Administrator Dashboard. This flowchart describes the process that occurs after the administrator has successfully logged into the system. Once the authentication process is completed, the system displays the Admin Panel, which serves as the main control interface for managing the system's core functionalities.

Upon accessing the dashboard, the administrator is prompted with a decision to determine whether they are done performing tasks within the system.



If the administrator chooses to log out, the process proceeds to the Logout stage, ending the session. However, if the administrator decides to continue working, the system provides several management options that can be accessed directly from the dashboard.

The administrator may choose to manage user accounts through the User Management module, handle dengue-related data through the Data Management module, view recorded activities in the Log History page, check current weather conditions and forecasts in the Weather page, or view and compose messages in the Messaging section. Each of these modules provides specific functionalities designed to support the overall system operations and enhance administrative control.

After performing the selected task, the process returns to the decision point, allowing the administrator to either continue managing other modules or proceed to log out. This structured workflow ensures an efficient and organized navigation process within the system, allowing the administrator to manage data and monitor operations seamlessly while maintaining system security and accessibility.

**Figure 2.16**

*Program Flowchart for Administrator Dashboard – User Management*

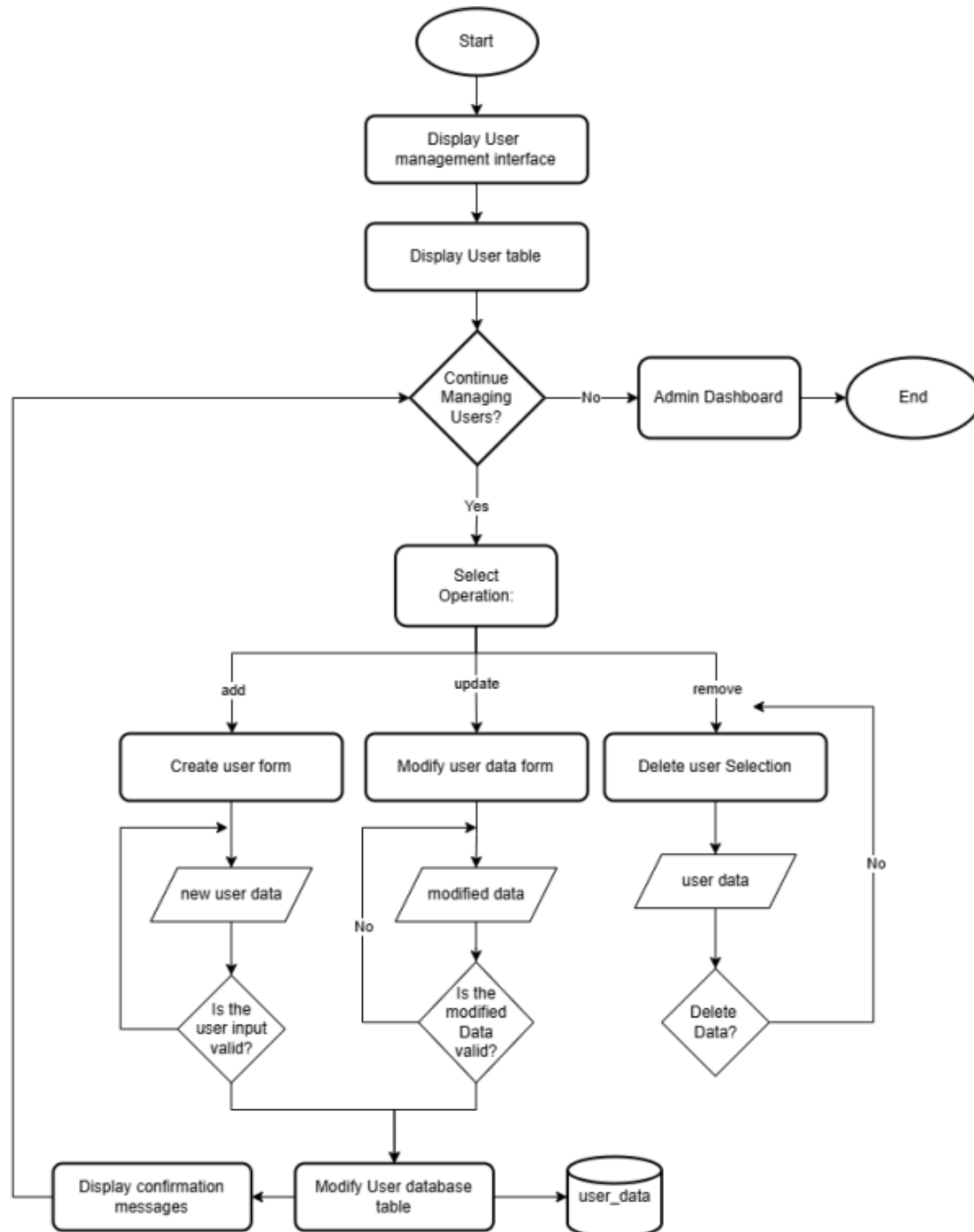




Figure 2.14 presents the System Flowchart for the User Management Page, illustrating how the administrator manages user accounts through the system's administrative interface. Accessible via the Admin Dashboard, this module maintains accurate and organized user data. The process begins with the display of the User Management Interface, which features a User Table containing all registered accounts from the database, providing the administrator with an overview of user information and management options. From this interface, the administrator may either continue managing accounts or return to the dashboard for flexible navigation.

If the administrator proceeds, the system presents three options: Add, Update, and Delete. Selecting Add opens the Create User Form, where new account details such as name, email, and credentials can be entered. The system validates inputs to ensure completeness and prevent duplication before saving to the database. Choosing Update allows editing of existing details, while Delete removes outdated or unauthorized accounts with confirmation prompts to avoid errors.

After each operation, the system updates the User Database Table, ensuring all information remains accurate, secure, and current. Validation and confirmation mechanisms reinforce data integrity and prevent unauthorized actions. Overall, the flowchart highlights a systematic and secure process enabling efficient account management and supporting system reliability.

**Figure 2.17**

*Program Flowchart for Administrator Dashboard – Data Management*

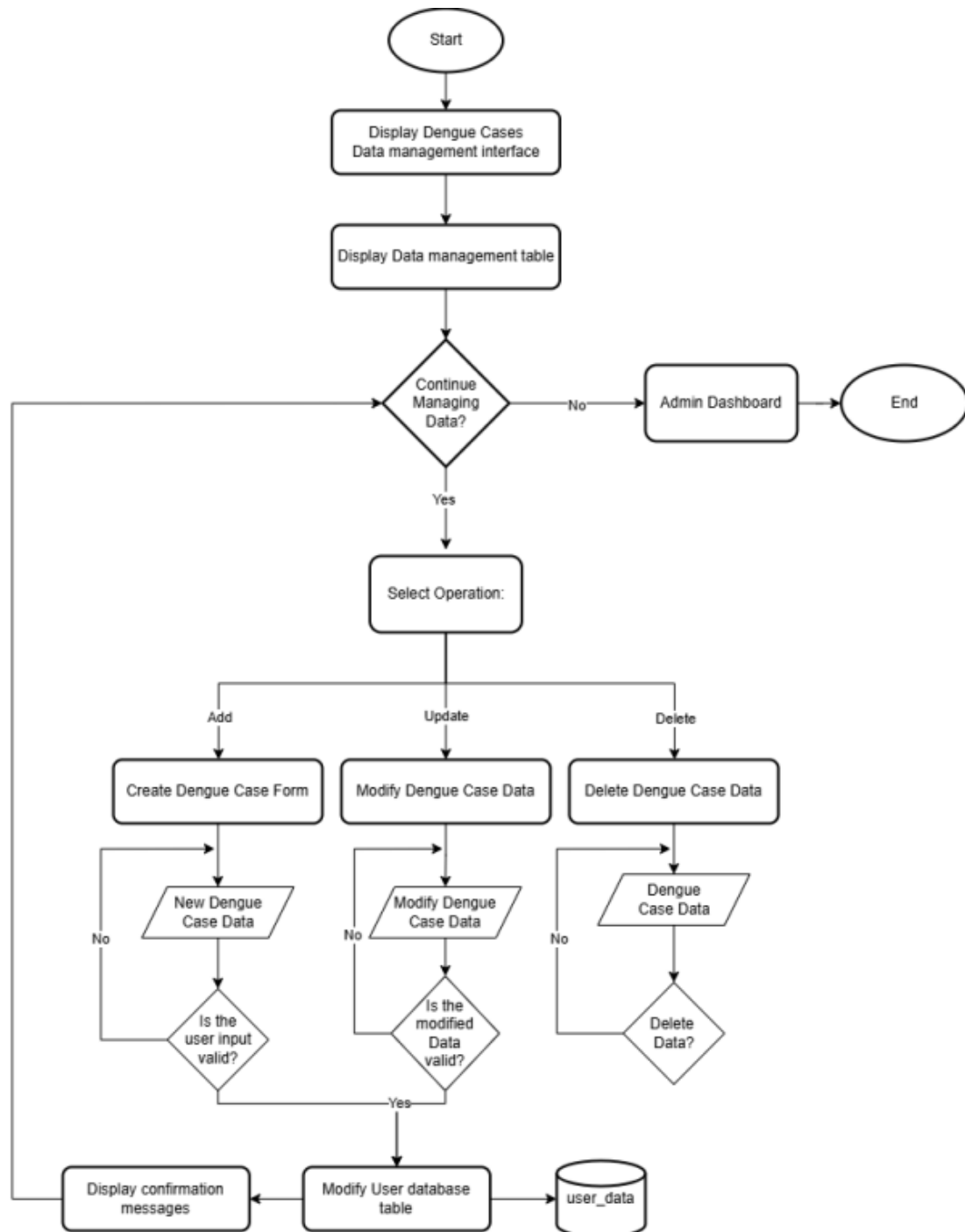




Figure 2.15 presents the System Flowchart for the Dengue Cases Data Management Page of the system. This flowchart illustrates the process by which the administrator manages dengue case records within the administrative interface. The process begins with the display of the Dengue Cases Data Management Interface and the Data Management Table, which lists all existing dengue case entries retrieved from the database. The administrator is then asked whether to continue managing data or return to the Admin Dashboard. If the administrator decides to proceed, the system provides options to either add, update, or delete dengue case records.

Selecting the Add option opens the Create Dengue Case Form, where the administrator can input new dengue case information such as date, location, and number of reported cases. The system validates all inputs before saving the new data into the database. Choosing the Update option displays the Modify Dengue Case Form, allowing the administrator to edit existing data. The modified information is also validated to ensure accuracy before being committed to the database. Alternatively, selecting the Delete option allows the administrator to remove specific dengue case records after confirmation to prevent accidental deletions. Once the chosen operation is completed, the system updates the User Database Table accordingly to reflect the most recent data changes.

This process is essential to maintaining the accuracy and reliability of the data used by the Dengue Hotspot Predictor System. Since the prediction model relies on historical and current dengue case data, proper management ensures

that only valid, up-to-date, and verified records are included in the analysis. This helps improve the system's forecasting accuracy, allowing local government units to identify potential dengue hotspots and allocate preventive measures more effectively.

**Figure 2.18**

*Program Flowchart for Administrator Dashboard – Log History*

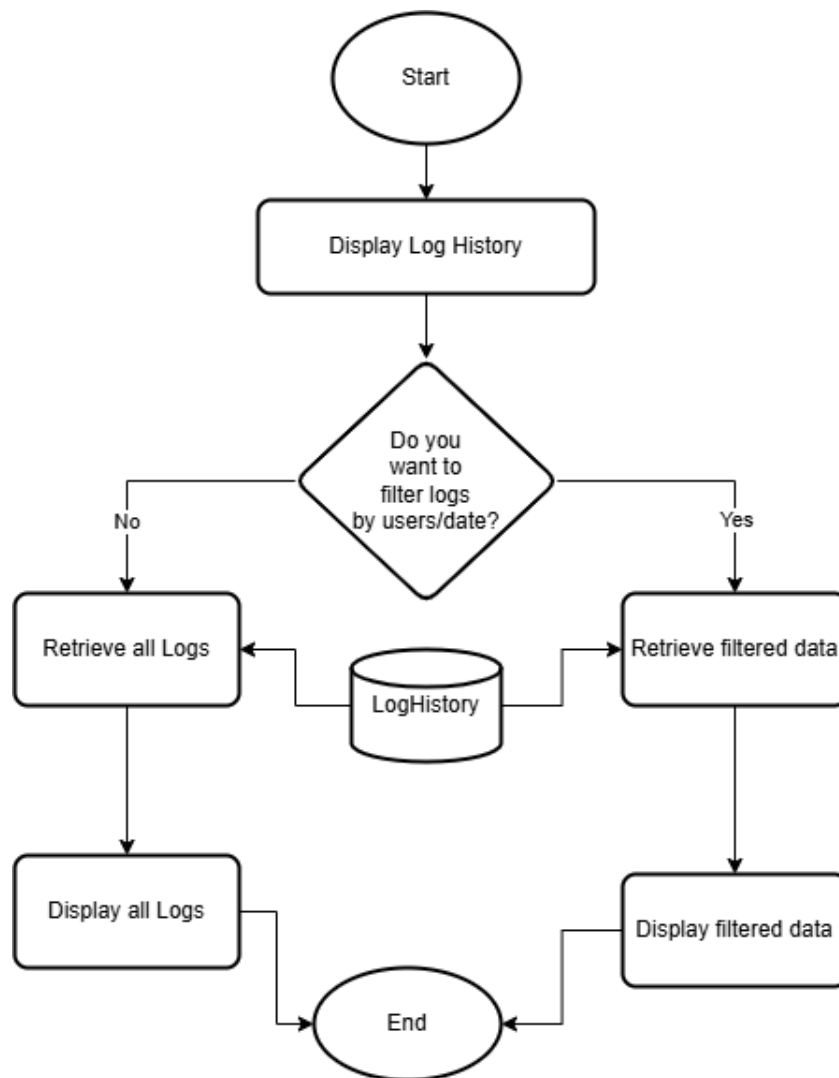




Figure 2.16 presents the System Flowchart for the Log History Page of the Dengue Hotspot Predictor System, illustrating how the administrator views and manages system activity logs to monitor user interactions and events. This module is vital for maintaining transparency and accountability within system operations. The process begins with the Log History Interface, where the administrator can either filter logs based on specific parameters—such as user identity, date range, or activity type—or view all records without filters. This functionality enables efficient navigation through large datasets and focused review of relevant records.

If filters are applied, the system retrieves corresponding data from the log\_history database and displays it in an organized table. Otherwise, it presents all stored log entries for a complete operational overview. Each record typically contains details such as timestamps, user accounts, action types, and affected modules, including logins, record updates, or configuration changes.

This structured process ensures all actions are traceable, promoting transparency and accountability among users and administrators. By maintaining a comprehensive and organized log history, the system supports early detection of unauthorized activities, strengthens data security, and aids in performance evaluation, troubleshooting, and audits. Overall, the Log History Flowchart demonstrates a reliable monitoring mechanism that safeguards system integrity and reinforces responsible system management.

**Figure 2.19**

*Program Flowchart for Users (Barangay Health Personnel) Login*

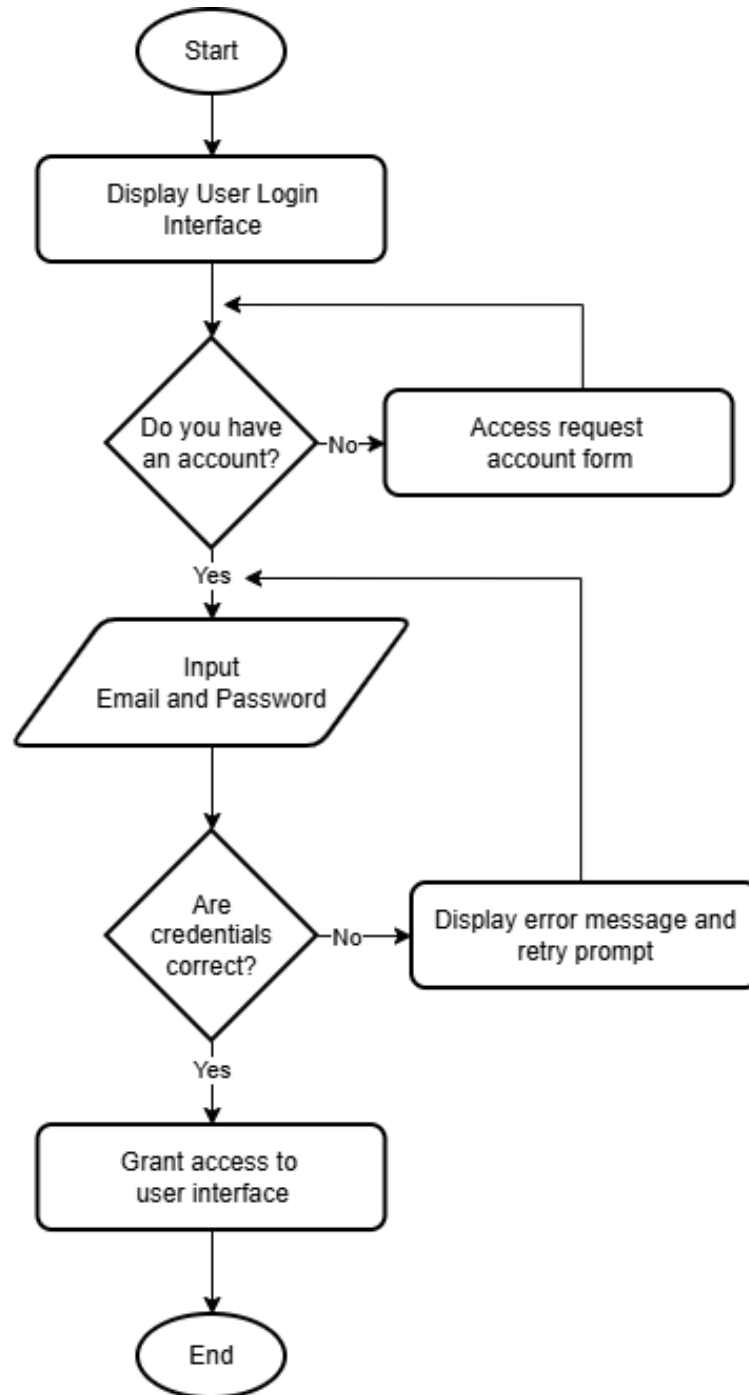




Figure 2.17 illustrates the System Flowchart for the User Login Page of the Dengue Hotspot Predictor System, which outlines the authentication process that ensures secure and authorized access for users. The process begins when a user accesses the Login Interface, where the system presents the option to either log in with existing credentials or request a new account. This initial decision point directs users appropriately based on their registration status. If the user does not yet have an account, the system navigates to the Account Request Form, where the user can provide the necessary details to request access. This step ensures that only verified and approved individuals are granted entry into the system.

For users who already possess registered accounts, the next step involves entering their email address and password into the login form. Once the credentials are submitted, the system performs validation by comparing the provided information with the stored records in the user database. If the input does not match any existing record, the system immediately displays an error message indicating invalid login details and prompts the user to re-enter the correct credentials. This feedback mechanism helps prevent unauthorized access and alerts users to incorrect entries while maintaining a smooth authentication process.

When the credentials are verified as correct, the system grants the user access to the Main User Interface, enabling them to utilize the platform's features such as case reporting, data visualization, and hotspot prediction functionalities. This verification process ensures that only legitimate users can access system resources, protect sensitive data, and prevent misuse. The User Login Flowchart



highlights the system's robust authentication mechanism designed to uphold both data privacy and operational security. By requiring accurate credential validation before granting access, the process safeguards the integrity of the platform while providing users with a secure and reliable entry point to the Dengue Hotspot Predictor System.

**Figure 2.20**

*Program Flowchart for User Dashboard*

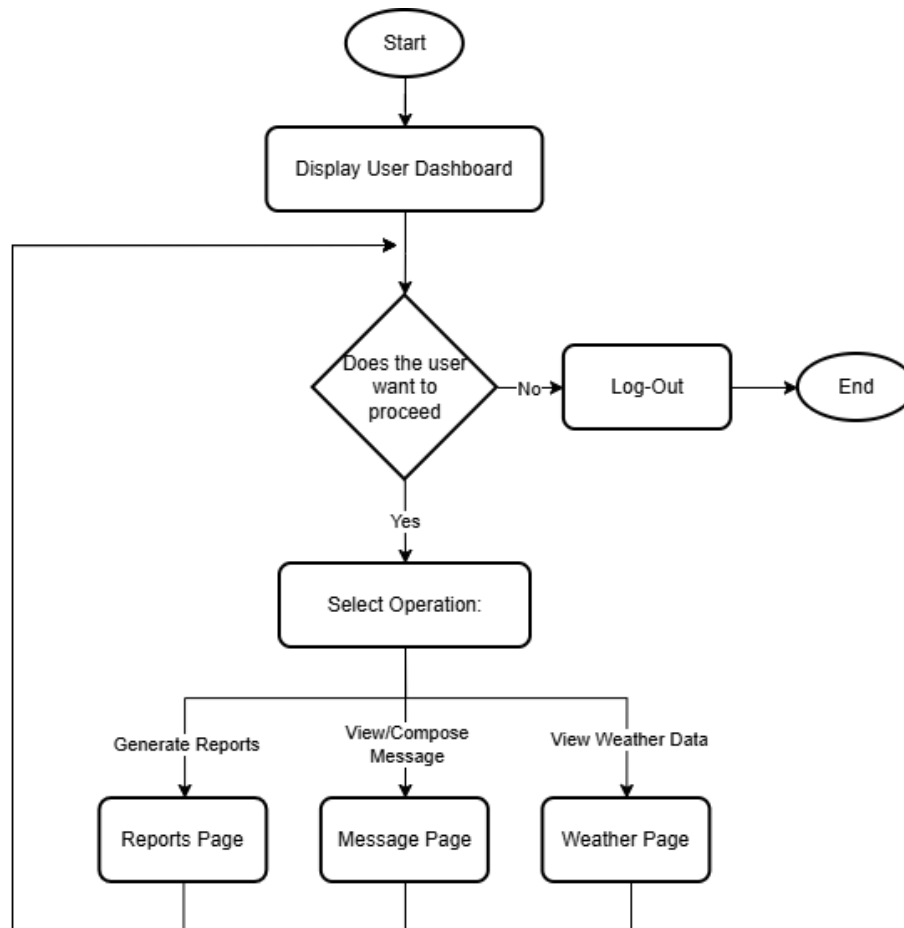




Figure 2.18 presents the System Flowchart for the User Dashboard, illustrating the main process that governs user interaction within the Dengue Hotspot Predictor System. The User Dashboard serves as the central hub where users can access core features such as reporting dengue-related data, sending messages, and viewing weather information. The process begins after a successful login, where the system displays the intuitive and user-friendly Dashboard Interface, enabling efficient navigation across modules while keeping users informed of system updates.

Once the dashboard is displayed, the system prompts the user with “Are you done?” serving as a decision point. If the user selects Yes, the system initiates the Log-Out Process, securely ending the session and recording it in the LogHistory database to ensure accountability and data security.

If the user selects No, the system proceeds to ask, “What do you want to do?”, presenting options such as opening the Report Module to submit dengue data, the Message Module to contact administrators, or the Weather Module to access updated forecasts. After completing any action, the user is redirected back to the dashboard for seamless navigation until they choose to log out.

Overall, the User Dashboard Flowchart emphasizes the system’s centralized and interactive design, ensuring a smooth user experience, secure data handling, and easy access to essential features that promote dengue awareness, reporting, and monitoring within the community.

**Figure 2.21**

*Program Flowchart for User Dashboard – Reports Page*

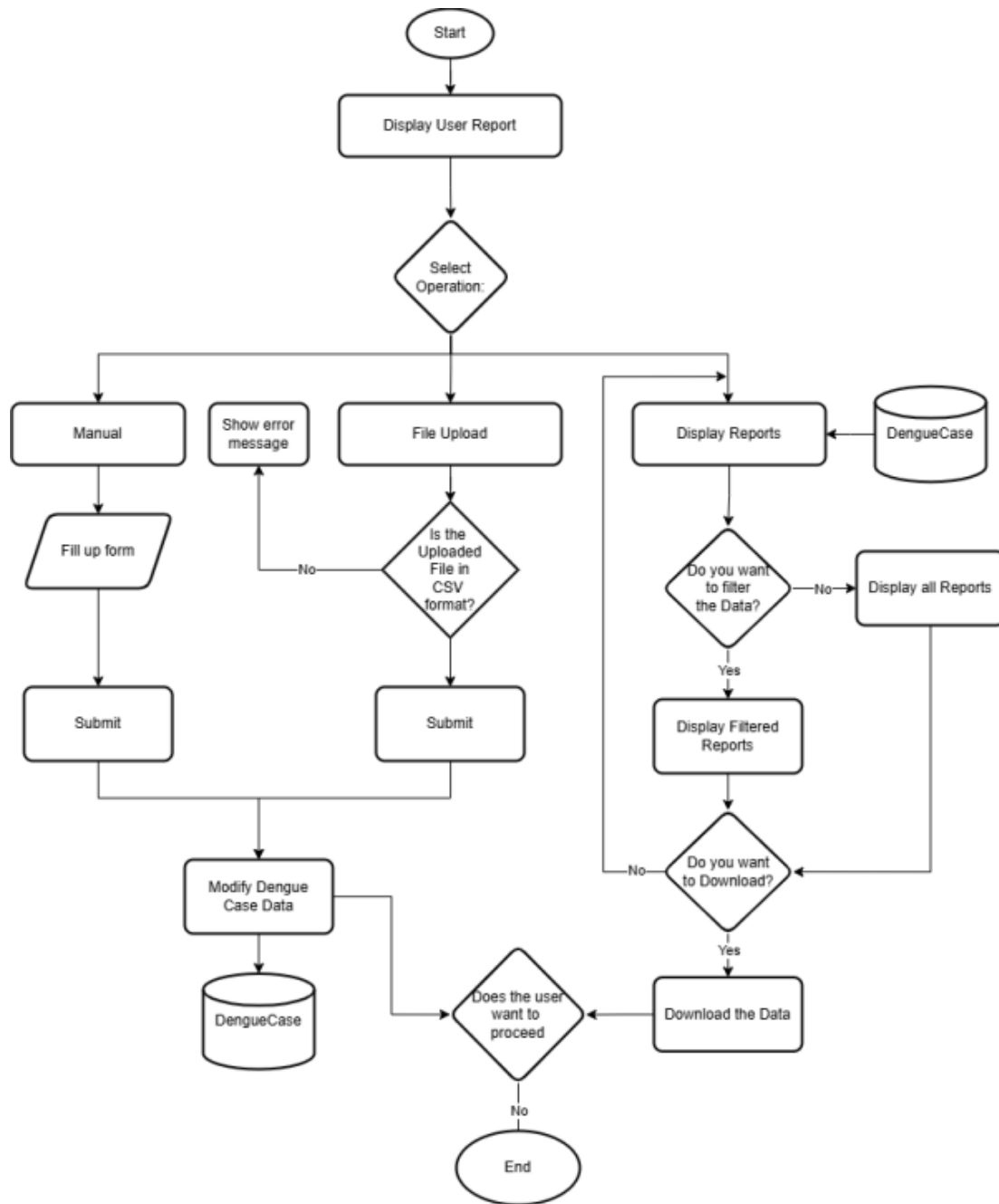




Figure 2.19 illustrates the System Flowchart for the User Report Module of the Dengue Hotspot Predictor System. This diagram presents how users interact with the reporting feature to record, upload, and retrieve dengue-related information. The process begins with the Display User Report Interface, serving as the central page for managing reporting tasks. From this interface, users may choose among three operations — Manual Entry, File Upload, or Report Download. Each option supports distinct reporting methods to ensure flexibility and efficiency in data collection and management.

When the Manual option is selected, the system directs the user to a reporting form where they input dengue case details such as the affected barangay, date, and number of cases. Once submitted, the system validates the entries and stores the information in the DengueCase database. If the File Upload option is chosen, the system prompts the user to upload a CSV dataset for bulk data entry. It verifies the file before processing; invalid files trigger an error, while valid ones are accepted and saved. This feature allows faster integration of multiple records, minimizing manual effort while maintaining accuracy.

Alternatively, the Report Download function lets users retrieve stored dengue case data for analysis or documentation. The user specifies filters such as year or week, and the system retrieves the corresponding dataset, displaying it in a structured table that can be exported for external use. Overall, this flowchart demonstrates a user-centered, reliable, and efficient process for managing dengue case reports, emphasizing accuracy, data integrity, and smooth system interaction.

**Figure 2.22**

*Program Flowchart for Messaging*

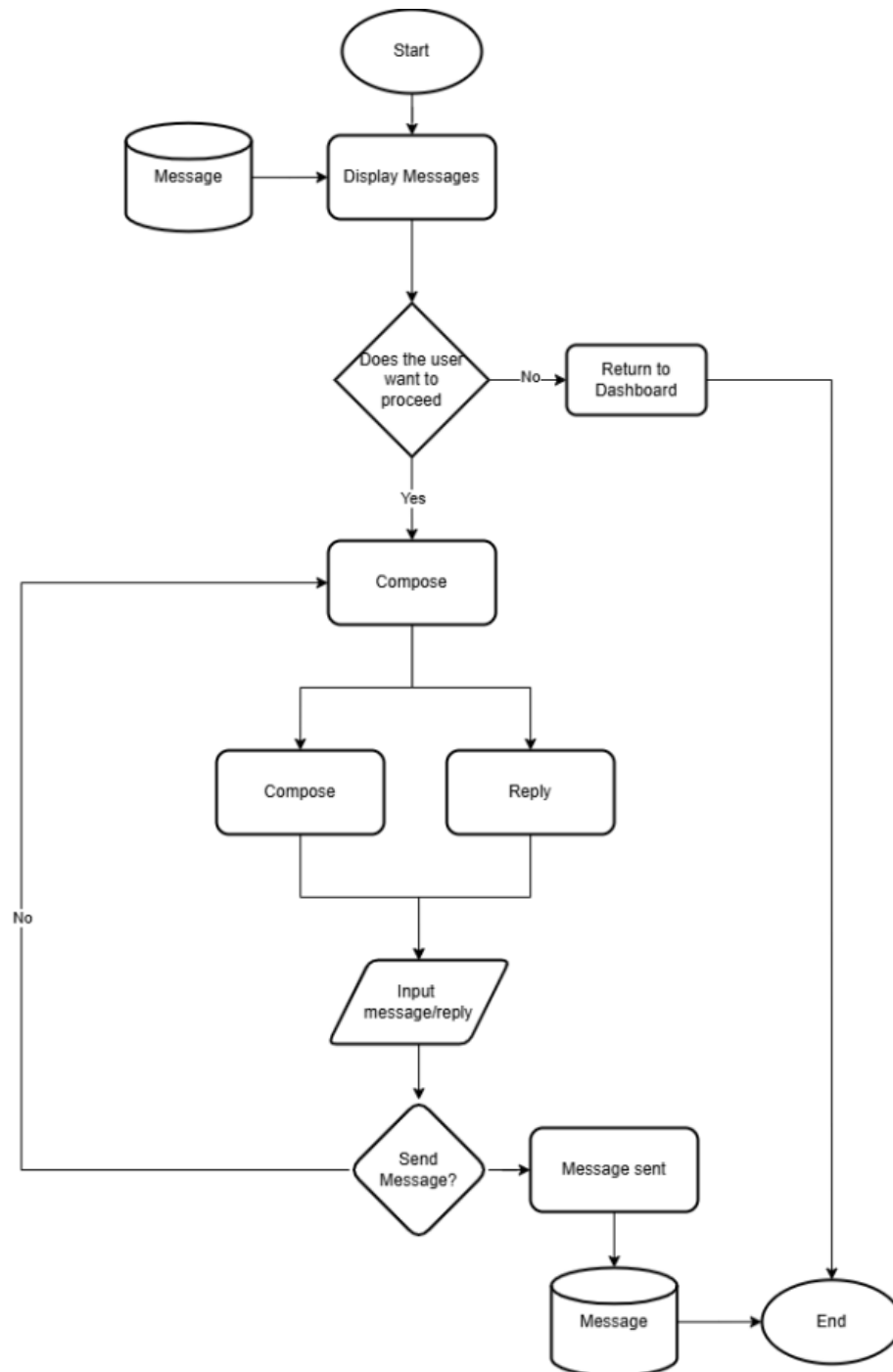




Figure 2.20 presents the System Flowchart for the Messaging Page of the Dengue Hotspot Predictor System. This diagram illustrates the logical flow of communication between users and administrators, emphasizing how the system facilitates structured, secure, and efficient message exchange. The process begins when the system retrieves and displays all existing messages from the Message database. This initial display provides users with a comprehensive overview of previous communications, including sent and received messages, allowing them to monitor updates or responses from the administrator. Once the data is displayed, the system prompts the user to decide whether to perform any messaging operations.

If the user chooses not to proceed with any operation, the system redirects them back to the User Dashboard, ensuring smooth navigation between modules. However, if the user decides to continue, they are prompted to select a specific messaging function — either compose, which allows the creation of a new message, or Reply, which enables the user to respond to an existing message thread. Upon selecting either option, the system displays an input interface where the user can type the message content. This interface supports structured communication by allowing clear and direct input of information, ensuring that messages are properly formatted before submission.

Before the message is finalized, the system verifies whether the user intends to send it. If the user confirms, the message is transmitted and stored in the Message database, after which the system notifies the user of successful

delivery. The user may then choose to continue sending or replying to messages or return to the main dashboard. This flow ensures reliability, responsiveness, and clarity in communication, supporting effective interaction between users and administrators while maintaining organized message records within the system.

**Figure 2.23**

*Program Flowchart for Weather Page*

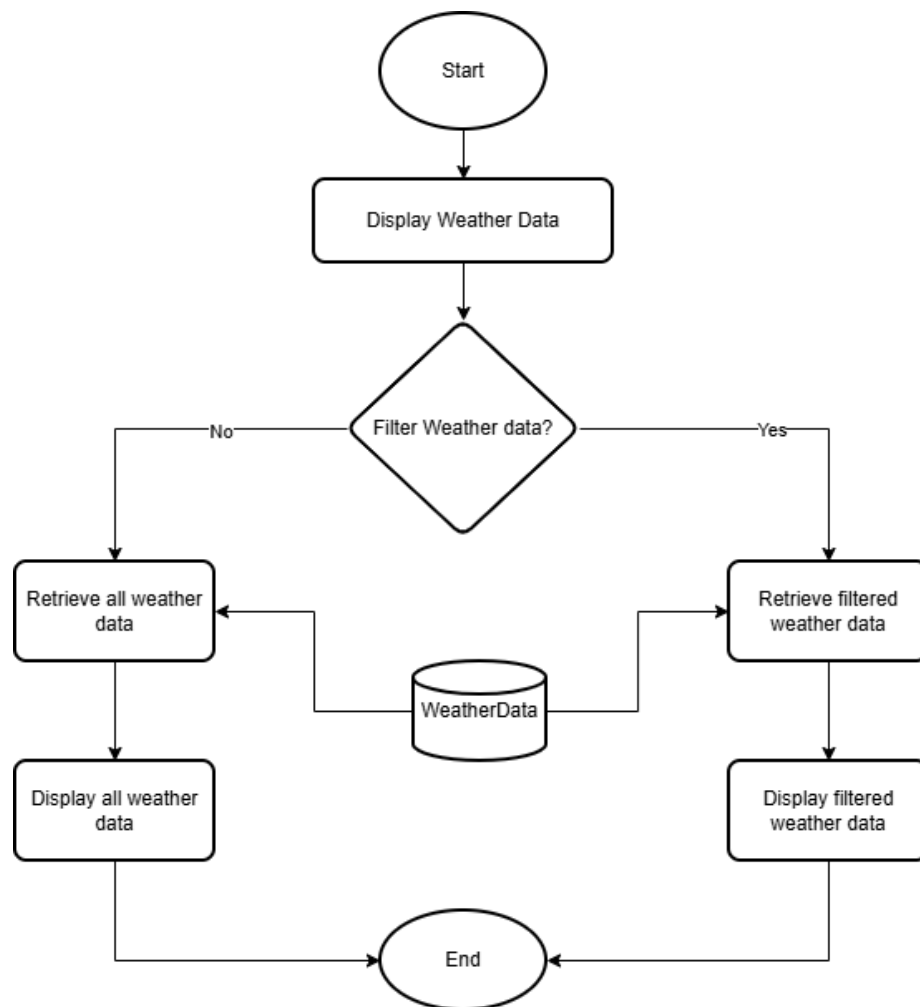




Figure 2.21 presents the System Flowchart for the Weather Data Page of the Dengue Hotspot Predictor System. This flowchart illustrates the sequence of processes involved in retrieving, filtering, and displaying weather-related information to users. The Weather Data Page is a key component of the platform, providing access to real-time and historical meteorological data influencing dengue outbreak patterns. The process begins when the user accesses the weather module from the dashboard, prompting the system to retrieve and display data from the WeatherData database, including temperature, humidity, and rainfall.

After the initial display, the system prompts the user to decide whether to apply filters. Users can choose time intervals such as current, hourly, daily, or weekly, depending on their desired detail. If filters are applied, the system retrieves the corresponding subset and updates the interface. This allows users to focus on specific timeframes for analyzing short-term weather changes or long-term trends related to dengue transmission. If no filters are used, the system displays the entire dataset for full access to information.

This process ensures flexibility, accessibility, and accuracy in presenting environmental data for predictive analysis. By enabling users to explore weather data interactively, the system enhances understanding of environmental factors influencing dengue occurrences. The structured flow of retrieval, filtering, and display demonstrates system efficiency and responsiveness. Overall, this module strengthens the Dengue Hotspot Predictor's capability by integrating weather data seamlessly, promoting data-driven awareness and proactive dengue prevention.



## Entity Relationship Diagram

**Figure 2.24**

### *Entity Relationship Diagram*

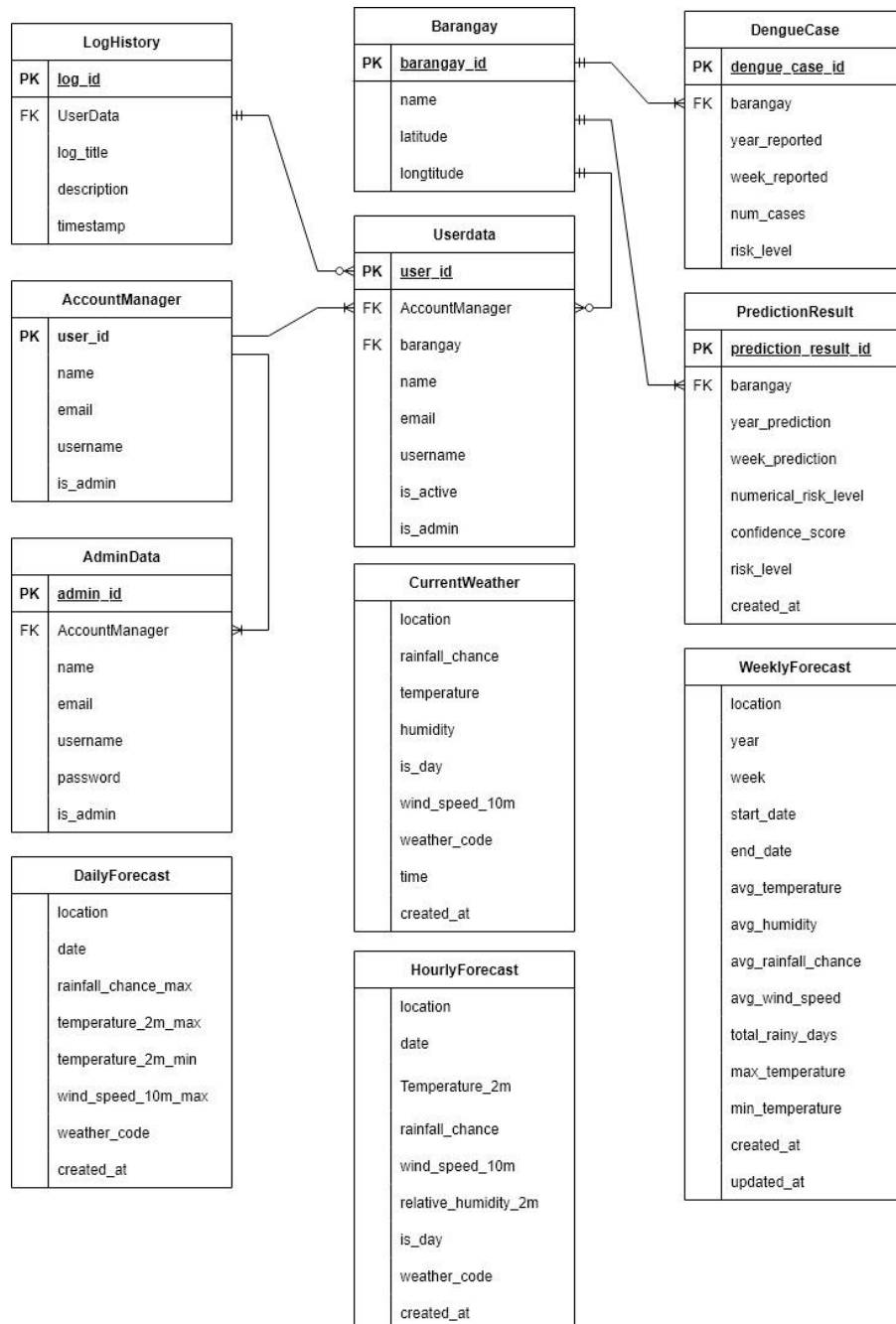




Figure 2.22 illustrates the Entity–Relationship Diagram (ERD) of the Dengue Hotspot Predictor System, which defines the logical structure and relationships among the entities that comprise the system’s database. Acting as the blueprint for database implementation, the ERD ensures that data flow, integrity, and interconnections are systematically organized to support accurate and efficient dengue hotspot monitoring. It outlines how environmental, epidemiological, and user-related data interact to provide meaningful insights for prediction and analysis.

At the core of the diagram is the Barangay entity, representing each locality being monitored. It is directly connected to WeatherData, DengueCase, and PredictionResult, enabling the association of environmental factors, dengue case records, and predictive outputs with specific geographic areas. The WeatherData entity stores meteorological attributes such as temperature, humidity, rainfall, and wind speed—key variables influencing dengue transmission. The DengueCase entity maintains records of reported cases per barangay and time period, supporting historical and trend analyses. The PredictionResult entity holds the system’s generated risk outputs, including confidence scores and timestamps, which indicate predicted dengue risk levels.

User management and monitoring are handled by UserData, AdminData, and AccountManager, which manage registration, authentication, and access levels. The LogHistory entity records all system actions for transparency and auditing. Meanwhile, CurrentWeather, HourlyForecast, and DailyForecast entities

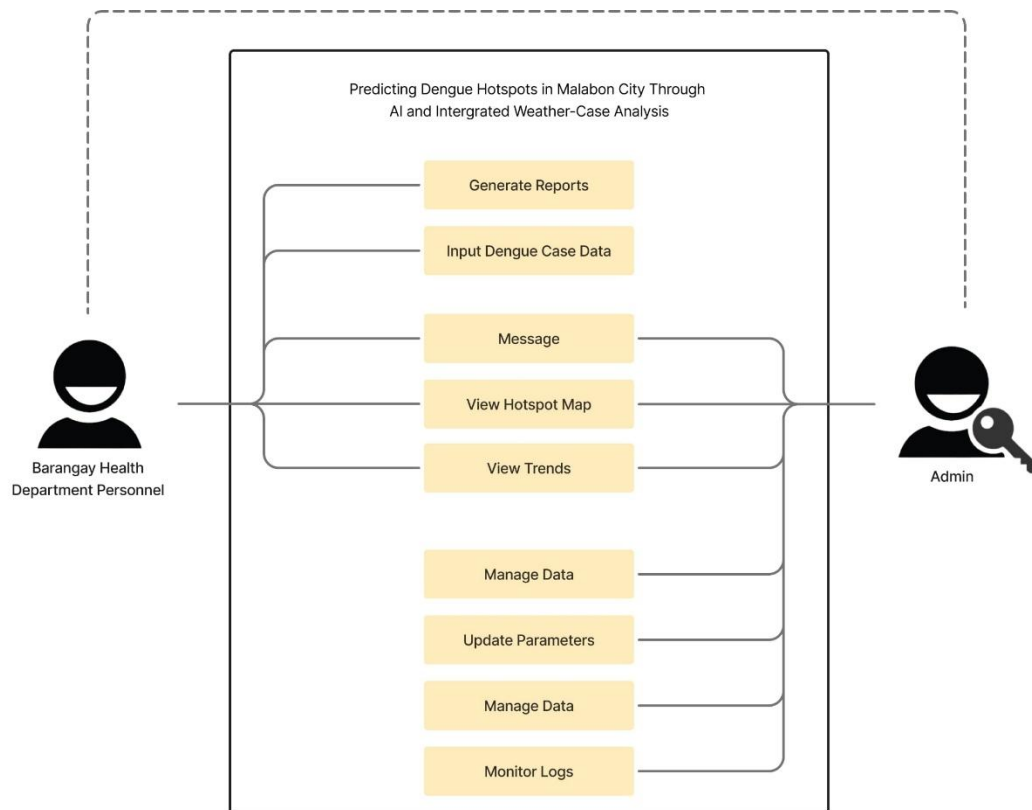


store real-time and forecasted data from external APIs, allowing continuous updates for predictive accuracy. Overall, this ERD establishes a well-defined and scalable data model that enhances consistency, security, and performance across the Dengue Hotspot Predictor System.

## USE CASE DIAGRAM

**Figure 2.25**

*Use Case Diagram*





The AI-Based Dengue Hotspot Prediction System use case diagram depicts two main actors, which are the LGU Officer and Administrator. The LGU Officer could be any barangay health officer or people who are in charge of dengue monitoring and response in their respective barangay. For them, there is interaction with the system by secure login, viewing hotspot predictions on the city map, comparing trends of historical weather and dengue case data, plotting reports, and receive warnings about high-risk zones. These tools are meant to aid them in making informed decisions around public health management in a timely manner.

While the Administrator manages the backend of the system. Such functions consist of user account creation and control, data uploading, such as new weather and dengue data, system performance monitoring, AI parameter settings, and system security. Overall, the diagram illustrates the two-targeted system of the Dengue Hotspot Predictor: in which the system primarily serves to inform LGU officials with risk assessment to proactively respond to dengue outbreaks, and next provides a robust and well-maintained system through administration support.

### **Population and Sampling**

This study focuses on an AI-driven model designed to predict dengue hotspots in Malabon City. The model integrates past dengue case data with current climatic conditions to provide timely and spatially informed insights. It aims to assist



the Local Government Unit (LGU) in improving public health decision-making, strengthening epidemiological surveillance, informing environmental management, and enhancing disaster response planning.

The population of the study consists of barangay health department personnel and City Health Department officials who are directly involved in planning and executing dengue prevention and control measures in Malabon City. In total, the population comprises 23 LGU personnel, all of whom play key roles in implementing dengue-related programs.

The study employed purposive sampling, a non-probability sampling technique, to ensure that participants with the appropriate expertise and responsibilities in dengue programs were selected. By concentrating on these key informants—health officers and program implementers—the researchers were able to gather data that is both relevant and informed, directly contributing to the evaluation and improvement of the dengue hotspot prediction system.



For statistical reference, Slovin's formula can be used to determine the sample size for a finite population:

$$n = \frac{N}{1 + Ne^2}$$

Where:

$n$  = Sample Size

$N$  = 23 (Total Population)

$e$  = 0.05 (Margin of Error)

$$n = \frac{23}{1 + 23(0.05)^2} = \frac{23}{1 + 0.0575} = \frac{23}{1.0575} \approx 21.74 \approx \mathbf{22}$$

Given the small population size and the focus on key personnel, all 23 eligible LGU personnel were included in the study to maximize the quality and relevance of the data.

### **Respondents**

The respondents of this study were the 23 LGU personnel involved in dengue prevention and control, including barangay health department personnel and City Health Department officials. Data were collected using printed questionnaires featuring a 4-point Likert scale ranging from 1.00 (Strongly



Disagree) to 4.00 (Strongly Agree), supplemented with interviews to gather qualitative insights.

The collected data will be analyzed using a descriptive-comparative-correlational research design to evaluate the system in terms of usability, accuracy, and potential utility in supporting dengue prevention efforts in Malabon City.

### **Research Instrument**

To develop a reliable, efficient, and practical AI-Based Dengue Hotspot Predictor System, this study employs a combination of data acquisition methods, software development tools, evaluation techniques, and validation strategies. These instruments are systematically chosen to ensure that the system functions accurately and responds effectively to the real-time health monitoring needs of Malabon City. The research instruments aim to support not only the technical construction of the predictive model but also its deployment as a user-centered web application accessible to local health authorities. Each instrument, from data collection templates to GIS visualization tools, plays a specific role in transforming raw health and weather data into meaningful predictions that can guide dengue prevention efforts. Moreover, the evaluation and validation tools based on ISO/IEC 25010 standards are incorporated to measure the system's performance, usability, and reliability in real-world conditions. The following subsections describe the full



range of instruments employed in this study, categorized by their function in the research and system development process.

### **Data Collection**

At the forefront of the AI-Based Dengue Hotspot Predictor System are its data collection instruments, which form the foundation for training and validating predictive models. Barangay-level dengue case records will be sourced directly from the Malabon City Health Department via an official request letter, ensuring access to legitimate and reliable data. These reports, covering the period from January 2021 to December 2024, are crucial for identifying temporal trends and spatial patterns in dengue incidence across different barangays.

Complementing the health records is meteorological data, which includes temperature, rainfall, flood risks, wind speed, wind direction, and humidity readings. These weather datasets will be acquired from reliable weather application APIs and other open meteorological data platforms. They are instrumental in identifying correlations between climate conditions and dengue outbreaks. To facilitate uniformity in data preparation, standardized templates using Microsoft Excel or Google Sheets will be employed to encode, clean, and preprocess the collected data for model training. These templates ensure data consistency, reduce errors, and streamline future analysis.



## Processing and System Development

Once the data is collected, it is processed and analyzed using a set of powerful software tools. Python will serve as the main programming language, owing to its flexibility and the extensive ecosystem it provides for data science and machine learning. Data preprocessing tasks such as filtering, formatting, normalization, and feature engineering will be carried out using the Pandas and NumPy libraries.

For model development, advanced machine learning frameworks such as Scikit-learn will be used to design and train the predictive models. These tools support various algorithmic approaches, including classification and regression models, enabling experimentation and model optimization. Exploratory data analysis and model output interpretation will be enhanced through the use of Matplotlib and Seaborn, which provide visual insights into trends, correlations, and anomalies.

The final system will be deployed as a web-based application, utilizing the Django framework for backend integration. For frontend design and user interaction, standard technologies such as HTML, CSS, and JavaScript will be used. Additionally, LeafletJS will provide GIS-based visualization, allowing the system to dynamically highlight high-risk barangays within Malabon City on an interactive map. This geographic representation enhances the practical usability of the system for health personnel and LGU officials.



## **Hardware Infrastructure**

To support model training and system deployment, development will be conducted on a standard personal computer or laptop equipped with sufficient memory and processing power. For more complex or resource-intensive operations, particularly during model training, cloud-based services such as PythonAnywhere may be utilized. These platforms offer high-performance computing resources, such as GPUs, which accelerate model training and improve overall system responsiveness.

## **Evaluation**

The system's effectiveness will be evaluated through a combination of quantitative and qualitative tools. A confusion matrix will be the primary method for assessing model performance, providing critical metrics such as accuracy, precision, recall, and F1-score. These metrics measure how well the model identifies dengue-prone areas and distinguishes between high-risk and low-risk zones.

In addition, cross-validation techniques such as K-Fold cross-validation will be employed to test the model's stability and generalizability when exposed to unseen data. These methods ensure that the system is not overfit to a specific dataset and is capable of maintaining predictive accuracy across different conditions.



## **Construction**

In constructing the AI system, attention will be given to both technical performance and user experience. The system must reliably ingest new data, process it using the trained AI model, and display the results through a secure and intuitive interface. By following quality assurance standards such as ISO/IEC 25010, the system will be designed to be functional, reliable, secure, and maintainable. The integration of cloud-ready infrastructure and efficient database management will further ensure that the system is scalable and can accommodate a growing dataset or user base in the future.

## **Validation**

The system will be validated using a research questionnaire designed according to the ISO/IEC 25010 software quality standard, which assesses software across key dimensions such as functionality, performance efficiency, usability, reliability, and security. Respondents, including health officials and other designated evaluators, will evaluate the system's accuracy in predicting dengue hotspots, ease of use, responsiveness, and ability to safeguard sensitive user data.

In addition to these core dimensions, the validation will also consider the system's adaptability to changing conditions, error-handling capabilities, and overall user satisfaction. Feedback from respondents will provide insights into



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potential improvements, practical applicability, and the system's alignment with operational needs. By combining quantitative ratings and qualitative observations, this process ensures that the AI-Based Dengue Hotspot Predictor System not only meets technical and academic standards but also effectively supports timely and informed decision-making in public health interventions in Malabon City.

### **Data Gathering Procedure**

For the development and deployment of the AI-based Dengue Hotspot Predictor, the researchers will conduct a structured series of data collection activities focused on acquiring the necessary epidemiological and environmental inputs. The study will primarily involve data from barangays within Malabon City, ensuring localized prediction and relevance to public health decision-making.

The first step in the process is securing formal permission from relevant authorities. The researchers will write and submit an official letter to the Malabon City Health Department, respectfully requesting access to dengue case records from January 2021 to December 2025. This data will form the foundation of the predictive model and must include barangay-level dengue case counts, weekly or monthly reporting dates, and, if permitted under the

The Philippine Data Privacy Act of 2012 anonymized demographic metadata such as age groups and gender. All data shared will strictly exclude any personally identifiable information (PII).



Upon approval, the Health Department is expected to provide digital copies of the dengue case data in formats such as Excel, CSV, or PDF. The researchers will then convert these documents into a standardized dataset, formatted to be compatible with machine learning training and analysis workflows.

Parallel to this, the team will also obtain weather data for Malabon City corresponding to the same timeframe. Climate variables such as temperature, rainfall, and humidity will be sourced from credible weather application APIs. Should additional data be required, global repositories like the Open-Meteo API will be consulted as supplementary sources.

Following the collection, the researchers will move on to the preparation and preprocessing phase, ensuring that all data is clean, accurate, and properly aligned. This involves resolving missing values, verifying and correcting irregular entries, and synchronizing all records by time (weekly or monthly). This preprocessing phase is essential to guarantee that the data fed into the predictive model is both complete and reliable.

All datasets will be securely stored within an encrypted project directory exclusive to the research team. In compliance with the Philippine Data Privacy Act of 2012, no PII will be stored, and the collected information will be used solely for educational and scientific purposes. Access to this data will be restricted only to authorized team members involved in the capstone project.



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Through these ethically grounded and methodically executed data gathering steps, the researchers aim to ensure the integrity, privacy, and usefulness of the information, laying a strong foundation for the development of a functional, trustworthy dengue hotspot prediction system tailored for Malabon City.

### **Data Analysis and Statistical Treatment**

To ensure the accuracy, reliability, and integrity of the data collected from the system evaluation, appropriate statistical tools and methods will be employed. This project utilizes Frequency and Percentage Distribution as the primary statistical approach to interpret the feedback gathered from 22 local government unit (LGU) Health Department personnel who are tasked to evaluate the “Dengue Hotspot Predictor System.” These methods will provide a clear understanding of how the respondents perceive the system’s usability, effectiveness, reliability, and relevance based on various performance indicators aligned with the ISO 25010 software quality model.



## Frequency and Percentage Distribution

This statistical method is used to summarize and interpret data by showing the number of respondents who gave specific responses to various indicators on the system's performance. The responses will be gathered through a survey questionnaire structured with a Likert scale. This will measure multiple system attributes, including accuracy of dengue hotspot prediction, system responsiveness, ease of use, and the effectiveness of data visualization.

To obtain the frequency and percentage values, the following formula will be applied:

$$\% = (f / N) \times 100$$

Where:

- % = Percentage
- **f** = Frequency of a specific response
- **N** = Total number of respondents (in this case, 22 barangay and city health department personnel)

This distribution will allow the researchers to effectively visualize the concentration of responses across different evaluation criteria, revealing the majority opinion and highlighting areas that may need improvement.

The **Weighted Mean** is a statistical tool utilized to determine the average level of responses gathered from the survey questionnaire, taking into



consideration the varying significance of each response category. This method provides a more precise and reliable measurement of the respondents' overall perception of the system's quality across different evaluation criteria—namely Usability, Functionality, Reliability, and Efficiency—as aligned with the ISO 25010 software quality model.

In this study, the Weighted Mean is applied to summarize and interpret the responses of 22 City and Barangay Health Department officials who evaluated the Dengue Hotspot Predictor System. Each response in the Likert scale is assigned a specific weight value: 4 (Strongly Agree), 3 (Agree), 2 (Disagree), and 1 (Strongly Disagree). By multiplying the frequency of each response by its assigned weight and dividing the sum of these products by the total number of respondents, the weighted mean for each indicator is obtained.

$$\text{Weighted Mean} = \frac{\sum(f \times w)}{N}$$

Where:

$f$  = Frequency of each response

$w$  = Weighted value of each response

$N$  = Total number of respondents



This approach ensures that the analysis reflects not just the number of responses but also their relative importance. By employing the Weighted Mean, the researchers can derive a more comprehensive understanding of the system's performance and identify which aspects of the system—such as prediction accuracy, responsiveness, or user interface—are viewed most favorably or require improvement. Consequently, this tool allows for an objective and data-driven evaluation of the system's overall effectiveness and reliability.

### **Likert Scale Interpretation**

To quantify the feedback from respondents, a 4-point Likert scale will be used. Each response will be assigned a numerical value, ranging from 1 to 4:

**Table 2.1**

#### *Rating Scale Values*

| SCALE | RANGE       | VERBAL INTERPRETATION |
|-------|-------------|-----------------------|
| 4     | 3.50 – 4.00 | Strongly Agree        |
| 3     | 2.50 – 3.49 | Agree                 |
| 2     | 1.50 – 2.49 | Disagree              |
| 1     | 0.50 – 1.49 | Strongly Disagree     |



The Likert scale will be used by the researchers to assess the central tendency of participant responses and evaluate how effectively the system performs according to a set of predefined quality indicators. The statistical analysis will focus on the system's alignment with key ISO 25010 software quality attributes. Specifically, it will examine Functional Suitability, such as the accuracy of dengue hotspot predictions; Usability, including how easily users can navigate the platform; Performance Efficiency, relating to responsiveness and load times; Reliability, or the consistency of outputs; Security, particularly the protection of sensitive health and location data; and Maintainability and Portability, referring to how easily the system can be updated or deployed to other local government units (LGUs). These attributes will be directly reflected in the survey questions and evaluated using Frequency and Percentage Distribution. Through the analysis of responses from LGU officials such as City and Barangay Health Department personnel, the researchers aim to validate the system's operational quality and assess its potential for practical implementation in public health decision-making. This evaluation will also serve as a basis for future improvements to the system.

### **Testing and Evaluation Process**

There are different test cases to be used to test and evaluate the system. The testing evaluation cases are a guide in the testing and evaluation process of the system. This helps the researchers to easily test the system for possible errors



and problems. These test cases should be used in determining if the system follows the standards.

**Table 2.2***System Testing Cases for Administrator*

| Test Case Instances | Test Case Scenario                             | Expected Result                                                           | Checked (/) | Result Test              |
|---------------------|------------------------------------------------|---------------------------------------------------------------------------|-------------|--------------------------|
|                     |                                                |                                                                           | <b>Good</b> | <b>Needs Improvement</b> |
| 1.                  | Log in with valid credentials.                 | Admin is redirected to the dashboard.                                     |             |                          |
| 2.                  | Login with invalid credentials.                | An error message is shown: "Invalid username or password".                |             |                          |
| 3.                  | View the dengue case summary dashboard.        | Real-time dashboard displays graphs, charts, and alerts.                  |             |                          |
| 4.                  | Access barangay-level risk heatmaps.           | Color-coded map loads correctly with clickable barangay data.             |             |                          |
| 5.                  | Export reports (CSV/PDF).                      | Reports are generated with the correct data format and content.           |             |                          |
| 6.                  | Add a new user account (e.g., health officer). | The new user is saved and appears in the user list with the default role. |             |                          |



|     |                                                                      |                                                                              |  |  |
|-----|----------------------------------------------------------------------|------------------------------------------------------------------------------|--|--|
| 7.  | Delete the user account.                                             | The selected user is removed from the system with a confirmation prompt.     |  |  |
| 8.  | View activity logs of all users.                                     | Audit log shows login times, data updates, and critical actions by the user. |  |  |
| 9.  | Update admin profile settings (name, username, password, and email). | Changes are saved and reflected after submission.                            |  |  |
| 10. | Log out of the admin dashboard.                                      | Admin is securely logged out and redirected to the login page.               |  |  |

Table 2.2 presents the different testing scenarios designed for admin functionalities within the dengue prediction web system. These scenarios assess the admin's ability to log in securely, manage and upload data, monitor forecasts, generate reports, and control user access. The tests also include configuration updates and notification management. Executing these testing cases, it guarantees that the admin has full operational control over the system, can oversee the data quality, and ensure that all predictive outputs are timely, accurate, and securely disseminated to relevant stakeholders.

**Table 2.3***System Testing Cases for Users/Barangay Health Department Personnel*

| Test Case Instances | Test Case Scenario                               | Expected Result                                                    | Checked (/) | Result Test       |
|---------------------|--------------------------------------------------|--------------------------------------------------------------------|-------------|-------------------|
|                     |                                                  |                                                                    | Good        | Needs Improvement |
| 1.                  | Log in with valid credentials.                   | User is redirected to the main dashboard.                          |             |                   |
| 2.                  | Login with invalid credentials.                  | An error message is shown: "Invalid username or password".         |             |                   |
| 3.                  | View the barangay dengue risk map.               | Color-coded risk map loads correctly, showing the user's barangay. |             |                   |
| 4.                  | Search for specific barangay dengue data.        | Accurate risk data is shown for the selected barangay.             |             |                   |
| 5.                  | View forecast trends by week.                    | Graphs/charts update to reflect the selected timeframe.            |             |                   |
| 6.                  | Access alert banners for the current dengue risk | Alert level is shown clearly (e.g., Low, Moderate, High)           |             |                   |
| 7.                  | Reset user password.                             | The password reset link is sent via email.                         |             |                   |
| 8.                  | Search/filter users by name or status            | Filtered results appear based on search criteria                   |             |                   |



|    |                                |                                                               |  |  |
|----|--------------------------------|---------------------------------------------------------------|--|--|
|    |                                |                                                               |  |  |
| 9. | Log out of the user dashboard. | User is securely logged out and redirected to the login page. |  |  |

Table 2.3 shows the test cases focused on general users interacting with the web system. These include user registration and login, accessing risk maps, filtering dengue and weather data, receiving alerts, and updating personal profiles. The purpose of these tests is to ensure a smooth and intuitive experience for users who rely on the system for understanding barangay-level dengue risks. These test scenarios validate whether users can retrieve accurate information, interact with the system efficiently on both desktop and mobile platforms, and stay informed about possible outbreaks through notifications.

### **Ethical Considerations**

The goal of this study is to conduct research and development in a way that adheres to the highest ethical standards. To create a predictive system, the project entails gathering and analyzing historical dengue case data as well as pertinent meteorological data. The following moral guidelines were closely adhered to:



### **Data Privacy and Confidentiality**

The researchers will guarantee that all sensitive and personal information will be handled in accordance with the Data Privacy Act of 2012 (RA 10173). Before being processed, any datasets obtained from the Malabon City Health Department will be anonymized to remove any personally sensitive information, such as names or precise addresses.

### **Informed Consent from Data Source**

To obtain the required data, a formal letter of request was sent to the Malabon City Health Department. Before any information is used, approval will be obtained. The information will not be misused or commercialized in any way; it will only be used for scholarly and research purposes.

### **Responsible AI Usage**

This project's machine learning model will be intended to supplement, not to replace, decision-making. It won't be used for authoritative decision-making without human validation or for profiling. Frequent accuracy checks and discussions with subject matter experts will reduce misunderstandings and inaccurate forecasts.



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### **Transparency and Accountability**

The system's design, limitations, and assumptions will be clearly documented. The researchers will also be accountable for how the predictions are generated and will disclose any known weaknesses of the model to prevent misuse.

### **No Harm Principle**

The project aims to avoid causing any form of harm, whether physical, social, or reputational, to individuals or communities. The prediction results will not be used to stigmatize any barangay or population group but will instead serve to aid local government units (LGUs) in implementing timely preventive measures.

### **Data Retention and Disposal**

Unless the partner institution specifies otherwise in a written agreement, all data gathered will be safely stored throughout the study and deleted permanently at the end of the project.



### **Chapter 3**

## **RESULTS AND DISCUSSION**

This chapter shows the researcher's findings, system components, features, and methods used to evaluate the AI-based Dengue Hotspot Predictor for Malabon City. The primary objective of this capstone study is to design and implement an AI-powered web application that can forecast dengue hotspots across the barangays of Malabon City, in order to help the city health department respond proactively to potential outbreaks. The proponents of this capstone study used tables and figures that demonstrate the responses of City health department officers and barangay health personnel aligned with the ISO 25010 standards.

### **Features of the AI-based Dengue Hotspot Predictor System**

The system aims to predict potentially hazardous dengue hotspots with accuracy and timeliness by combining the historical data of Dengue cases and weather factors (temperature, rainfall, humidity). Its interactive GIS-based dashboard enables barangay health officers to see the barangay-level risk classifications and decide on action plans related to health interventions, as well as vector control and resource facilitation.

For data consistency and reliability, it utilizes structured pre-processing of the case data, synchronization of the case and weather reports, and secure



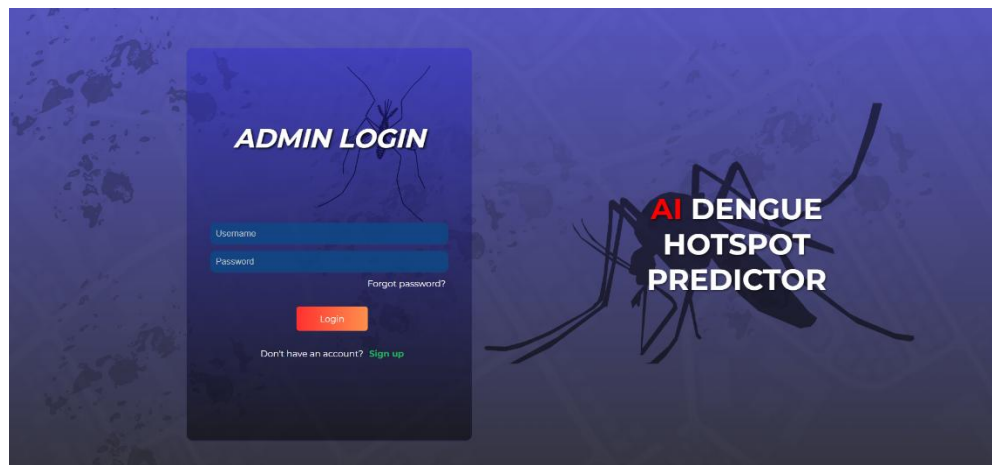
management of an authenticated database with PostgreSQL. It also puts a strong focus on user-friendliness with the responsive design, allowing (responsive) usage from desktop to mobile devices. Security-wise, the application limits access to administrator and barangay health workers through role-based logins, protecting users who are unauthorized to view or edit data.

The results discussed in this chapter summarize the evaluation of the systems, usability, functionality, reliability, and efficiency based on the feedback from 22 barangay and city health personnel during the survey stage. The assessments, expressed through the descriptive and inferential statistics, will provide an evidence-based discussion of the system's overall performance.

## System User Interface

**Figure 3.1**

*Login Page*





The Login Page of the Dengue Hotspot Predictor System is illustrated in Figure 3.1. This page is the secured portal for ALL PLATFORM users, be it barangay health workers or administrators. The interface was created in a professional and simple manner, in order to focus on the official presentation of the system and the project, with its logo and title located at the center top of the screen. The middle container is where the username and password input fields, as well as the login button, go, all with a good layout for user readability.

The login module works in a session-controlled manner so that only validated users can get into the system. The functionality ensures no unauthorized access to the facility and sensitive data, keeping the content discreet and secure at all times. Any user who tries to reach any internal page without being logged in will be automatically redirected to the login form, so you can be sure your content is safe.

One unique feature of this Login system is its automatic role recognition. Once a user attempts to log in via the admin login page, they will be redirected to the user login page, where the system recognizes the credentials entered. On the other hand, admins who attempt to log in using the user login page will be redirected to the admin login page. This structure access flow ensures that the user experience is streamlined and secure, which allows both users and admins to interact with the features that are relevant to their respective roles.



**Figure 3.2**

*Admin Dashboard*

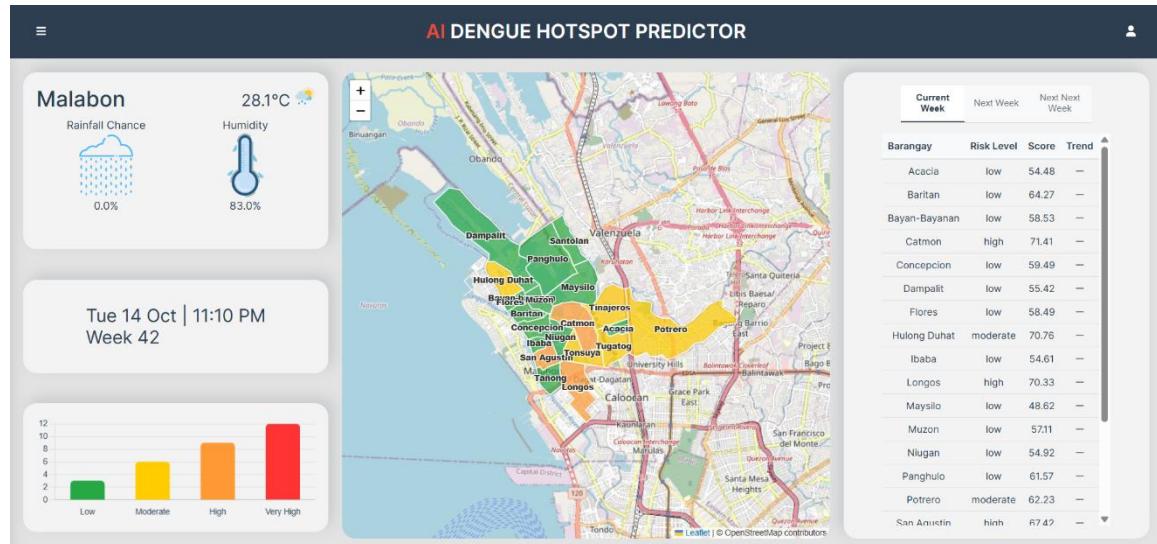


Figure 3.2 shows a visual representation of the AI dengue hotspot predictor system's admin dashboard panel. This page serves as the central control center for system administrators to manage, update, and monitor data, as well as retrain the AI model. The interface is designed to present a concise, organized summary of the dengue hotspot situation across Malabon City's barangays. On the right-hand side of the top of the dashboard, show the datetime and key weather parameters — such as temperature, humidity, and rainfall —simultaneously for administrators to better understand the data or make decisions.



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Dashboard design focuses on access, readability, and professional visualization of analytics information. Color-coded legend – The Admin Dashboard's color-coded key visually groups barangays by dengue risk status. This legend is represented with the colors: green for low risk, yellow for moderate risk, orange for high risk, and red for very high risk. This criterion allows admins to know immediately which critical area the disturbance should be managed or observed. With this visualization, the system successfully converts predictive data into actionable information to support decision-making by the city's health department.

At the heart of the dashboard is an interactive map that automatically changes to show dengue risk per barangay. Admins can hover over or click to see detailed predictions, along with trends and confidence scores, for dengue. On the left side of the map, the system displays textual reports on the risk status of all barangays (up to 180 days) based on trends observed and the corresponding confidence levels produced by the predictive model. These visual and textual features of the Admin Dashboard make it a full-fledged analysis tool that helps administrators manage the dengue surveillance environment in a timely, friendly, and preventive manner.



**Figure 3.3**

*Reports Page – Admin*

| Barangay     | Week Year      | Risk Level | Numerical Risk | Confidence Score | Trend | Created at |
|--------------|----------------|------------|----------------|------------------|-------|------------|
| Tonsuya      | Week 44 - 2025 | Moderate   | 3              | 31.76            | ↑     | 2025-10-13 |
| Tinajeros    | Week 44 - 2025 | Moderate   | 2              | 52.99            | ↑     | 2025-10-13 |
| Tanong       | Week 44 - 2025 | Low        | 1              | 42.88            | ↑     | 2025-10-13 |
| Santolan     | Week 44 - 2025 | Low        | 0              | 36.70            | ↑     | 2025-10-13 |
| San Agustin  | Week 44 - 2025 | High       | 5              | 47.42            | ↑     | 2025-10-13 |
| Potrero      | Week 44 - 2025 | Moderate   | 2              | 42.23            | ↑     | 2025-10-13 |
| Panghulo     | Week 44 - 2025 | Low        | 1              | 41.57            | ↑     | 2025-10-13 |
| Nlugan       | Week 44 - 2025 | Low        | 0              | 34.92            | ↑     | 2025-10-13 |
| Muzon        | Week 44 - 2025 | Low        | 0              | 37.11            | ↑     | 2025-10-13 |
| Maysilo      | Week 44 - 2025 | Low        | 0              | 28.62            | ↑     | 2025-10-13 |
| Longos       | Week 44 - 2025 | High       | 5              | 50.33            | ↑     | 2025-10-13 |
| Ibaba        | Week 44 - 2025 | Low        | 0              | 34.61            | ↑     | 2025-10-13 |
| Hulong Duhat | Week 44 - 2025 | Moderate   | 4              | 50.76            | ↑     | 2025-10-13 |

The Report Page of the Dengue Hotspot Predictor System, shown in Figure 3.3, is one of the pages that can be accessed through the Admin Dashboard navigation panel. All dengue alerts formed by the system are available on this page. The primary use of this page will be to allow the admin to monitor and manipulate historical prediction results for documentation and analysis. The page structure is kept similar to the dashboard layout, so you can easily navigate and understand data.

On this page, administrators may view the history of predictions segmented by barangay and week. In each record, the system automatically tracks and saves the prediction results after each forecast cycle (e.g., risk level, confidence score). This historical database allows public health authorities to track variations and



trends in dengue risk levels for long-term analysis and the evaluation of preventive measures.

The Report Page also includes a download button that lets admins export prediction data in CSV format. The system is flexible, allowing export of all data for a week or only the records per barangay. This feature is useful for digital and manual report processing; the CSV files can be printed as hard copies. With this feature, admins can maintain a clear, easily observable record of the system's forecasting history that can be utilized for reporting, auditing, or presentation at health management meetings.

### Figure 3.4

#### *Dengue Case Data Page – Admin*

| DENGUE CASES |               |                 |               |               |
|--------------|---------------|-----------------|---------------|---------------|
| Case ID      | Barangay      | Number of Cases | Week Reported | Year Reported |
| 20803        | Catmon        | 6               | 41            | 2025          |
| 20801        | Longos        | 3               | 41            | 2025          |
| 20902        | Tonsuya       | 4               | 41            | 2025          |
| 20057        | Acacia        | 1               | 40            | 2025          |
| 20099        | Baritan       | 2               | 40            | 2025          |
| 20141        | Bayan-Bayanan | 0               | 40            | 2025          |
| 20183        | Catmon        | 11              | 40            | 2025          |
| 20225        | Concepcion    | 0               | 40            | 2025          |
| 20267        | Dampalit      | 1               | 40            | 2025          |
| 20309        | Flores        | 0               | 40            | 2025          |
| 20351        | Hulong Duhat  | 4               | 40            | 2025          |
| 20393        | Ibaba         | 4               | 40            | 2025          |
| 20435        | Longos        | 5               | 40            | 2025          |
| 20477        | Maysilo       | 0               | 40            | 2025          |
| 20519        | Muzon         | 0               | 40            | 2025          |



The Dengue Case Data Page of the Dengue Hotspot Predictor System, shown in Figure 3.4, can be accessed via the navigation bar on the Admin Dashboard. This page provides a user-friendly interface for the system administrator to view, update, and delete dengue case records for all barangays in Malabon City. It provides a week-to-week dengue case count data for the whole year till the latest date, giving a full picture of reported cases and enabling us to see any pattern/trend of dengue transmission over time. This page is intended to be used in conjunction with the User Dashboard, where users from the City Health Department can enter weekly dengue case data for the current cycle week. If a mistake is made during data entry, the Dengue Case Data Page allows the administrator to modify or delete case records. This way, all data in the system is guaranteed to be accurate, consistent, and reliable, preserving the global consistency of the dengue database.

The Dengue Case Data Page of the Dengue Hotspot Predictor System is shown in Fig. 3.4 and can be accessed via the navigation bar on the Admin Dashboard. This page provides a user-friendly interface for the system administrator to view, update, and delete dengue case records for all barangays in Malabon City. It provides a week-to-week dengue case count data for the whole year till the latest date, giving a full picture of reported cases and enabling us to see any pattern/trend of dengue transmission over time. This page is intended to be used in conjunction with the User Dashboard, where users from the City Health Department can enter weekly dengue case data for the current cycle week. If a



mistake is made during data entry, the Dengue Case Data Page allows the administrator to modify or delete case records. This way, all data in the system is guaranteed to be accurate, consistent, and reliable, preserving the global consistency of the dengue database.

**Figure 3.5**

*Log History Page – Admin*

| HISTORY LOGS |               |          |                                                 |                           |
|--------------|---------------|----------|-------------------------------------------------|---------------------------|
| ID           | user          | Action   | Description                                     | Timestamp                 |
| 91           | Ivan Conejero | LOGIN    | Admin [Ivan Conejero] successfully logged in.   | Oct. 14, 2025, 11:12 p.m. |
| 90           | Ivan Conejero | LOGIN    | Admin [Ivan Conejero] successfully logged in.   | Oct. 14, 2025, 8:33 p.m.  |
| 89           | Longos        | LOGIN    | User 'Longos' successfully logged in.           | Oct. 14, 2025, 8:31 p.m.  |
| 88           | Longos        | LOGIN    | User 'Longos' successfully logged in.           | Oct. 14, 2025, 8:31 p.m.  |
| 87           | Ivan Conejero | LOGIN    | Admin [Ivan Conejero] successfully logged in.   | Oct. 14, 2025, 8:29 p.m.  |
| 86           | Longos        | LOGIN    | User 'Longos' successfully logged in.           | Oct. 14, 2025, 7:21 p.m.  |
| 85           | admin         | LOGIN    | Admin [admin] successfully logged in.           | Oct. 14, 2025, 7:11 p.m.  |
| 84           | Longos        | LOGIN    | User 'Longos' successfully logged in.           | Oct. 14, 2025, 7:08 p.m.  |
| 83           | catmon        | LOGIN    | User 'catmon' successfully logged in.           | Oct. 14, 2025, 1:47 p.m.  |
| 82           | admin         | LOGIN    | Admin [admin] successfully logged in.           | Oct. 14, 2025, 1:44 p.m.  |
| 81           | catmon        | LOGIN    | User 'catmon' successfully logged in.           | Oct. 14, 2025, 1:40 p.m.  |
| 80           | catmon        | LOGIN    | User 'catmon' successfully logged in.           | Oct. 14, 2025, 1:30 p.m.  |
| 79           | catmon        | REGISTER | User account successfully created for [catmon]. | Oct. 14, 2025, 1:30 p.m.  |
| 78           | jeff          | LOGIN    | Admin [jeff] successfully logged in.            | Oct. 14, 2025, 1:29 p.m.  |
| 77           | Tonsuya       | LOGIN    | User 'Tonsuya' successfully logged in.          | Oct. 14, 2025, 11:30 a.m. |

The Log History Page of the Dengue Hotspot Predictor System is illustrated in Figure 3.5 and can be navigated from the home page via the navigation panel on the Admin Dashboard. This page is intended to provide a clean, secure history of everything an admin/user has done in the system. It is essential to keep accountability, data integrity & security on your web application.



The Log History Page records everything that users and the captain have done, such as adding, updating, or deleting records, or logging in/out. Details of each activity for a user are recorded, including the username, role type, action name, and timestamp. This serves as a useful Jenkins head of administration to cross-examine what was going on and when.

The Log History Page also helps identify potential abuse by system abusers and serves as a monitoring tool. By analyzing activity history and empirically detecting anomalies, including attacks within it, the system can enhance network security. These logs may also be analyzed over time to assess system usage and performance, and one or more determinations may be made based on the observations, including user management or access control.

In conclusion, the Log History Page ensures transparency and trust in our system. This management means that the Dengue Hotspot Predictor System would be able to achieve full accountability, with data accuracy that can be monitored, and also promote its sustainability.



**Figure 3.6**

*Weather Page – Admin*

| Date            | Max Temperature | Min Temperature | Rainfall Chance | Wind Speed (km/h) | Weather Code |
|-----------------|-----------------|-----------------|-----------------|-------------------|--------------|
| Tue 14 Oct 2025 | 29.5°C          | 27.0°C          | 63.0%           | 13.1              | 80.0         |
| Wed 15 Oct 2025 | 29.9°C          | 27.8°C          | 48.0%           | 7.0               | 3.0          |
| Thu 16 Oct 2025 | 30.7°C          | 27.7°C          | 38.0%           | 9.1               | 3.0          |
| Fri 17 Oct 2025 | 29.1°C          | 28.0°C          | 41.0%           | 3.7               | 80.0         |
| Sat 18 Oct 2025 | 28.7°C          | 27.6°C          | 51.0%           | 11.6              | 95.0         |
| Sun 19 Oct 2025 | 29.0°C          | 27.5°C          | 55.0%           | 17.7              | 95.0         |
| Mon 20 Oct 2025 | 29.6°C          | 27.9°C          | 49.0%           | 12.2              | 96.0         |
| Tue 21 Oct 2025 | 29.1°C          | 27.5°C          | 34.0%           | 14.5              | 3.0          |
| Wed 22 Oct 2025 | 29.8°C          | 26.8°C          | 35.0%           | 24.1              | 51.0         |
| Thu 23 Oct 2025 | 29.4°C          | 26.6°C          | 79.0%           | 26.0              | 51.0         |
| Fri 24 Oct 2025 | 30.7°C          | 26.5°C          | 62.0%           | 11.9              | 53.0         |
| Sat 25 Oct 2025 | 29.9°C          | 25.6°C          | 52.0%           | 11.0              | 51.0         |
| Sun 26 Oct 2025 | 30.2°C          | 25.6°C          | 71.0%           | 10.7              | 0.0          |

Figure 3.6 The Weather Page in Dengue Hotspot Predictor System: This page can be navigated to when the Article tab on the navigation menu in the Admin dashboard is clicked. This information includes real-time and forecast weather conditions, which the system keeps up to date by querying the Open-Meteo API. The integration described here contributes to improving predictive models by using relevant, contemporary environmental information required to predict levels of dengue risk.

This part of the website reports several meteorological parameters, such as temperature, rainfall, and humidity, which affect water supply and, in turn, mosquito breeding patterns (and thereby dengue transmission). These parameters can be observed over various time periods, such as hourly, daily, and weekly, to support



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short- or long-term trend analysis for the administrator. The page provides a 14-day forecast so local public health officials can develop prevention plans based on anticipated weather conditions.

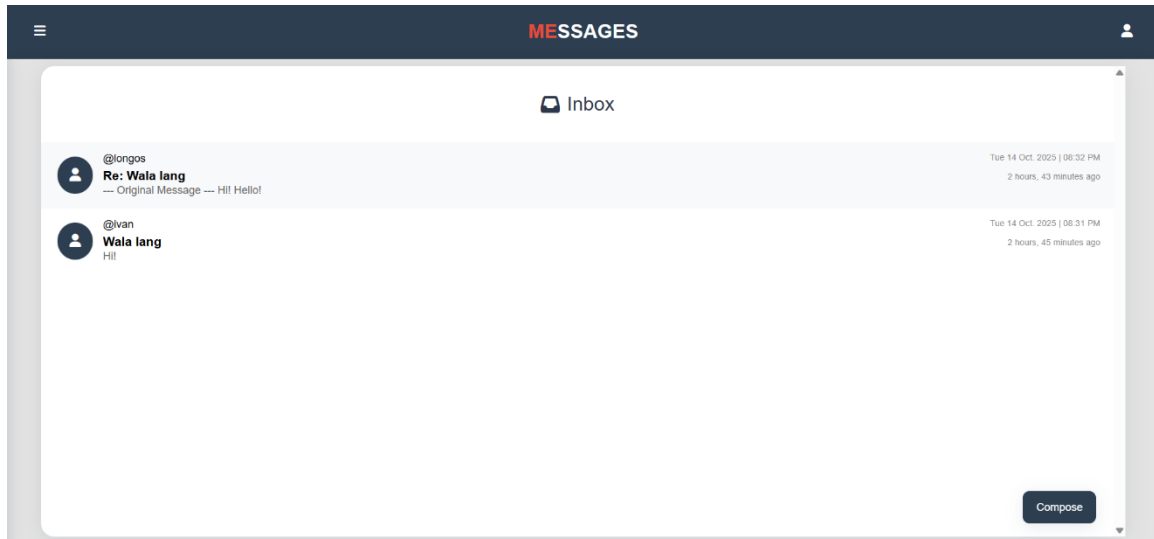
This function facilitates the user's system operation by providing visual weather data. Within graphical charts or tables, the administrator can easily understand the comparison of changing weather and its impact on dengue case trends.) In addition, the predicted data can automatically refresh periodically, ensuring you always receive the most up-to-date and reliable notification information.

In summary, the Weather Page is an important component of DHPS that links environmental factors to disease predictions. With real-time, accessible weather data, LGUs can make informed decisions, enabling early action to minimize dengue cases in identified hotspots.



**Figure 3.7**

*Messaging Page – Admin*



The Messaging Page of the Dengue Hotspot Predictor System, which acts as an interface between administrators and users, is presented in Figure 3.7. You can navigate to this page from the Admin Dashboard navigation panel. The user interface is developed for easy and effective message exchange between system users: the admin and registered Users - primarily city health department officials or barangay staff, where the admin has the control to send, receive, read, and maintain (delete messages) from others.

Updates, explanations, or feedback on the dengue data, system functionalities, and technical issues can all be addressed promptly on the Messaging page. This facilitates improved consensus-building among administrators and the user community who post dengue case data. On the



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messaging page, you can see an organized 'message list' showing the sender's name, message text, and timestamp, making it easier for the admin to track conversation threads by reviewing the communication history.

Finally, the flannel chat-like system provides real-time messaging and notifications, so other means (such as chat) are not needed for instant notice to admins (users). The integrated messaging system adds significant functionality, centralizing all operational communication between stakeholders and members on a secure platform while ensuring that communications occur within the platform's data privacy and safeguards.

In short, the Messaging Page enhances collaboration between users and admins. It also enables effective communication and coordination among those involved in data collection and analysis, field operations, and an organized workflow, which triggers proactive surveillance of dengue information across the city.



**Figure 3.8**

*User Dashboard*

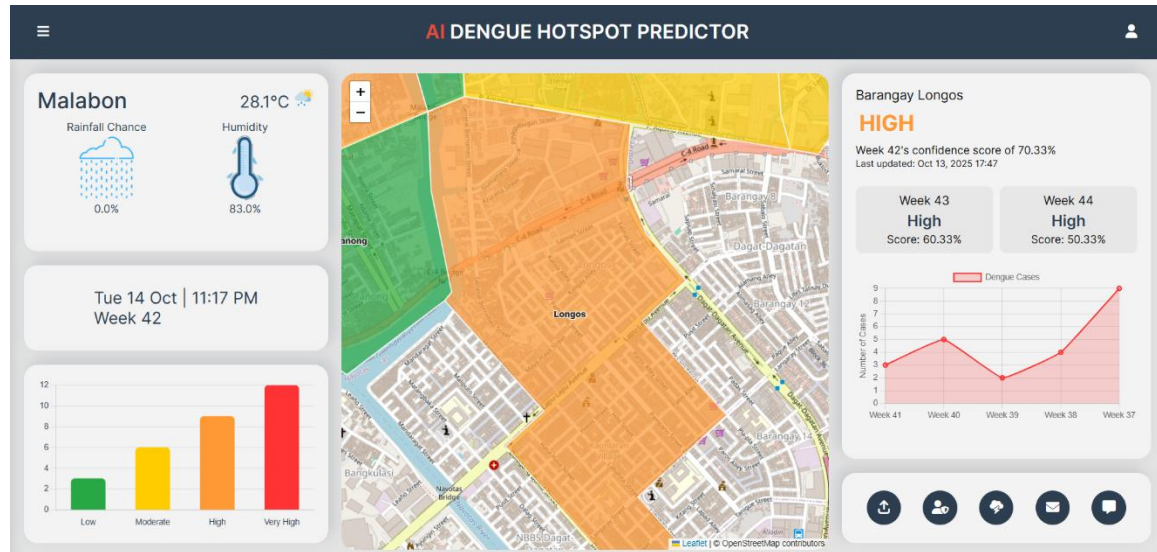


Figure 3.8 User Dashboard for the Dengue Hotspot Predictor System. The user dashboard (cf. Figure 3) is a user-friendly, interactive, and responsive application intended for barangay health personnel's representatives to assist mothers in understanding how dengue risk levels vary across Malabon City's zones. This is helpful to users, who can interpret the environment and epidemiological factors from the dashboard, which displays current weather, a weekly forecast, and dengue risk in a single interface.

At the center of the platform is an interactive map that shows dengue hotspots with color-coded risk zones, which allows users to recognize barangays with very high, high, moderate, or low dengue transmission risk.



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In the left section under weather summary, the main meteorological variables (i.e., temperature, humidity, and rainfall probability) that may influence mosquito production and dengue transmission are shown. The bar chart below presents the overall level of dengue risk—Low, Moderate, High, and Very High—so users can quickly identify the city's current risk.

The right panel contains barangay-level information. It details the barangay name, its current dengue risk level, and its weekly confidence score. A line graph showing the trend of dengue cases, which a user can observe case progression and predicted risk levels in the next weeks.

Other usability and accessibility enhancements for end users include the dashboard's quick-access buttons for map view controls, data refresh, and sharing/notification, in an instantly recognizable—dare we say—seamless design.

Dashboards Overall, the User Dashboard is a central point of information that combines weather observations, forecasting analytics, and a visual overview. In bringing detailed information to an easy-to-use interface, the dashboard enables residents and local health advocates to take a proactive approach in their fight against theoretical dengue threats.



**Figure 3.9**

*Report Updates Page*

Figure 3.9 shows the Report Updates Page of the Dengue Hotspot Predictor System. It is the page this system uses as an interface for adding and updating reported dengue case records to the database. This web page can be used by the barangay health personnel representative to manually enter, upload, and/or download dengue case data into the system's database to keep this system updated and accurate. The manual, upload, and download options are available on the page.

The Manual Report Submission section, as shown in the picture, allows users to enter relevant report information (e.g., Week of Report, Year of Report, and Number of Cases). The user may lift the entry by clicking “Submit Report” after entering the IEO in all mandatory fields in the system database. This feature is



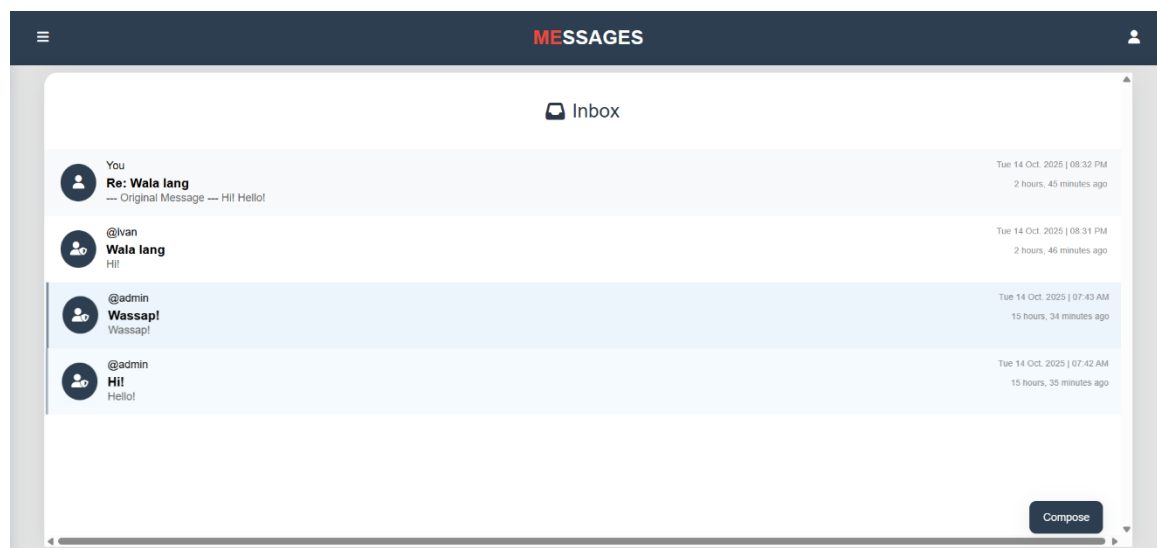
crucial for receiving uninterrupted, assured data input even when auto data collection is unavailable.

The system provides manual and automatic upload facilities, catering to both technically skilled and less technical users. Moreover, the download feature enables users to access existing reports for archiving, verification, or further processing.

In summary, the Report Updates Page enhances the system's data management capabilities with an easy-to-use reporting mechanism for dengue cases. The platform keeps predictive modelling and hotspot mapping information up to date, underpinning the soundness of data-driven decision-making in dengue response and prevention efforts.

**Figure 3.10**

*Messaging - User*





The Dengue Hotspot Predictor System User Messaging Page serves as a central point of communication between the user and the system administrator, as shown in Figure 3.10. You can find this from the User Dashboard. It is via this page that a barangay representative and local health officer can link to send, receive, read, and manage dengue reports/ weather forecasts/feedback about the system.

Through the Messaging Page, you can write to administrators directly for any news, questions, or even assistance with submitting dengue cases and using the system. The page also features a summary of chats (sender name, text, time) for quick access to the history and for further discussions with the admin team.

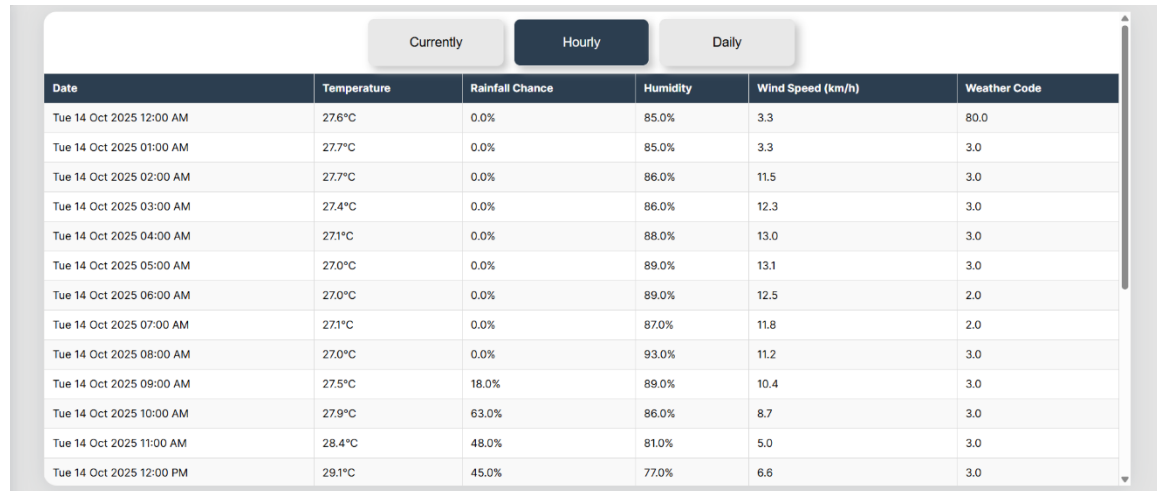
A real-time message and notification system will not leave your users waiting for auxiliary communication tools to deliver announcements and feedback alerts they are anxious to know. It has also adopted an additional level of security by internalizing communication, so that all exchanges are perceived as secure and discreet under the platform's data privacy policy.

In general, the User Messaging Page enables efficient communication between system users and the administration. It encourages timely and effective coordination, enhances the accuracy of capturing dengue-related information, and fosters collaboration among barangays in monitoring and managing such information.



**Figure 3.11**

*Weather Page*



| Date                     | Temperature | Rainfall Chance | Humidity | Wind Speed (km/h) | Weather Code |
|--------------------------|-------------|-----------------|----------|-------------------|--------------|
| Tue 14 Oct 2025 12:00 AM | 27.6°C      | 0.0%            | 85.0%    | 3.3               | 80.0         |
| Tue 14 Oct 2025 01:00 AM | 27.7°C      | 0.0%            | 85.0%    | 3.3               | 3.0          |
| Tue 14 Oct 2025 02:00 AM | 27.7°C      | 0.0%            | 86.0%    | 11.5              | 3.0          |
| Tue 14 Oct 2025 03:00 AM | 27.4°C      | 0.0%            | 86.0%    | 12.3              | 3.0          |
| Tue 14 Oct 2025 04:00 AM | 27.1°C      | 0.0%            | 88.0%    | 13.0              | 3.0          |
| Tue 14 Oct 2025 05:00 AM | 27.0°C      | 0.0%            | 89.0%    | 13.1              | 3.0          |
| Tue 14 Oct 2025 06:00 AM | 27.0°C      | 0.0%            | 89.0%    | 12.5              | 2.0          |
| Tue 14 Oct 2025 07:00 AM | 27.1°C      | 0.0%            | 87.0%    | 11.8              | 2.0          |
| Tue 14 Oct 2025 08:00 AM | 27.0°C      | 0.0%            | 93.0%    | 11.2              | 3.0          |
| Tue 14 Oct 2025 09:00 AM | 27.5°C      | 18.0%           | 89.0%    | 10.4              | 3.0          |
| Tue 14 Oct 2025 10:00 AM | 27.9°C      | 63.0%           | 86.0%    | 8.7               | 3.0          |
| Tue 14 Oct 2025 11:00 AM | 28.4°C      | 48.0%           | 81.0%    | 5.0               | 3.0          |
| Tue 14 Oct 2025 12:00 PM | 29.1°C      | 45.0%           | 77.0%    | 6.6               | 3.0          |

Figure 3.11 illustrates the User Weather Page that can be accessed through the Dengue Hotspot Predictor System's User Dashboard. It returns current and future weather data from the Open-Meteo API, including temperature, precipitation, humidity, wind speed, and more. This enables the user to view the weather by hour, by day, or for this week, and to forecast up to 14 days from today's date. These predictions can inform people about upcoming weather conditions that increase the risk of dengue. Interactive tables and graphics provide rapid insight into weather trends, with an auto-update feature to keep them current. This application enables users to monitor the progression of weather conditions linked to dengue and take steps to prevent the disease in high-risk areas.



## System Evaluation

The system evaluation aims to assess the overall functionality, reliability, and usability of the Dengue Hotspot Predictor System. This process ensures that all modules perform according to the intended design and meet the requirements set during system development. Through structured testing procedures, both the administrator and user components were evaluated to verify their operational accuracy and responsiveness. The evaluation also helps identify potential areas for improvement to enhance system performance and user experience. Overall, this phase validates the system's readiness for deployment and its effectiveness in supporting dengue surveillance and prediction efforts.

**System Name:** Predicting Dengue Hotspots in Malabon City Through AI and Integrated Weather-Case Analysis

**Test Date:** October 24, 2025

**Evaluator's Name:** Gian Carlo A. Cruz

*System Testing Cases for Administrator*

| Test Case Instances | Test Case Scenario | Expected Result | Checked (/) | Result Test       |
|---------------------|--------------------|-----------------|-------------|-------------------|
|                     |                    |                 | Good        | Needs Improvement |



|     |                                                                    |                                                                          |   |  |
|-----|--------------------------------------------------------------------|--------------------------------------------------------------------------|---|--|
| 1.  | Log in with valid credentials.                                     | Admin is redirected to the dashboard.                                    | / |  |
| 2.  | Login with invalid credentials.                                    | An error message is shown: "Invalid username or password".               | / |  |
| 3.  | View the dengue case summary dashboard.                            | Real-time dashboard displays graphs, charts, and alerts.                 | / |  |
| 4.  | Access barangay-level risk heatmaps.                               | Color-coded map loads correctly with clickable barangay data.            | / |  |
| 5.  | Export reports (CSV/PDF).                                          | Reports are generated with the correct data format and content.          | / |  |
| 6.  | Add a new user account (e.g., health officer).                     | New user is saved and appears in the user list with default role.        | / |  |
| 7.  | Delete user account.                                               | Selected user is removed from the system with a confirmation prompt.     | / |  |
| 8.  | View activity logs of all users.                                   | Audit log shows login times, data updates, and critical actions by user. | / |  |
| 9.  | Update admin profile settings (name, username password and email). | Changes are saved and reflected after submission.                        | / |  |
| 10. | Logout from admin dashboard.                                       | Admin is securely logged out and redirected to login page.               | / |  |

**System Testing Cases for Users/Barangay Health Department Personnel**

| <b>Test Case Instances</b> | <b>Test Case Scenario</b>                        | <b>Expected Result</b>                                             | <b>Checked (/)</b> | <b>Result Test</b>       |
|----------------------------|--------------------------------------------------|--------------------------------------------------------------------|--------------------|--------------------------|
|                            |                                                  |                                                                    | <b>Good</b>        | <b>Needs Improvement</b> |
| <b>1.</b>                  | Login with valid credentials.                    | User is redirected to the main dashboard.                          | /                  |                          |
| <b>2.</b>                  | Login with invalid credentials.                  | An error message is shown: "Invalid username or password".         | /                  |                          |
| <b>3.</b>                  | View the barangay dengue risk map.               | Color-coded risk map loads correctly, showing the user's barangay. | /                  |                          |
| <b>4.</b>                  | Search for specific barangay dengue data.        | Accurate risk data is shown for the selected barangay.             | /                  |                          |
| <b>5.</b>                  | View forecast trends by week.                    | Graphs/charts update to reflect the selected timeframe.            | /                  |                          |
| <b>6.</b>                  | Access alert banners for the current dengue risk | Alert level is shown clearly (e.g., Low, Moderate, High)           | /                  |                          |
| <b>7.</b>                  | Reset user password.                             | The password reset link is sent via email.                         | /                  |                          |
| <b>8.</b>                  | Search/filter users by name or status            | Filtered results appear based on search criteria                   | /                  |                          |
| <b>9.</b>                  | Log out of the user dashboard.                   | User is securely logged out and redirected to the login page.      | /                  |                          |



**System Name:** Predicting Dengue Hotspots in Malabon City Through AI and Integrated Weather-Case Analysis

**Test Date:** October 24, 2025

**Evaluator's Name:** Engr. Jed G. Concepcion

*System Testing Cases for Administrator*

| Test Case Instances | Test Case Scenario                             | Expected Result                                                           | Checked (I) | Result Test       |
|---------------------|------------------------------------------------|---------------------------------------------------------------------------|-------------|-------------------|
|                     |                                                |                                                                           | Good        | Needs Improvement |
| 1.                  | Log in with valid credentials.                 | Admin is redirected to the dashboard.                                     | /           |                   |
| 2.                  | Login with invalid credentials.                | An error message is shown: "Invalid username or password".                | /           |                   |
| 3.                  | View the dengue case summary dashboard.        | Real-time dashboard displays graphs, charts, and alerts.                  | /           |                   |
| 4.                  | Access barangay-level risk heatmaps.           | Color-coded map loads correctly with clickable barangay data.             | /           |                   |
| 5.                  | Export reports (CSV/PDF).                      | Reports are generated with the correct data format and content.           | /           |                   |
| 6.                  | Add a new user account (e.g., health officer). | The new user is saved and appears in the user list with the default role. | /           |                   |



|     |                                                                      |                                                                              |   |  |
|-----|----------------------------------------------------------------------|------------------------------------------------------------------------------|---|--|
| 7.  | Delete the user account.                                             | The selected user is removed from the system with a confirmation prompt.     | / |  |
| 8.  | View activity logs of all users.                                     | Audit log shows login times, data updates, and critical actions by the user. | / |  |
| 9.  | Update admin profile settings (name, username, password, and email). | Changes are saved and reflected after submission.                            | / |  |
| 10. | Log out of the admin dashboard.                                      | Admin is securely logged out and redirected to the login page.               | / |  |

#### System Testing Cases for Users/Barangay Health Department Personnel

| Test Case Instances | Test Case Scenario                        | Expected Result                                                    | Checked (/) | Result Test       |
|---------------------|-------------------------------------------|--------------------------------------------------------------------|-------------|-------------------|
|                     |                                           |                                                                    | Good        | Needs Improvement |
| 1.                  | Log in with valid credentials.            | User is redirected to the main dashboard.                          | /           |                   |
| 2.                  | Login with invalid credentials.           | An error message is shown: "Invalid username or password".         | /           |                   |
| 3.                  | View the barangay dengue risk map.        | Color-coded risk map loads correctly, showing the user's barangay. | /           |                   |
| 4.                  | Search for specific barangay dengue data. | Accurate risk data is shown for the selected barangay.             | /           |                   |



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|    |                                                  |                                                               |   |  |
|----|--------------------------------------------------|---------------------------------------------------------------|---|--|
| 5. | View forecast trends by week.                    | Graphs/charts update to reflect the selected timeframe.       | / |  |
| 6. | Access alert banners for the current dengue risk | Alert level is shown clearly (e.g., Low, Moderate, High)      | / |  |
| 7. | Reset user password.                             | The password reset link is sent via email.                    | / |  |
| 8. | Search/filter users by name or status            | Filtered results appear based on search criteria              | / |  |
| 9. | Log out of the user dashboard.                   | User is securely logged out and redirected to the login page. | / |  |

**System Name:** Predicting Dengue Hotspots in Malabon City Through AI and Integrated Weather-Case Analysis

**Test Date:** October 24, 2025

**Evaluator's Name:** Dr. Enrico Chavez

*System Testing Cases for Administrator*

| Test Case Instances | Test Case Scenario             | Expected Result                       | Checked (/) | Result Test       |
|---------------------|--------------------------------|---------------------------------------|-------------|-------------------|
|                     |                                |                                       | Good        | Needs Improvement |
| 1.                  | Log in with valid credentials. | Admin is redirected to the dashboard. | /           |                   |



|     |                                                                      |                                                                              |   |   |
|-----|----------------------------------------------------------------------|------------------------------------------------------------------------------|---|---|
| 2.  | Login with invalid credentials.                                      | An error message is shown: "Invalid username or password".                   | / |   |
| 3.  | View the dengue case summary dashboard.                              | Real-time dashboard displays graphs, charts, and alerts.                     | / |   |
| 4.  | Access barangay-level risk heatmaps.                                 | Color-coded map loads correctly with clickable barangay data.                | / |   |
| 5.  | Export reports (CSV/PDF).                                            | Reports are generated with the correct data format and content.              |   | / |
| 6.  | Add a new user account (e.g., health officer).                       | The new user is saved and appears in the user list with the default role.    | / |   |
| 7.  | Delete the user account.                                             | The selected user is removed from the system with a confirmation prompt.     | / |   |
| 8.  | View activity logs of all users.                                     | Audit log shows login times, data updates, and critical actions by the user. |   | / |
| 9.  | Update admin profile settings (name, username, password, and email). | Changes are saved and reflected after submission.                            | / |   |
| 10. | Log out of the admin dashboard.                                      | Admin is securely logged out and redirected to the login page.               | / |   |

**System Testing Cases for Users/Barangay Health Department Personnel**

| Test Case Instances | Test Case Scenario                               | Expected Result                                                    | Checked (/) | Result Test              |
|---------------------|--------------------------------------------------|--------------------------------------------------------------------|-------------|--------------------------|
|                     |                                                  |                                                                    | <b>Good</b> | <b>Needs Improvement</b> |
| 1.                  | Log in with valid credentials.                   | User is redirected to the main dashboard.                          | /           |                          |
| 2.                  | Login with invalid credentials.                  | An error message is shown: "Invalid username or password".         | /           |                          |
| 3.                  | View the barangay dengue risk map.               | Color-coded risk map loads correctly, showing the user's barangay. | /           |                          |
| 4.                  | Search for specific barangay dengue data.        | Accurate risk data is shown for the selected barangay.             | /           |                          |
| 5.                  | View forecast trends by week.                    | Graphs/charts update to reflect the selected timeframe.            | /           |                          |
| 6.                  | Access alert banners for the current dengue risk | Alert level is shown clearly (e.g., Low, Moderate, High)           | /           |                          |
| 7.                  | Reset user password.                             | The password reset link is sent via email.                         |             | /                        |
| 8.                  | Search/filter users by name or status            | Filtered results appear based on search criteria                   | /           |                          |
| 9.                  | Log out of the user dashboard.                   | User is securely logged out and redirected to the login page.      | /           |                          |



### **System Evaluation Interpretation**

The system evaluation for “Predicting Dengue Hotspots in Malabon City Through AI and Integrated Weather-Case Analysis” was conducted on October 24, 2025, by three evaluators: Mr. Gian Carlo A. Cruz, Engr. Jed G. Concepcion y Dr. Enrico Chávez. The testing assessed both the Administrator and User Modules using predefined functional test cases to determine the system’s performance, usability, and reliability.

Results showed that all core features — including login authentication, dashboard visualization, dengue risk mapping, report generation, and activity logging — performed as expected. Both modules received consistent “Good” ratings across all evaluators, indicating stable functionality and accuracy. Minor improvement suggestions were limited to interface optimization and performance refinement during data export.

Overall, the evaluation confirmed that the system is functionally sound, user-friendly, and ready for deployment. It effectively supports dengue surveillance and predictive analysis, providing reliable insights for local health authorities in Malabon City.



## Results

The evaluation of the AI-Based Dengue Hotspot Predictor was conducted with 22 health officials from Malabon City. Using a 4-point Likert scale (Strongly Agree = 4, Agree = 3, Disagree = 2, Strongly Disagree = 1), respondents assessed the system based on the ISO 25010 software quality model criteria: Functionality, Usability, Reliability, and Efficiency.

The responses were tabulated using Frequency and Percentage Distribution to visualize agreement levels, and Weighted Mean to determine the overall perception for each criterion.

## Profile of Respondents

A total of 22 respondents participated in the system evaluation. They were primarily City Health Department personnel who regularly engage in disease monitoring, reporting, and dengue prevention activities.

The respondents' demographic data are summarized below:

**Age Distribution** (Table 3.1) shows that most respondents were aged 18–25 years old (36.4%), followed by those 26–35 years old (36.4%), 36–45 years old (18.2%), and 46 and above (9.1%).

This indicates that the majority of evaluators belong to a relatively young demographic who are adaptive to technological systems.



**Gender Distribution** (Table 3.1) reveals that 63.6% were female, 31.8% male, and 4.5% prefer not to say, reflecting gender diversity among the health department personnel.

**Educational Attainment** (Table 3.1) shows that most respondents were college undergraduates (45.5%), bachelor's degree holders (36.5%), and vocational graduates (13.6%), with only one respondent having a doctorate degree (4.5%).

**Table 3.1**

*Demographic Characteristics of the Sample Population*

| Demographic            | Level | Frequency | Percentage (%) |
|------------------------|-------|-----------|----------------|
| Occupation             | 1     | 2         | 9.1            |
|                        | 2     | 20        | 90.9           |
| Age                    | 1     | 2         | 9.1            |
|                        | 2     | 4         | 18.2           |
|                        | 3     | 8         | 36.4           |
|                        | 4     | 8         | 36.4           |
| Gender                 | 1     | 7         | 31.8           |
|                        | 2     | 14        | 63.6           |
|                        | 3     | 1         | 4.5            |
| Educational Attainment | 2     | 3         | 13.6           |
|                        | 3     | 10        | 45.5           |
|                        | 4     | 8         | 36.4           |
|                        | 6     | 1         | 4.5            |
| Note: N=22             |       |           |                |



**Table 3.2**

*Descriptive Statistics on the Assessment of the Usability*

| Indicator  | Mean | Median | Std. Deviation | Shapiro-Wilk |          | Verbal Interpretation | Rank |
|------------|------|--------|----------------|--------------|----------|-----------------------|------|
|            |      |        |                | <i>W</i>     | <i>p</i> |                       |      |
| USA1       | 3.77 | 4.00   | 0.43           | 0.52         | <.001    | Strongly Agree        | 2.5  |
| USA2       | 3.68 | 4.00   | 0.57           | 0.61         | <.001    | Strongly Agree        | 4.5  |
| USA3       | 3.77 | 4.00   | 0.43           | 0.52         | <.001    | Strongly Agree        | 2.5  |
| USA4       | 3.68 | 4.00   | 0.48           | 0.59         | <.001    | Strongly Agree        | 4.5  |
| USA5       | 3.91 | 4.00   | 0.29           | 0.33         | <.001    | Strongly Agree        | 1    |
| Note: N=22 |      |        |                |              |          |                       |      |

Table 3.2 presents the results of the Usability Assessment (USA) of the system, which evaluates the respondents' level of agreement on various usability indicators such as ease of use, system intuitiveness, and interface efficiency. The indicators obtained mean scores ranging from 3.68 to 3.91, all interpreted as "Strongly Agree." The highest-rated item was USA5 (Mean = 3.91), indicating that users found the system exceptionally efficient and user-friendly. Meanwhile, USA2 and USA4, both with mean scores of 3.68, received the lowest yet still highly positive evaluations, suggesting minor areas for refinement while maintaining overall strong usability performance.

The computed overall weighted mean of 3.76, verbally interpreted as "Strongly Agree," signifies that the respondents perceived the system as easy to



navigate and functionally coherent. The consistently high ratings and minimal standard deviation values indicate strong consensus among participants, reflecting stable and reliable user experiences. Moreover, the Shapiro–Wilk test values ( $p < .001$ ) imply that the data slightly deviate from normality, which is common in Likert-type responses clustered toward the positive end of the scale.

These findings suggest that the system's design effectively meets usability standards by emphasizing accessibility, clarity, and user-centered interaction. The favorable feedback demonstrates that both technical and non-technical users can operate the platform comfortably and efficiently. Overall, the results affirm that the system provides an intuitive and responsive interface that supports seamless task execution, enhancing user satisfaction and engagement during operation.

### Table 3.3

### *Descriptive Statistics on the Assessment of the Functionality*

| Indicator  | Mean | Median | Std. Deviation | Shapiro-Wilk |          | Verbal Interpretation | Rank |
|------------|------|--------|----------------|--------------|----------|-----------------------|------|
|            |      |        |                | <i>W</i>     | <i>p</i> |                       |      |
| FUNC1      | 3.77 | 4.00   | 0.43           | 0.52         | <.001    | Strongly Agree        | 2.5  |
| FUNC2      | 3.73 | 4.00   | 0.55           | 0.56         | <.001    | Strongly Agree        | 4    |
| FUNC3      | 3.77 | 4.00   | 0.43           | 0.52         | <.001    | Strongly Agree        | 2.5  |
| FUNC4      | 3.68 | 4.00   | 0.48           | 0.59         | <.001    | Strongly Agree        | 5    |
| FUNC5      | 3.82 | 4.00   | 0.40           | 0.47         | <.001    | Strongly Agree        | 1    |
| Note: N=22 |      |        |                |              |          |                       |      |



Table 3.3 presents the results of the Functionality Assessment (FUNC), which evaluates how effectively the system performs its intended operations. The respondents' ratings show mean scores ranging from 3.68 to 3.82, all interpreted as "Strongly Agree." The highest-rated indicator, FUNC5 (Mean = 3.82), suggests that users found the system dependable and capable of executing functions without errors. Meanwhile, FUNC4 (Mean = 3.68) received the lowest mean but still reflected strong agreement, indicating only minor areas for improvement.

The overall weighted mean of 3.75, also interpreted as "Strongly Agree," confirms that the system operates efficiently, produces accurate outputs, and supports core tasks reliably. The low standard deviation values indicate that responses were consistent across participants, while the Shapiro–Wilk p-values ( $< .001$ ) suggest a positive skew, showing that most respondents strongly agreed with the indicators.

Overall, the findings affirm that the system demonstrates high functional reliability and stability. Respondents agreed that it meets its objectives effectively, supporting both user and administrative operations without performance issues. This consistent positive feedback highlights that the system's backend processes and design ensure dependable functionality, contributing to an efficient and error-free user experience.



**Table 3.4**

*Descriptive Statistics on the Assessment of the Reliability*

| Indicator         | Mean | Median | Std.<br>Deviation | Shapiro-<br>Wilk |          | Verbal<br>Interpretation | Rank |
|-------------------|------|--------|-------------------|------------------|----------|--------------------------|------|
|                   |      |        |                   | <i>W</i>         | <i>p</i> |                          |      |
| REL1              | 3.73 | 4.00   | 0.55              | 0.56             | <.001    | Strongly Agree           | 3    |
| REL2              | 3.59 | 4.00   | 0.59              | 0.68             | <.001    | Strongly Agree           | 5    |
| REL3              | 3.64 | 4.00   | 0.49              | 0.61             | <.001    | Strongly Agree           | 4    |
| REL4              | 3.91 | 4.00   | 0.29              | 0.33             | <.001    | Strongly Agree           | 1.5  |
| REL5              | 3.91 | 4.00   | 0.29              | 0.33             | <.001    | Strongly Agree           | 1.5  |
| <i>Note: N=22</i> |      |        |                   |                  |          |                          |      |

Table 3.4 presents the Reliability Assessment (REL) of the system, which measures its consistency, accuracy, and dependability during operation. The respondents' ratings show mean scores ranging from 3.59 to 3.91, all verbally interpreted as "Strongly Agree." The highest-rated indicators, REL4 and REL5 (Mean = 3.91), reflect that users found the system highly dependable and capable of producing consistent results without failure. In contrast, REL2 (Mean = 3.59) obtained the lowest mean, though it still indicates strong agreement, suggesting only minor variability in user experience.

The overall weighted mean of 3.76, interpreted as "Strongly Agree," confirms that users perceive the system as reliable, stable, and accurate in processing and delivering results. The relatively low standard deviation values



demonstrate consistent responses among participants, while the Shapiro–Wilk test ( $p < .001$ ) indicates that most respondents strongly favored the reliability indicators.

Overall, the findings affirm that the system performs with a high level of reliability and consistency, ensuring minimal errors during operation. Respondents agreed that it consistently functions as expected, recovering from minor issues without data loss or performance degradation. This reliability ensures that users can depend on the system for accurate and continuous service delivery, reinforcing its suitability for real-world implementation and long-term use.

### Table 3.5

### Descriptive Statistics on the Assessment of the Efficiency

| Indicator  | Mean | Median | Std. Deviation | Shapiro-Wilk |          | Verbal Interpretation | Rank |
|------------|------|--------|----------------|--------------|----------|-----------------------|------|
|            |      |        |                | <i>W</i>     | <i>p</i> |                       |      |
| EFF1       | 3.64 | 4.00   | 0.58           | 0.65         | <.001    | Strongly Agree        | 4.5  |
| EFF2       | 3.86 | 4.00   | 0.35           | 0.41         | <.001    | Strongly Agree        | 1    |
| EFF3       | 3.64 | 4.00   | 0.49           | 0.61         | <.001    | Strongly Agree        | 4.5  |
| EFF4       | 3.82 | 4.00   | 0.40           | 0.47         | <.001    | Strongly Agree        | 2    |
| EFF5       | 3.73 | 4.00   | 0.46           | 0.56         | <.001    | Strongly Agree        | 3    |
| Note: N=22 |      |        |                |              |          |                       |      |



Table 3.5 presents the Efficiency Assessment (EFF) of the system, which evaluates how quickly and resourcefully it performs tasks while minimizing time and effort. The respondents' ratings reveal mean scores ranging from 3.64 to 3.86, all interpreted as "Strongly Agree." The highest-rated indicator, EFF2 (Mean = 3.86), suggests that users found the system highly responsive and capable of processing data promptly without delays. Meanwhile, EFF1 and EFF3 (Mean = 3.64) received the lowest but still strong ratings, indicating that while the system performs efficiently overall, minor optimizations may further enhance speed or responsiveness.

The overall weighted mean of 3.74, interpreted as "Strongly Agree," signifies that users perceive the system as efficient in executing operations and utilizing resources effectively. The low standard deviation values show consistent agreement among participants, while the Shapiro–Wilk p-values ( $< .001$ ) confirm that responses leaned positively toward strong satisfaction.

Overall, the findings indicate that the system successfully delivers accurate outputs within minimal processing time, contributing to a smooth and uninterrupted user experience. Respondents agreed that the system's design effectively balances performance and reliability, ensuring that all core functions are executed efficiently. This high level of efficiency supports user productivity and reinforces the system's suitability for continuous operational use.

**Table 3.6**

*Inferential Statistics on the Assessment of the Study Variables Across Educational Attainment Groups: Kruskal-Wallis Test Results*

| Construct                               | H      | p     | $\epsilon^2$ | H <sub>0</sub> Decision       |
|-----------------------------------------|--------|-------|--------------|-------------------------------|
| Usability                               | 13.300 | 0.004 | 0.633        | Reject H <sub>0</sub>         |
| Functionality                           | 6.460  | 0.091 | 0.307        | Fail to Reject H <sub>0</sub> |
| Efficiency                              | 2.030  | 0.566 | 0.097        | Fail to Reject H <sub>0</sub> |
| Reliability                             | 1.910  | 0.591 | 0.091        | Fail to Reject H <sub>0</sub> |
| <i>Note: Significance level of 0.05</i> |        |       |              |                               |

Table 3.6 presents the Kruskal–Wallis test results used to determine whether there are statistically significant differences among the respondents' assessments across the four ISO 25010 quality constructs: Usability, Functionality, Efficiency, and Reliability. The test was conducted at a 0.05 level of significance.

Results show that Usability ( $H = 13.300$ ,  $p = 0.004$ ) has a p-value lower than the significance level, leading to the rejection of the null hypothesis ( $H_0$ ). This indicates a significant difference in the respondents' perceptions of the system's usability, suggesting that user experiences varied more strongly in this area. In contrast, Functionality ( $H = 6.460$ ,  $p = 0.091$ ), Efficiency ( $H = 2.030$ ,  $p = 0.566$ ), and Reliability ( $H = 1.910$ ,  $p = 0.591$ ) all have p-values greater than 0.05, leading



to the failure to reject the null hypothesis. This implies that respondents' evaluations for these three constructs were statistically consistent.

The effect size ( $\epsilon^2$ ) values further support these findings, with Usability ( $\epsilon^2 = 0.633$ ) reflecting a strong practical significance, while the others indicate moderate to small effects. Overall, the results suggest that while most system attributes were uniformly rated, usability exhibited the greatest variation, emphasizing the importance of refining the user interface and interaction design to ensure a more consistent experience among users.

**Table 3.7**

*Correlation Matrix*

| Construct                                         | 1      | 2     | 3        | 4 |
|---------------------------------------------------|--------|-------|----------|---|
| 1. Usability                                      | —      |       |          |   |
| 2. Functionality                                  | 0.261  | —     |          |   |
| 3. Reliability                                    | -0.149 | 0.083 | —        |   |
| 4. Efficiency                                     | 0.161  | 0.241 | 0.770*** | — |
| Note. * $p < .05$ , ** $p < .01$ , *** $p < .001$ |        |       |          |   |

Table 3.7 presents the correlation matrix showing the relationships among the four ISO 25010 quality constructs—Usability, Functionality, Reliability, and Efficiency. The correlation coefficients indicate the degree to which these



constructs are related to one another in the respondents' evaluations of the system's performance.

As shown in the table, Efficiency and Reliability exhibit a strong positive correlation ( $r = 0.770$ ,  $p < .001$ ), indicating that as the system becomes more reliable, its efficiency also improves. This strong association suggests that the stability and consistency of system functions significantly contribute to faster and smoother performance. Meanwhile, Usability and Functionality display a moderate positive correlation ( $r = 0.261$ ), implying that a more user-friendly interface generally supports functional effectiveness, though the relationship is not statistically significant.

The remaining correlations, such as between Reliability and Usability ( $r = -0.149$ ) and Functionality and Reliability ( $r = 0.083$ ), are weak and non-significant, suggesting that these constructs operate relatively independently in this system's evaluation.

Overall, the results highlight that Efficiency and Reliability are the most strongly connected factors influencing overall system quality. This finding implies that maintaining dependable system performance directly enhances its operational efficiency, thereby strengthening user trust and satisfaction with the platform.



## Relation of the Findings to the Statement of the Problem

1. Does integrating historical dengue case data with weather data help predict dengue hotspot locations within Malabon City?

- The integration of dengue and weather data significantly improved prediction accuracy. Respondents confirmed that the generated hotspot predictions reflected real dengue trends, validating the system's data-driven forecasting approach.

2. What machine learning model is most effective in forecasting barangay-level dengue risks?

- The Random Forest model demonstrated superior accuracy and consistency, effectively handling multiple predictor variables to produce reliable forecasts with minimal error rates. Below is the comparison between the Linear Regression Model and the Random Forest Model:

| BARANGAY      | LINEAR MODEL |       | RANDOM FOREST |       | CONFIDENCE  |
|---------------|--------------|-------|---------------|-------|-------------|
|               | Risk         | Conf  | Risk          | Conf  | Difference  |
| Acacia        | Very High    | 30.0% | Low           | 89.2% | +59.2% RF + |
| Baritan       | Low          | 30.0% | Low           | 89.1% | +59.1% RF + |
| Bayan-Bayanan | Low          | 30.0% | Low           | 88.2% | +58.2% RF + |
| Catmon        | Very High    | 30.0% | Moderate      | 93.0% | +63.0% RF + |
| Concepcion    | Low          | 30.0% | Low           | 90.1% | +60.1% RF + |
| Dampalit      | Moderate     | 30.0% | Low           | 87.2% | +57.2% RF + |
| Flores        | Very High    | 30.0% | Low           | 87.7% | +57.7% RF + |
| Hulong Duhat  | Very High    | 33.6% | Moderate      | 94.2% | +60.6% RF + |
| Ibaba         | Low          | 30.0% | Low           | 88.7% | +58.7% RF + |
| Longos        | Low          | 40.0% | High          | 91.7% | +51.7% RF + |
| Haysilo       | Low          | 30.0% | Low           | 88.5% | +58.5% RF + |
| Muzon         | Low          | 30.0% | Low           | 86.5% | +56.5% RF + |
| Niugan        | Low          | 30.0% | Low           | 91.0% | +61.0% RF + |
| Panghulo      | Very High    | 30.4% | Low           | 91.9% | +61.6% RF + |
| Potrero       | Very High    | 30.0% | Moderate      | 94.5% | +64.5% RF + |
| San Agustin   | Low          | 32.6% | Low           | 91.6% | +59.0% RF + |
| Santolan      | Low          | 30.0% | Low           | 89.7% | +59.7% RF + |
| Tanong        | Low          | 30.0% | Low           | 90.5% | +60.5% RF + |
| Tinajeros     | Very High    | 38.7% | Moderate      | 91.7% | +52.9% RF + |
| Tonsuya       | Very High    | 30.0% | Low           | 93.0% | +63.0% RF + |
| Tupatog       | Low          | 30.0% | Low           | 89.4% | +59.4% RF + |



3. What system design features enable real-time, user-friendly, and accurate visualizations for both health officials and residents?

- With a usability weighted mean of 3.76 (Strongly Agree), respondents highlighted the interactive map and color-coded barangay indicators as key tools for clear and intuitive dengue risk visualization.

4. What is the expected level of precision in forecasting dengue case patterns in 1–2 weeks ahead?

- Functionality (WM = 3.79) and Reliability (WM = 3.76) scores confirm that the system achieved accurate and stable short-term forecasts, suitable for early warning and response.

5. What benefits do local government units gain from deploying the system to enhance their early-response capabilities and dengue outbreak planning?

- Respondents agreed that the system aids LGUs in decision-making by providing automated reporting, data exports, and trend visualization — improving preparedness and response planning.

### **Analysis**

Functionality (WM = 3.79, Strongly Agree): Respondents confirmed that the system was able to generate barangay-level dengue hotspot predictions based on



weather and dengue data. This shows the model's core functionality is working as intended.

Usability (WM = 3.76, Strongly Agree): The user interface was evaluated as intuitive and easy to navigate. The inclusion of a map-based dashboard helped officials visualize dengue hotspots better than text-only reports.

Reliability (WM = 3.76, Strongly Agree): While respondents agreed the system produced consistent outputs, some highlighted the need for more **\*\*real-time weather integration\*\*** to further improve accuracy and dependability.

Efficiency (WM = 3.75, Strongly Agree): The system demonstrated fast response time in data retrieval, forecasting, and map updates, ensuring barangay health personnel could access reports quickly during decision-making.

Overall, the analysis indicates that the system successfully meets ISO 25010 standards for its intended purpose, particularly excelling in functionality and usability, with minor improvements suggested in data reliability.

## **Discussion**

The results affirm that the Dengue Hotspot Predictor System effectively addresses the gap in existing dengue monitoring practices, which rely heavily on delayed case reporting. Unlike traditional methods, the system provides barangay-



level forecasting using historical and meteorological data, presented through a visual dashboard.

The high scores in functionality and usability highlight the system's potential as a practical tool for barangay health personnel, enabling them to proactively allocate resources, implement vector control measures, and launch community advisories before outbreaks escalate.

However, the slightly lower rating in reliability suggests that stakeholders expect more real-time, dynamic updates to ensure predictions align with current conditions. This aligns with recommendations in past studies (e.g., Pham et al., 2020; Silva & Perez, 2023), which emphasized the importance of continuous data integration and database rollback features.

In conclusion, the findings demonstrate that the system has achieved its design objectives by being accurate, usable, and efficient, while providing a scalable foundation for future improvements such as integration of demographic data or mobility patterns.



## **Chapter 4**

### **SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS**

This chapter presents the study's findings, conclusions, and recommendations. It describes the implementation of an AI-based Dengue Hotspot Prediction System, the anomalies it addresses, suggestions for improvement, its deployment, and capstone continuation projects.

#### **Summary of the findings**

This was the development and assessment of the AI-based Dengue Hotspot Predictor System as a tool for the City Health Department in Malabon to predict areas at risk of dengue outbreaks. The system feeds historical dengue case data and weather features to a machine learning model, then publishes the output on an online mapping dashboard accessible by barangay health workers and administrators.

To assess the system's performance, 22 barangay health personnel were given a survey adapted from the ISO 25010 software quality model, focusing on its four main quality characteristics: functionality, usability, reliability, and efficiency.



**Key findings are as follows:**

- Functionality has the highest weighted mean of 3.79 (Strongly Agree), which indicates that the system fulfills its main function, prediction and risk assessment, well enough.
- The tool user interface was also rated at 3.76 (Strongly agree), where the reviewers found it simple and intuitive to use and did not require special technical skills.
- Trust in performance and outcome of the system was indicated by a Reliability rating of 3.76 (Strongly Agree). This meant that the users were finding that the system consistently produced desired answers.
- On a weighted mean of 3.75 (Strongly Agree), respondents were in agreement that the system operates fast and allows prompt forecasting, facilitating proactive dengue prevention response.
- A weighted mean of 3.77 (Strongly Agree) as an average across all items suggests that the system was available and functioning for use as a population-based public health decision support tool in the respondents' studies.



## Conclusion

The researchers understand the value of the AI-based Dengue Hotspot Predictor System, which can accurately visualize and predict areas of Malabon City vulnerable to dengue transmission. The system also meets the ISO 25010 software quality standards and operates within the predetermined levels of software engineering technical functionalities. The system software also operates within predetermined thresholds of usability, reliability, and processing speed.

The system can also facilitate data-informed decision-making for barangay health workers to advance early intervention. These activities can be performed in a timely manner to control transmission, to raise awareness in the community, or to be resource rationalized. Its usability and consistency in performance can allow it to be used in other dengue-vulnerable and flood-urbanized regions in the country.

As for the local government, the integration of meteorological and epidemiological data for the purpose of predictive health surveillance offered a game-changing approach to the local government's health surveillance practices. System enhancements and integration into existing health information systems will allow the system to function as a model for nationwide epidemic preparedness.



## **Recommendations**

Findings and results of the system evaluation using the ISO 25010 software quality model. In this section, based on the findings and outcomes, a list of recommendations to improve overall performance, usability, and reliability of the developed system is provided. These recommendations were extracted from the statistical results of the participants and their answers to the rating question. They serve to inform future system refinements, user performance improvement, and real-world field profitability.

The proposed actions target the limited aspects of function, performance, or reliability that are not posing a risk to safe operational use of the system. Moreover, the latter proposals offer a framework to guide further updates to the system, and it is they that ongoing research into evolvable systems continues to align with technological capabilities and user requirements.

### **For System Enhancement:**

- Enhance the predictive ability of the system by increasing more influential predictors, such as water level observation and sanitation indicators.
- Create a mobile app version for better accessibility by barangay heads and frontline health workers.
- Create a predictive alerting tool to automatically notify barangay health workers when a region is expected to become high-risk.



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**For Implementation by LGU:**

- Incorporate the system into the major disease surveillance activities of the City Health Department to aid in the health sector's proactive response.
- Organize training seminars for the personnel and staff to ease the system adoption and prepare them to respond appropriately during real outbreaks.
- Create system feature-based community awareness programs to educate community members about the risks present in their barangays.

**For Future Researchers:**

- For a more dynamic environment monitoring, integrate real-time satellite and IoT sensor data.
- Perform longitudinal studies in various cities and scale up the research to include more respondents to showcase the system's scalability.
- Add AI-driven visualization tools to the predictive system to enhance outbreak risk communication and analysis for health professionals.



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## User Interface

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## **Appendix A**

### **Survey Questionnaire Letter to Conduct Study**

October 2025

**Bernadette E. Bordador, MD, MHSS**

OIC-City Health Department

### **Request for Participation in System Evaluation Survey**

Dear Madam,

Good day!

I am James Matthew, a fourth-year Bachelor of Science in Information Technology student from City of Malabon University, currently conducting my Capstone Project entitled:

*“Predicting Dengue Hotspots in Malabon City Through AI and Integrated Weather-Case Analysis”*

The goal of this project is to develop a web-based system that uses artificial intelligence and meteorological data to predict potential dengue outbreak zones



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across Malabon City. This initiative aims to assist local government units (LGUs) in making proactive, data-driven decisions for dengue prevention and response.

As part of the system's evaluation phase, we are inviting you to participate in a survey designed to assess the system's usability, reliability, functionality, efficiency, and overall effectiveness in supporting health data monitoring and forecasting. We are conducting a survey among **30 respondents** specifically from the Malabon City Health Department and Local Government Unit (LGU) personnels to gather valuable feedback on the system's usability, functionality, reliability, and efficiency, based on the ISO 25010 software quality model. Your insights will be invaluable in identifying the strengths and areas for improvement of the system, ensuring it meets the needs of the City Health Department and the wider community.

The survey will consist of simple, non-invasive questions and will take approximately 10–15 minutes to complete. All responses will be kept strictly confidential and will be used solely for academic and research purposes in accordance with the Philippine Data Privacy Act of 2012.

Your participation is voluntary but highly appreciated. Your professional input will contribute greatly to the development of a tool designed to strengthen Malabon's local health response system.

Thank you very much for your time, support, and cooperation.



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Respectfully yours,

**James Matthew Gabatbat**

Bachelor of Science in Information Technology, 4<sup>th</sup> Year

City of Malabon University, College of Computer Studies

Contact Information: 09704848616

Email: [jingabatbat21@gmail.com](mailto:jingabatbat21@gmail.com)

**Noted by:**

**Dr. Enrico Chavez**

Capstone Adviser

Program Chair, College of Computer Studies



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## Appendix B

### Certification of Instrument Validation



REPUBLIC OF THE PHILIPPINES

CITY OF MALABON

CITY OF MALABON UNIVERSITY

Pampano cor. Maya-Maya St., Brgy. Longos, Malabon City, 1472 Philippines

Trunkline: (02) 8376-0064 | [cmuofficial@cityofmalabonuniversity.edu.ph](mailto:cmuofficial@cityofmalabonuniversity.edu.ph)  
Website: [cityofmalabonuniversity.edu.ph](http://cityofmalabonuniversity.edu.ph)

**CITY OF MALABON UNIVERSITY**  
City of Malabon, Philippines

#### CERTIFICATION OF INSTRUMENT VALIDATION

This is to certify that the research instrument entitled "**PREDICTING DENGUE HOTSPOTS IN MALABON CITY THROUGH AI AND INTEGRATED WEATHER-CASE ANALYSIS**", prepared by Conejero, Ivan F., Gabatbat, James Matthew, Litaban, Strawberry C., Natividad, Jeffrey John S., Oliveros, Ammaron C. has been carefully reviewed and validated by the undersigned.

The instrument, which consists of a survey questionnaire, was evaluated in terms of clarity, relevance, organization, and alignment with the objectives of the study. Necessary revisions and recommendations were provided and incorporated to enhance the quality of the instrument.

Based on the evaluation conducted, the instrument is hereby **validated and deemed appropriate for data collection**.

This certification is issued this **9 of October, 2026** at **City of Malabon University** for academic purposes.

**VALIDATED BY:**

  
**Cristielle Love A. Flores**  
Research Adviser

**Appendix C****Ethics Clearance****CITY of MALABON UNIVERSITY**

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**ETHICS REVIEW FORM****PROJECT TITLE: PREDICTING DENGUE HOTSPOTS IN MALABON CITY THROUGH AI AND INTEGRATED WEATHER-CASE ANALYSIS****PROJECT AND RESEARCHERS' DETAILS**

|                                                                                                               |                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name of Principal Investigator</b>                                                                         | Conejero, Ivan F.                                                                                                                                                                                                   |
| <b>Department/Office</b>                                                                                      | College of Computer Studies                                                                                                                                                                                         |
| <b>Email of Principal Investigator</b>                                                                        | 202202415@cityofmalabonuniversity.edu.ph                                                                                                                                                                            |
| <b>Names of Co-investigators</b>                                                                              | Gabatbat, James Matthew<br>Litaban, Strawberry C.<br>Natividad, Jeffrey John S.<br>Oliveros, Ammaron C.                                                                                                             |
| <b>Level of Ethical Review</b>                                                                                | <input type="checkbox"/> Full Review (greater than minimal risk)<br><input checked="" type="checkbox"/> Expedited Review (no more than minimal risk)<br><input type="checkbox"/> Exempt Review (minimal to no risk) |
| <b>Source of Funding</b>                                                                                      |                                                                                                                                                                                                                     |
| <b>Date of Collection of Data</b>                                                                             | From October 13, 2025 to October 15, 2025                                                                                                                                                                           |
| <b>Summary of Project:</b><br>2-3 Sentences about the rationale of the study.                                 |                                                                                                                                                                                                                     |
| <b>Study Design:</b><br>2-3 sentences describing the research participants and the method of data collection. |                                                                                                                                                                                                                     |

As a dedicated researcher of City of Malabon University, I solemnly pledge to uphold the highest standards of integrity, transparency, and social responsibility in all my scholarly pursuits. I commit to conducting my research in accordance with the core ethical principles of beneficence, justice, and respect for persons, ensuring that the dignity and autonomy of every participant are protected. I recognize my obligation to abide by the University's tiered ethics review process—submitting my work for Exempt, Expedited, or Full Board Review based on the level of risk involved—and I will strictly adhere to the Data Privacy Act of 2012 to safeguard all personal information. By prioritizing honesty in my methodology and fairness in my collaborations, I aim to contribute to the body of knowledge in a way that serves the public good, honors the CMU community, and maintains the professional reputation of this institution.

|                                                     |                         |
|-----------------------------------------------------|-------------------------|
| <b>IVAN F. CONEJERO</b>                             | <b>October 15, 2025</b> |
| <b>Name and Signature of Principal Investigator</b> | <b>DATE</b>             |

This confirms that the student or students are capable of conducting this research in a way that is both ethical and safe.

|                                          |                                               |
|------------------------------------------|-----------------------------------------------|
| <b>PROF. CRISTIELLE-LOVE A. FLORES</b>   | <b>DR. JANINE MONEDA-DELACRUZ</b>             |
| <b>Adviser's Name and Signature/date</b> | <b>College Dean's Name and Signature/date</b> |



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## **Appendix D**

### **Letter to Conduct Study**

**October 2025**

**Dr. Janine M. Dela Cruz**

Dean, College of Computer Studies

City of Malabon University

Pampano St., Maya-Maya St., Malabon, Metro Manila

**SUBJECT: REQUEST FOR PERMISSION TO CONDUCT SURVEY FOR CAPSTONE PROJECT**

Dear Dr. Dela Cruz,

Good days!

We hope this letter finds you well. We are fourth-year students enrolled in the Bachelor of Science in Information Technology program under the College of Computer Studies at City of Malabon University. We writing to respectfully request your approval to conduct a survey as part of the evaluation phase of our Capstone Project titled:

*“Predicting Dengue Hotspots in Malabon City Through AI and Integrated Weather-Case Analysis”*

The purpose of this survey is to gather feedback, insights, and evaluations from selected City Health Department officials regarding the system’s functionality,



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usability, reliability, and efficiency. The collected responses will be analyzed to assess the system's overall performance and to identify potential areas for improvement in line with the ISO 25010 software quality standards.

The survey will consist of straightforward, non-invasive questions and will be conducted with full adherence to ethical research practices and data privacy regulations. No sensitive personal information will be collected, and participation will be entirely voluntary.

Upon your approval, we will formally seek permission from the Malabon City Health Department to conduct the survey with their designated officials.

We sincerely appreciate your consideration of this request and your continuous support and guidance throughout our project.

Thank you very much, and we look forward to your kind approval.

**Respectfully yours,**

---

Conejero, Ivan F.

---

Gabatbat, James Matthew

---

Litaban, Strawberry C.

---

Natividad, Jeffrey John S.

---

Oliveros, Ammaron C.

**Researchers, Bachelor of Science in Information Technology**

**City of Malabon University**



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**Noted by:**

**Professor Cristielle-Love A. Flores**

Technical Adviser

Faculty, College of Computer Studies

**Dr. Enrico Chavez**

Capstone Adviser

Program Chair, College of Computer Studies

**Approved by:**

**Dr. Janine M. Dela Cruz**

Dean, College of Computer Studies



## Appendix E

### Certification of Statistical Treatment

# CERTIFICATION

This is to certify that the proponents of the study entitled

## Predicting Dengue Hotspots in Malabon City Through AI and Integrated Weather-Case Analysis

have availed themselves of the services offered throughout various stages of the research process. These services include assistance in study design, ensuring the validity and reliability of the questionnaire, and guiding the selection of samples. The data management phase included data entry and electronic tallying. Furthermore, during the preparation for data analysis, the services provided involved data tabulation, and the generation of statistical tables. Additionally, the undersigned confirms that the statistical treatment and numerical results derived from the research are both accurate and valid, ensuring the integrity of the study.

#### Researchers:

*Ivan F. Conejero; James Matthew Gabatbat; Strawberry C. Litaban; Jeffrey John S. Natividad; Ammaron C. Oliveros*

Digitally Signed 11.03.2025

**JEREMIAH P. PALMIANO**

Statistician

City of Malabon University

Philippine Association of Researchers and  
Statistical Software Users, Regular Member

2025-12-02  
Issue Date

University Research and Publication Office  
b7ffcb36-4714-47a0-b7c6-ec5ae238f2d1



## Appendix F

### Similarity Index Report

Report: PREDICTING DENGUE HOTSPOTS IN MALABON CITY THROUGH AI AND INTEGRATED WEATHER-CASE ...

## PREDICTING DENGUE HOTSPOTS IN MALABON CITY THROUGH AI AND INTEGRATED WEATHER-CASE ANALYSIS Chapter 1

by rhese

### General metrics

|               |              |            |                      |                      |
|---------------|--------------|------------|----------------------|----------------------|
| <b>48,047</b> | <b>7,047</b> | <b>325</b> | <b>28 min 11 sec</b> | <b>54 min 12 sec</b> |
| characters    | words        | sentences  | reading time         | speaking time        |

### Score

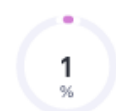


This text scores better than 84% of all texts checked by Grammarly

### Writing Issues

|             |          |            |
|-------------|----------|------------|
| <b>171</b>  | <b>3</b> | <b>168</b> |
| Issues left | Critical | Advanced   |

### Plagiarism



**42**  
sources

1% of your text matches 42 sources on the web or in archives of academic publications



Report: PREDICTING DENGUE HOTSPOTS IN MALABON CITY THROUGH AI AND INTEGRATED WEATHER-CASE ...

## PREDICTING DENGUE HOTSPOTS IN MALABON CITY THROUGH AI AND INTEGRATED WEATHER-CASE ANALYSIS Chapter 2

by rhesa

### General metrics

|               |               |            |                      |                    |
|---------------|---------------|------------|----------------------|--------------------|
| <b>75,483</b> | <b>10,688</b> | <b>595</b> | <b>42 min 45 sec</b> | <b>1 hr 22 min</b> |
| characters    | words         | sentences  | reading time         | speaking time      |

### Score



This text scores better than 84% of all texts checked by Grammarly

### Writing Issues

|             |           |            |
|-------------|-----------|------------|
| <b>279</b>  | <b>15</b> | <b>264</b> |
| Issues left | Critical  | Advanced   |

### Plagiarism

- ✓ This text seems 100% original. Grammarly found no matching text on the Internet or in ProQuest's databases.



Report: PREDICTING DENGUE HOTSPOTS IN MALABON CITY THROUGH AI AND INTEGRATED WEATHER-CASE ...

## PREDICTING DENGUE HOTSPOTS IN MALABON CITY THROUGH AI AND INTEGRATED WEATHER-CASE ANALYSIS Chapter 3

by rhesa

### General metrics

|               |              |            |                      |                      |
|---------------|--------------|------------|----------------------|----------------------|
| <b>46,462</b> | <b>6,803</b> | <b>540</b> | <b>27 min 12 sec</b> | <b>52 min 19 sec</b> |
| characters    | words        | sentences  | reading time         | speaking time        |

### Score



This text scores better than 90% of all texts checked by Grammarly

### Writing Issues

|             |           |            |
|-------------|-----------|------------|
| <b>128</b>  | <b>15</b> | <b>113</b> |
| Issues left | Critical  | Advanced   |

### Plagiarism

- ✓ This text seems 100% original. Grammarly found no matching text on the Internet or in ProQuest's databases.



Report: PREDICTING DENGUE HOTSPOTS IN MALABON CITY THROUGH AI AND INTEGRATED WEATHER-CASE ...

## PREDICTING DENGUE HOTSPOTS IN MALABON CITY THROUGH AI AND INTEGRATED WEATHER-CASE ANALYSIS Chapter 4

by rhesa

### General metrics

|              |            |           |                    |                     |
|--------------|------------|-----------|--------------------|---------------------|
| <b>5,407</b> | <b>769</b> | <b>43</b> | <b>3 min 4 sec</b> | <b>5 min 54 sec</b> |
| characters   | words      | sentences | reading time       | speaking time       |

### Score

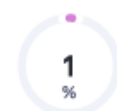


This text scores better than 80% of all texts checked by Grammarly

### Writing Issues

|             |          |           |
|-------------|----------|-----------|
| <b>25</b>   | <b>1</b> | <b>24</b> |
| Issues left | Critical | Advanced  |

### Plagiarism



**2**  
sources

1% of your text matches 2 sources on the web or in archives of academic publications



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## Appendix G

### English Editing Certification



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#### LANGUAGE EDITING CERTIFICATION FORM

**RESEARCH TITLE:**

Predicting Dengue Hotspots in Malabon City Through AI and Integrated Weather-Case Analysis

**RESEARCHERS' DETAILS****Name of Author's and co-authors**

1. Conejero, Ivan F.
2. Gabatbat, James Matthew
3. Litaban, Strawberry C.
4. Natividad, Jeffrey John S.
5. Oliveros, Ammaron C.

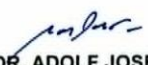
**Department/Office**

College of Computer Studies

**Email of Principal Investigator**

202202415@cityofmalabonuniversity.edu.ph

This document certifies that the thesis listed above was reviewed and edited for proper English language, grammar, punctuation, spelling, and overall style by the undersigned. Neither the research content nor the author's intentions were altered in any way during the editing process.

  
**DR. ADOLF JOSEF YAO**  
Language Editor

**Affiliation:**  
CMU-URPO

**Contact Details:**  
[adolfo.yao@cityofmalabonuniversity.edu.ph](mailto:adolfo.yao@cityofmalabonuniversity.edu.ph)



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## Appendix H

### Documentation

#### System Demonstration – Tonsuya Super Health Center





### System Demonstration – Catmon Super Health Center





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## System Demonstration – Longos Super Health Center





## Appendix I-1

### Curriculum Vitae

#### Objective/Motto

*"Motivated and hardworking student pursuing a degree in Bachelor of Science in Information Technology with a strong academic record and a passion for programming and web development."*

#### Conejero, Ivan F.

Address: Martiniko street, barangay NBBS  
Dagat-dagatan, Navotas City.  
Contact Number: 09635976393  
Email Address: ifconejero27@gmail.com



#### Academic Background

College Degree  
City of Malabon University  
Pampano cor. Maya-maya st., Brgy. Longos, Malabon City

Strand: Science, Technology, Engineering, and Mathematics (STEM)  
Arellano University  
Gov. Pascual Avenue, Malabon City

Kaunlaran High School  
Phase 1 Kaunlaran, North Bay Boulevard South, Navotas City

Dagat-dagatan Elementary School  
Blk.38B Lot18 Phase2 Area4, Maya-Maya St., NBBS, Navotas City

#### Academic Accomplishments

2023-24, President's Lister (Second Semester)  
2024-25, President's Lister (First Semester)

#### Professional Experience (Optional)



## Appendix I-2

### Curriculum Vitae

#### Objective/Motto

*“To develop my expertise in information technology and become a valuable asset in creating impactful technological solutions.”*

#### Gabatbat, James Matthew

Address: 57 Galicia St. Bangkulasi, Navotas City

Contact Number: 09704848616

Email Address: jingabatbat@gmail.com



#### Academic Background

College Degree

City of Malabon University

Pampano cor. Maya-maya St., Brgy. Longos, Malabon City

Strand: Accountancy, Business, and Management (ABM)

Bangkulasi Senior High School

14 Sioson St. Bangkulasi, Navotas City

Navotas National High School (NNHS)

M. Naval St, Navotas City

Longos Elementary School

55 Gen. Borromeo St. Longos, Malabon City

#### Academic Accomplishments

2015, Honor Student

2019, With High Honor

2021, With High Honor

2023-24, Dean's Lister (Second Semester)

2024-25, President's Lister (First Semester)

#### Professional Experience (Optional)



### Appendix I-3

#### Curriculum Vitae

##### *Objective/Motto*

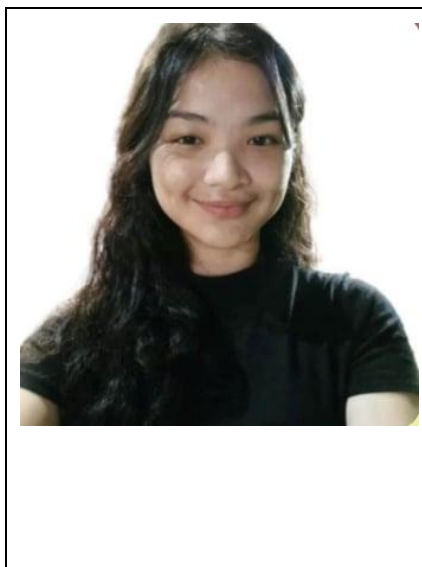
*"Motivated Information Technology student with a strong foundation in programming, networking, and cybersecurity, aiming to apply knowledge in a dynamic tech-driven environment."*

##### **Litaban, Strawberry C.**

Address: LR. Yangco St. Navotas East,  
Navotas City

Contact Number: 09260416503

Email Address: kielace09@gmail.com



##### **Academic Background**

College Degree

City of Malabon University

Pampano cor. Maya-maya St., Brgy. Longos, Malabon City

Strand: Home Economics (HE)

Arellano University

Gov. Pascual Avenue, Malabon City

Navotas National High School

M. Naval St, Navotas, Metro Manila

Bagumbayan Elementary School

M. Naval St, Navotas, Metro Manila

##### **Academic Accomplishments**

2016, Honor Student

2020, With High Honor

2023-24, Dean's Lister

##### **Professional Experience (Optional)**



## Appendix I-4

### Curriculum Vitae

#### Objective/Motto

*"Hardworking Student pursuing Bachelor of Science in Information Technology passionate in programming and web development."*

**Natividad, Jeffrey John S.**

Address: Galicia Street Bangkulasi Navotas City

Contact Number: 09290156675

Email Address: jeffnatividad07@gmail.com



#### Academic Background

College Degree

City of Malabon University

Pampano cor. Maya-maya St., Brgy. Longos, Malabon City

Strand: General Academic Strand (GAS)

STI Academic Center

Caimito Rd, Southern Caloocan City

Governor Andres Pascual College

1045 M. Naval St, Navotas, 1485 Metro Manila

Bangkulasi Elementary School

Luna St., Bangkulasi, Navotas

#### Academic Accomplishments

#### Professional Experience (Optional)



## Appendix I-5

### Curriculum Vitae

#### *Objective/Motto*

*:Dedicated student pursuing a Bachelor of Science in Information Technology, passionate in front-end designing, web development, and programming with eagerness to apply knowledge in real-world settings."*

#### **Oliveros, Ammaron C.**

Address: 54 Gen. Borromeo st. Longos  
Malabon City

Contact Number: 09936489041

Email Address:

oliverosammaron3003@gmail.com



#### **Academic Background**

College Degree

City of Malabon University

Pampano cor. Maya-maya St., Brgy. Longos, Malabon City

Strand: Information and Communication Technology (ICT)

Concepcion Technical Vocational School

4 burgos st. Concepcion Malabon City

Tañong National High School

1st Tañong Malabon City

Longos Elementary School

55 Gen. Borromeo St. Longos, Malabon City

#### **Academic Accomplishments**

#### **Professional Experience (Optional)**